

19 Spontaneously Broken Global Symmetries

Much of the physics of this century has been built on principles of symmetry: first the spacetime symmetries of Einstein's 1905 special theory of relativity, and then internal symmetries, such as the approximate $SU(2)$ isospin symmetry of the 1930s. It was therefore exciting when in the 1960s it was discovered that there are more internal symmetries than could be guessed by inspection of the spectrum of elementary particles. There are exact or approximate symmetries of the underlying theory that are 'spontaneously broken,' in the sense that they are not realized as symmetry transformations of the physical states of the theory, and in particular do not leave the vacuum state invariant. The breakthrough was the discovery of a broken approximate global $SU(2) \cdot SU(2)$ symmetry of the strong interactions, which will be discussed in detail in Section 19.3. This was soon followed by the discovery of an exact but spontaneously broken local $SU(2) \cdot U(1)$ symmetry of the weak and electromagnetic interactions, which will be taken up along with more general broken local symmetries in Chapter 21. In this chapter we shall begin with a general discussion of broken global symmetries, and then move on to physical examples.

19.1 Degenerate Vacua

We do not have to look far for examples of spontaneous symmetry breaking. Consider a chair. The equations governing the atoms of the chair are rotationally symmetric, but a solution of these equations, the actual chair, has a definite orientation in space. Here we will be concerned not so much with the breaking of symmetries by *objects* like chairs, but rather with the symmetry breaking in the ground state of any realistic quantum field theory, the vacuum.

A spontaneously broken symmetry in field theory is always associated with a degeneracy of vacuum states. For instance, consider a symmetry transformation of the action, and of the measure used in integrating over fields, that acts linearly on a set of scalar fields $\phi_n(x)$:

$$\phi_n(x) \longrightarrow \phi'_n(x) = \sum_m L_{nm} \phi_m(x). \quad (19.1.1)$$

(The ϕ_n need not be elementary fields; they can be composite objects, like $\bar{\psi}\Gamma_n\psi$.) As we saw in Section 16.4, the quantum effective action $\Gamma[\phi]$ will then have the same symmetry

$$\Gamma[\phi] = \Gamma[L\phi]. \quad (19.1.2)$$

For the vacuum the expectation value of $\phi(x)$ must be at a minimum of the vacuum energy $-\Gamma[\phi]$, say at $\phi(x) = \bar{\phi}$ (a constant). But if $L\bar{\phi} \neq \bar{\phi}$, then this vacuum is not unique; $-\Gamma[\phi]$ has the same value at $\phi = L\bar{\phi}$ as it does at $\bar{\phi}$. In the simple special case where the symmetry transformation (19.1.1) is a reflection, $\phi \rightarrow -\phi$, if $-\Gamma(\phi)$ has a minimum at a non-zero value $\bar{\phi}$ of ϕ , then it has two minima, at $\bar{\phi}$ and $-\bar{\phi}$, each corresponding to a state of broken symmetry.

We are not yet ready to conclude that in such cases the symmetry is broken, because we have not yet ruled out the possibility that the true vacuum is a linear superposition of vacuum states in which ϕ_m has various expectation values, which would respect the

assumed symmetry. For instance, in a theory with a symmetry $\phi \rightarrow -\phi$, even if $\Gamma(\phi)$ has a minimum for some non-zero value $\bar{\phi}$ of ϕ , how do we know that the true vacuum is one of the states $|\Omega_{\pm}\rangle$ for which Φ has expectation values $\bar{\phi}$ and $-\bar{\phi}$, and not some linear combination like $|\Omega_{+}\rangle + |\Omega_{-}\rangle$ that *would* respect the symmetry under $\phi \rightarrow -\phi$? The assumed symmetry under the transformation $\phi \rightarrow -\phi$ tells us that the vacuum matrix elements of the Hamiltonian are

$$\langle \Omega_{+} | H | \Omega_{+} \rangle = \langle \Omega_{-} | H | \Omega_{-} \rangle = a$$

(with a real) and

$$\langle \Omega_{+} | H | \Omega_{-} \rangle = \langle \Omega_{-} | H | \Omega_{+} \rangle = b$$

(with b real), so the eigenstates of the Hamiltonian are $|\Omega_{+}\rangle \pm |\Omega_{-}\rangle$, with energies $a \pm |b|$. These energy eigenstates are invariant (or invariant up to a sign) under the symmetry $\phi \rightarrow -\phi$. In fact, the same issue also arises for chairs. The quantum mechanical ground state of an isolated chair is actually rotationally invariant; it is a state with zero angular momentum quantum numbers, and hence with no definite orientation in space.

Spontaneous symmetry breaking actually occurs only for idealized systems that are infinitely large. The appearance of broken symmetry for a chair arises because it has a macroscopic moment of inertia I , so that its ground state is part of a tower of rotationally excited states whose energies are separated by only tiny amounts, of order $\hbar^2/I$. This gives the state vector of the chair an exquisite sensitivity to external perturbations; even very weak external fields will shift the energy by much more than the energy difference of these rotational levels. In consequence, any rotationally asymmetric external field will cause the ground state or any other state of the chair with definite angular momentum numbers rapidly to develop components with other angular momentum quantum numbers. The states of the chair that are relatively stable with respect to small external perturbations are not those with definite angular momentum quantum numbers, but rather those with a definite orientation, in which the rotational symmetry of the underlying theory is broken.

For the vacuum also, the possibility of spontaneous symmetry breaking is again related to the large size of the system, specifically to the large volume of space. In the above example of a reflection symmetry, the off-diagonal matrix element b of the Hamiltonian involves an integration over field configurations that tunnel from the minimum at $\phi = \bar{\phi}$ to the one at $\phi = -\bar{\phi}$, so it is smaller than the diagonal matrix element a by a barrier penetration factor that for a spatial volume $\mathcal{V}$ is of the form $\exp(-C\mathcal{V})$, where C is a positive constant¹ depending on the microscopic parameters of the theory. The two energy eigenstates $|\Omega_{+}\rangle \pm |\Omega_{-}\rangle$ are thus essentially degenerate for any macroscopic volume, and so are strongly mixed by any perturbation that is an odd functional of ϕ . Even if such a perturbation H' is very weak, its diagonal elements $\langle \Omega_{\pm} | H' | \Omega_{\pm} \rangle$ will differ by much more than the exponentially suppressed off-diagonal elements of either H or the perturbation.

¹For instance, by analogy with the classic wave mechanical problem of barrier penetration, for a Lagrangian density of the form $-\frac{1}{2}\partial_{\mu}\phi\partial^{\mu}\phi - V(\phi)$, we have $C = \int_{-\bar{\phi}}^{+\bar{\phi}} \sqrt{2V(\phi)} d\phi$. We will not bother to calculate the off-diagonal matrix element b here, because we shall soon give a general argument that shows that it vanishes for infinite volume.

Thus the vacuum eigenstates of the perturbed Hamiltonian will be very close to either one of the broken symmetry states $|\Omega_{\pm}\rangle$ which diagonalize the perturbation, and *not* to the invariant states $|\Omega_+\rangle \pm |\Omega_-\rangle$. Which one of the states $|\Omega_{\pm}\rangle$ is the true vacuum for very small perturbations? This depends on the perturbation, but since these two states are related by a symmetry transformation of the original Hamiltonian, it doesn't matter; if the perturbation is sufficiently small, no observer will be able to tell the difference.

The vanishing of matrix elements between vacuum states with different field expectation values becomes exact in a space of infinite volume [1]. For infinite volume, a general vacuum state $|\Omega_v\rangle$ may be defined as a state with zero momentum

$$\mathbf{P}|\Omega_v\rangle = 0 \quad (19.1.3)$$

for which this is a *discrete* momentum eigenvalue. (This excludes single-particle or multiparticle states, for which the momentum value zero is always part of a continuum of momentum values in a space of infinite volume.) In general there may be a number of such states. They can usually be expanded in a discrete set, and our notation will treat them as if they were discrete. They will be chosen to be orthonormal

$$\langle\Omega_u|\Omega_v\rangle = \delta_{uv}. \quad (19.1.4)$$

Any matrix element of a product of local Hermitian operators at equal times between these states may be expressed as a sum over states:

$$\begin{aligned} \langle\Omega_u|A(\mathbf{x})B(\mathbf{0})|\Omega_v\rangle &= \sum_w \langle\Omega_u|A(\mathbf{0})|\Omega_w\rangle \langle\Omega_w|B(\mathbf{0})|\Omega_v\rangle \\ &+ \int d^3p \sum_N \langle\Omega_u|A(\mathbf{0})|N, \mathbf{p}\rangle \langle N, \mathbf{p}|B(\mathbf{0})|\Omega_v\rangle e^{-i\mathbf{p}\cdot\mathbf{x}}, \end{aligned} \quad (19.1.5)$$

where $|N, \mathbf{p}\rangle$ are a set of orthonormalized continuum states of definite three-momentum $\mathbf{p}$ that together with the $|\Omega_v\rangle$ span the whole physical Hilbert space. (Here N may include continuous as well as discrete labels. Also, we are dropping time arguments.) We assume without proof that because the $|N, \mathbf{p}\rangle$ belong to the continuous spectrum of the momentum operator $\mathbf{P}$, the dependence of matrix elements on $\mathbf{p}$ is smooth enough (that is, Lebesgue integrable) to allow the use of the Riemann–Lebesgue theorem [2], so that the integral over $\mathbf{p}$ vanishes as $|\mathbf{x}| \rightarrow \infty$. In this limit, we have then

$$\langle\Omega_u|A(\mathbf{x})B(\mathbf{0})|\Omega_v\rangle \xrightarrow{|\mathbf{x}| \rightarrow \infty} \sum_w \langle\Omega_u|A(\mathbf{0})|\Omega_w\rangle \langle\Omega_w|B(\mathbf{0})|\Omega_v\rangle. \quad (19.1.6)$$

Likewise,

$$\langle\Omega_u|B(\mathbf{0})A(\mathbf{x})|\Omega_v\rangle \xrightarrow{|\mathbf{x}| \rightarrow \infty} \sum_w \langle\Omega_u|B(\mathbf{0})|\Omega_w\rangle \langle\Omega_w|A(\mathbf{0})|\Omega_v\rangle. \quad (19.1.7)$$

But causality tells us that the equal-time commutator $[A(\mathbf{x}), B(\mathbf{0})]$ vanishes for $\mathbf{x} \neq \mathbf{0}$ (see Section 5.1), so the matrix elements (19.1.6) and (19.1.7) are equal, and thus the Hermitian matrices $\langle\Omega_u|A(\mathbf{0})|\Omega_v\rangle$, $\langle\Omega_u|B(\mathbf{0})|\Omega_v\rangle$, etc., must all commute with one another. It follows

that they can all be simultaneously diagonalized. Changing if necessary to this basis, we have then for every Hermitian local operator $A(x)$ of the theory

$$\langle \Omega_u \| A(\mathbf{0}) \| \Omega_v \rangle = \delta_{uv} a_v \quad (19.1.8)$$

with a_v a real number, the expectation value of A in the state $\| \Omega_v \rangle$. So for infinite volume any Hamiltonian constructed from local operators will have vanishing matrix elements between the different vacua $\| \Omega_v \rangle$. In the absence of off-diagonal terms in the Hamiltonian, any two $\| \Omega_v \rangle$ s connected by a symmetry operation will be degenerate. A symmetry-breaking perturbation built out of such local operators will be diagonal in the same basis, and will therefore yield a ground state that is one of the $\| \Omega_v \rangle$ s, rather than a linear combination of them.

It is reassuring that the vacuum states $\| \Omega_v \rangle$ which are stable against small field-dependent perturbations are also vacuum states in which the cluster decomposition condition (see Chapter 4) is satisfied. This principle requires that for the physical vacuum state $\| \Omega \rangle$

$$\langle \Omega \| A(\mathbf{x}) B(\mathbf{0}) \| \Omega \rangle \xrightarrow{|\mathbf{x}| \rightarrow \infty} \langle \Omega \| A(\mathbf{x}) \| \Omega \rangle \langle \Omega \| B(\mathbf{0}) \| \Omega \rangle. \quad (19.1.9)$$

This condition is satisfied if we take the vacuum state $\| \Omega \rangle$ to be any one of the states $\| \Omega_v \rangle$ in the basis defined by (19.1.8), but not if we take it to be a general linear combination of several of the $\| \Omega_v \rangle$ s.

19.2 Goldstone Bosons

We now specialize to the case of a spontaneously broken *continuous* symmetry. In this case there is a theorem, that (with one important exception, to be considered in Chapter 21) the spectrum of physical particles must contain one particle of zero mass and spin for each broken symmetry. Such particles, known as Goldstone bosons (or Nambu–Goldstone bosons) were first encountered in specific models by Goldstone [3] and Nambu [4]; two general proofs of their existence were then given by Goldstone, Salam, and myself [5]. This section will present both of these proofs, and then go on to consider the properties of the Goldstone bosons.

Suppose that the action and measure are invariant under a continuous symmetry, under which a set of Hermitian scalar fields $\phi_n(x)$ (either elementary or composite) are subjected to the linear infinitesimal transformations

$$\phi_n(x) \longrightarrow \phi_n(x) + i\epsilon \sum_m t_{nm} \phi_m(x), \quad (19.2.1)$$

with it_{nm} a finite real matrix. As we found in Section 16.4, the effective action is then also invariant under this transformation

$$\sum_{n,m} \int \frac{\delta \Gamma[\phi]}{\delta \phi_n(x)} t_{nm} \phi_m(x) d^4x = 0. \quad (19.2.2)$$

We shall specialize to the case of a translationally invariant theory with constant fields ϕ_n , where as we saw in Section 16.2, the effective action takes the form

$$\Gamma[\phi] = -\mathcal{V}V(\phi), \quad (19.2.3)$$

where $\mathcal{V}$ is the spacetime volume and $V(\phi)$ is known as the effective potential. Eq. (19.2.2) may then be written

$$\sum_{n,m} \frac{\partial V(\phi)}{\partial \phi_n} t_{nm} \phi_m = 0. \quad (19.2.4)$$

We will use this symmetry requirement in a form obtained by differentiating again with respect to ϕ_ℓ :

$$\sum_n \frac{\partial V(\phi)}{\partial \phi_n} t_{n\ell} + \sum_{n,m} \frac{\partial^2 V(\phi)}{\partial \phi_n \partial \phi_\ell} t_{nm} \phi_m = 0. \quad (19.2.5)$$

Now specialize to the case where ϕ_n is at the minimum of $V(\phi)$, that is, at the vacuum expectation value $\bar{\phi}_n$. Since $V(\phi)$ is stationary at its minimum, the first term in Eq. (19.2.5) vanishes, so

$$\sum_{n,m} \frac{\partial^2 V(\phi)}{\partial \phi_n \partial \phi_\ell} \bigg|_{\phi=\bar{\phi}} t_{nm} \bar{\phi}_m = 0. \quad (19.2.6)$$

The general results of Section 16.1 show that the second derivative in Eq. (19.2.6) is just the sum of all connected one-particle-irreducible momentum-space Feynman diagrams with external lines labelled n and ℓ and carrying zero four-momentum. As shown at the end of Section 16.1, it is related to the reciprocal of the momentum-space propagator by

$$\frac{\partial^2 V(\phi)}{\partial \phi_n \partial \phi_\ell} = \Delta_{n\ell}^{-1}(0), \quad (19.2.7)$$

so Eq. (19.2.6) gives

$$\sum_{n,m} \Delta_{n\ell}^{-1}(0) t_{nm} \bar{\phi}_m = 0. \quad (19.2.8)$$

Thus, if the symmetry is broken, so that $\sum_m t_{nm} \bar{\phi}_m$ is non-zero, then this is an eigenvector of $\Delta_{n\ell}^{-1}(0)$ with eigenvalue zero. The existence of such an eigenvector means that $\Delta_{n\ell}(q)$ has a pole at $q^2 = 0$. The rank of the residue of the pole at $q^2 = 0$ is equal to the dimensionality of the space of the vectors $t\bar{\phi}$, with t running over all the generators of continuous symmetries of the theory. Roughly speaking, there is one massless boson for every independent broken symmetry.

In the classic example of a broken symmetry, the Lagrangian involves a set of N real scalar fields ϕ_n , and takes the form

$$\mathcal{L} = -\frac{1}{2} \sum_n \partial_\mu \phi_n \partial^\mu \phi_n - \frac{\mathcal{M}^2}{2} \sum_n \phi_n \phi_n - \frac{g}{4} \left(\sum_n \phi_n \phi_n \right)^2. \quad (19.2.9)$$

This is invariant under the group $O(N)$, consisting of rotations of the N -vector with components ϕ_n . For constant fields the effective potential in the tree approximation is given simply by minus the non-derivative terms in the Lagrangian density

$$V(\phi) \simeq \frac{\mathcal{M}^2}{2} \sum_n \phi_n \phi_n + \frac{g}{4} \left(\sum_n \phi_n \phi_n \right)^2. \quad (19.2.10)$$

As usual we suppose that g is positive-definite. (Otherwise the minimum, if any, of $V(\phi)$ lies outside the range of validity of perturbation theory.) If $\mathcal{M}^2$ is also positive, the minimum of $V(\phi)$ is at the point $\phi = 0$, which is invariant under $O(N)$. On the other hand, for $\mathcal{M}^2 < 0$ the minimum is at points $\bar{\phi}_n$ at which

$$\sum_n \bar{\phi}_n \bar{\phi}_n = -\mathcal{M}^2/g. \quad (19.2.11)$$

The mass matrix in the tree approximation is then

$$\begin{aligned} M_{nm}^2 &= \left. \frac{\partial^2 V(\phi)}{\partial \phi_n \partial \phi_m} \right|_{\phi=\bar{\phi}} \\ &= \mathcal{M}^2 \delta_{nm} + g \delta_{nm} \sum_{\ell} \bar{\phi}_{\ell} \bar{\phi}_{\ell} + 2g \bar{\phi}_n \bar{\phi}_m \\ &= 2g \bar{\phi}_n \bar{\phi}_m. \end{aligned} \quad (19.2.12)$$

This has one eigenvector $\bar{\phi}_n$ with a non-zero eigenvalue:

$$m^2 = 2g \sum_n \bar{\phi}_n \bar{\phi}_n = 2|\mathcal{M}^2|, \quad (19.2.13)$$

and $N - 1$ eigenvectors perpendicular to $\bar{\phi}$ with eigenvalue zero. The reason for the appearance of only $N - 1$ Goldstone bosons is just that $O(N)$ is broken down to $O(N - 1)$ (the subgroup of $O(N)$ that leaves $\bar{\phi}$ invariant), and so the number of independent broken symmetries is the dimensionality of $O(N)$ minus the dimensionality of $O(N - 1)$, or

$$\frac{1}{2}N(N - 1) - \frac{1}{2}(N - 1)(N - 2) = N - 1. \quad (19.2.14)$$

Here is another proof of the existence of Goldstone bosons, one that makes no use of the effective action formalism. As we saw in Chapter 7, any continuous symmetry of the action leads to the existence of a conserved current J^μ :

$$\frac{\partial J^\mu(x)}{\partial x^\mu} = 0, \quad (19.2.15)$$

with a charge Q that induces the associated symmetry transformation

$$Q = \int d^3x J^0(\mathbf{x}, 0), \quad (19.2.16)$$

and

$$[Q, \phi_n(x)] = - \sum_m t_{nm} \phi_m(x). \quad (19.2.17)$$

Operator relations like Eqs. (19.2.15)–(19.2.17) are unaffected by spontaneous symmetry breaking, which is manifested in the properties of physical *states*. Now, consider the vacuum expectation value of the commutator of the current and field. Summing over intermediate states, this is

$$\langle \Omega | [J^\lambda(y), \phi_n(x)] | \Omega \rangle = (2\pi)^{-3} \int d^4p \left[\rho_n^\lambda(p) e^{ip \cdot (y-x)} - \tilde{\rho}_n^\lambda(p) e^{ip \cdot (y-x)} \right], \quad (19.2.18)$$

where, using translation invariance,

$$(2\pi)^{-3} i \rho_n^\lambda(p) = \sum_N \langle \Omega \| J^\lambda(0) \| N \rangle \langle N \| \phi_n(0) \| \Omega \rangle \delta^4(p - p_N), \quad (19.2.19)$$

$$(2\pi)^{-3} i \tilde{\rho}_n^\lambda(p) = \sum_N \langle \Omega \| \phi_n(0) \| N \rangle \langle N \| J^\lambda(0) \| \Omega \rangle \delta^4(p - p_N). \quad (19.2.20)$$

We usually take J^λ as well as ϕ_n to be Hermitian operators, in which case Eqs. (19.2.19) and (19.2.20) are complex conjugates

$$\rho_n^\lambda(p) = -\tilde{\rho}_n^{\lambda*}(p), \quad (19.2.21)$$

but this will not be assumed here.

Lorentz invariance tells us that ρ and $\tilde{\rho}$ must take the forms

$$\rho_n^\lambda(p) = p^\lambda \rho_n(-p^2) \theta(p^0), \quad (19.2.22)$$

$$\tilde{\rho}_n^\lambda(p) = p^\lambda \tilde{\rho}_n(-p^2) \theta(p^0). \quad (19.2.23)$$

(The factor $\theta(p^0)$, which is +1 for $p^0 > 0$ and zero otherwise, is required by the fact that p_N is the four-momentum of a physical state.) This gives

$$\begin{aligned} \langle \Omega \| [J^\lambda(y), \phi_n(x)] \| \Omega \rangle &= \frac{\partial}{\partial y_\lambda} \int d\mu^2 \left[\rho_n(\mu^2) \Delta_+(y - x; \mu^2) \right. \\ &\quad \left. + \tilde{\rho}_n(\mu^2) \Delta_+(x - y; \mu^2) \right], \end{aligned} \quad (19.2.24)$$

where Δ_+ is the familiar function

$$\Delta_+(z; \mu^2) = (2\pi)^{-3} \int d^4 p \theta(p^0) \delta(p^2 + \mu^2) e^{ip \cdot z}. \quad (19.2.25)$$

As remarked in Chapter 5, Lorentz invariance allows $\Delta_+(z; \mu^2)$ to depend only on z^2 , μ^2 , and $\theta(z^0)$, and for $z^2 > 0$ only on z^2 and μ^2 . Hence $\Delta_+(x - y; \mu^2)$ and $\Delta_+(y - x; \mu^2)$ are equal for $x - y$ spacelike, and so in this case

$$\langle \Omega \| [J^\lambda(y), \phi_n(x)] \| \Omega \rangle = \frac{\partial}{\partial y_\lambda} \int d\mu^2 [\rho_n(\mu^2) + \tilde{\rho}_n(\mu^2)] \Delta_+(y - x; \mu^2). \quad (19.2.26)$$

But for $x - y$ spacelike all commutators must vanish, so

$$\rho_n(\mu^2) = -\tilde{\rho}_n(\mu^2), \quad (19.2.27)$$

and therefore Eq. (19.2.24) gives for general x and y :

$$\langle \Omega \| [J^\lambda(y), \phi_n(x)] \| \Omega \rangle = \frac{\partial}{\partial y_\lambda} \int d\mu^2 \rho_n(\mu^2) [\Delta_+(y - x; \mu^2) - \Delta_+(x - y; \mu^2)]. \quad (19.2.28)$$

Where Eq. (19.2.21) applies, Eq. (19.2.27) also shows that $\rho_n(p)$ is *real*.

Now let us use the fact that $J^\lambda(y)$ is conserved. Applying the derivative $\partial/\partial y^\lambda$ to both sides of Eq. (19.2.28), and using the familiar equation

$$(\Box_y - \mu^2) \Delta_+(y - x; \mu^2) = 0, \quad (19.2.29)$$

we find, for *all* x and y

$$0 = \int d\mu^2 \mu^2 \rho_n(\mu^2) [\Delta_+(y-x; \mu^2) - \Delta_+(x-y; \mu^2)], \quad (19.2.30)$$

and so (since $\Delta_+(x-y)$ is *not* even for $x-y$ timelike or lightlike)

$$\mu^2 \rho_n(\mu^2) = 0. \quad (19.2.31)$$

Normally we would conclude from this that $\rho_n(\mu^2)$ vanishes for all μ^2 . However, this is not possible in the case of broken symmetry. Set $\lambda = 0$ and $x^0 = y^0 = t$ in Eq. (19.2.28):

$$\begin{aligned} \langle \Omega | [J^0(\mathbf{y}, t), \phi_n(\mathbf{x}, t)] | \Omega \rangle &= 2i(2\pi)^{-3} \int d\mu^2 \rho_n(\mu^2) \\ &\quad \cdot \int d^4p \sqrt{\mathbf{p}^2 + \mu^2} e^{i\mathbf{p} \cdot (\mathbf{y} - \mathbf{x})} \delta(p^2 + \mu^2) \\ &= i\delta^3(\mathbf{y} - \mathbf{x}) \int d\mu^2 \rho_n(\mu^2). \end{aligned}$$

Integrating and using Eqs. (19.2.16) and (19.2.17) gives

$$-\sum_m t_{nm} \langle \Omega | \phi_m | \Omega \rangle = i \int \rho_n(\mu^2). \quad (19.2.32)$$

Eqs. (19.2.31) and (19.2.32) can be reconciled only if

$$\rho_n(\mu^2) = i\delta(\mu^2) \sum_m t_{nm} \langle \Omega | \phi_m(0) | \Omega \rangle. \quad (19.2.33)$$

(This is real for Hermitian fields ϕ_n because in this case Eq. (19.2.1) requires t_{nm} to be imaginary.) Thus as long as the symmetry is broken, $\rho_n(\mu^2)$ cannot vanish, but rather consists entirely of a term proportional to $\delta(\mu^2)$. Such a term can obviously only arise in a theory that has massless particles, because otherwise the spectrum of center-of-mass squared energies $-p_N^2$ would not extend down to zero. Furthermore a delta-function $\delta(\mu^2)$ can only arise from single particle states of zero mass; multiparticle states would contribute a continuum extending down to $\mu^2 = 0$. The state $\phi_n(0)|\Omega\rangle$ is rotationally invariant, so $\langle N | \phi_n(0) | \Omega \rangle$ must vanish for any state N of non-zero helicity. Also $\langle \Omega | J^0 | N \rangle$ vanishes for any state N that has different intrinsic parity or (unbroken) internal quantum numbers from J^0 . We conclude then that *a broken symmetry with $t_{nm} \langle \Omega | \phi_m(0) | \Omega \rangle \neq 0$ requires the existence of a massless particle of spin zero and the same parity and internal quantum numbers as J^0 .* These are our Goldstone bosons.

The above argument breaks down when the spontaneously broken symmetry is a local rather than a global symmetry. Either we choose a Lorentz-invariant gauge, such as the Landau gauge with $\partial_\mu A^\mu = 0$, in which case as shown in Section 15.7 the positivity assumptions of quantum mechanics are violated, or we adopt a gauge like the axial gauge with $A_3^\mu = 0$, in which case the ordinary rules of quantum mechanics apply but manifest Lorentz invariance is lost. As we shall see in Chapter 21, this exception is not just a technicality; spontaneously broken local symmetries do not lead to Goldstone bosons.

It will be useful to look in a little more detail at how the coefficient of the delta-function in $\rho_N(\mu^2)$ is related to the properties of the Goldstone boson. For a spin zero boson B of four-momentum q^μ , Lorentz invariance requires the matrix element of the current between the vacuum and one-particle states to take the form

$$\langle \Omega \| J^\lambda(x) \| B \rangle = i \frac{F p_B^\lambda e^{i p_B \cdot x}}{(2\pi)^{3/2} \sqrt{2p_B^0}}, \quad (19.2.34)$$

where $\mathbf{p}_B$ is the momentum of B , $p_B^0 = |\mathbf{p}_B|$ and F is a constant coefficient with the dimensions of energy. (This is consistent with current conservation because $p_{B\lambda} p_B^\lambda = 0$.) Also, the matrix element of the scalar field $\phi_n(y)$ between a single-particle state and the vacuum is of the form

$$\langle B \| \phi_n(y) \| \Omega \rangle = \frac{Z_n e^{-i p_B \cdot y}}{(2\pi)^{3/2} \sqrt{2p_B^0}}, \quad (19.2.35)$$

where Z_n is a dimensionless constant. From Eqs. (19.2.34) and (19.2.35), we have then

$$\begin{aligned} (2\pi)^{-3} i \rho_n(-p^2) p^\lambda \theta(p^0) &= \int d^3 p_B \langle \Omega \| J^\lambda(0) \| B \rangle \langle B \| \phi_n(0) \| \Omega \rangle \delta^4(p - p_B) \\ &= \delta(p^0 - |\mathbf{p}|) (2\pi)^{-3} (2p^0)^{-1} p^\lambda F Z_n \\ &= \theta(p^0) \delta(-p^2) (2\pi)^{-3} p^\lambda i F Z_n, \end{aligned}$$

so

$$\rho_n(\mu^2) = F Z_n \delta(\mu^2). \quad (19.2.36)$$

Comparing with Eq. (19.2.33), this gives

$$i F Z_n = - \sum_m t_{nm} \langle \Omega \| \phi_m(0) \| \Omega \rangle. \quad (19.2.37)$$

More generally, we may have several broken symmetries with generators t_a and currents J_a^μ , which we can take independent in the sense that no linear combination of the t_a is unbroken. For each of these, there is a Goldstone boson $|B_a\rangle$, and we define Z_{an} and F_{ab} by

$$\langle \Omega \| J_a^\lambda(x) \| B_b \rangle = i \frac{F_{ab} p_B^\lambda e^{i p_B \cdot x}}{(2\pi)^{3/2} \sqrt{2p_B^0}}, \quad (19.2.38)$$

$$\langle B_a \| \phi_n(y) \| \Omega \rangle = \frac{Z_{an} e^{-i p_B \cdot y}}{(2\pi)^{3/2} \sqrt{2p_B^0}}. \quad (19.2.39)$$

Eq. (19.2.37) applies for each a , so

$$i \sum_b F_{ab} Z_{bn} = - \sum_m [t_a]_{nm} \langle \Omega \| \phi_m(0) \| \Omega \rangle. \quad (19.2.40)$$

For instance, in the $O(N)$ example discussed earlier, we can adapt our basis so that the vacuum expectation value points in the one-direction

$$\bar{\phi}_m \equiv \langle \Omega \| \phi_m(0) \| \Omega \rangle = v \delta_{m1}. \quad (19.2.41)$$

The $N - 1$ broken symmetry generators t_a (with $a = 2 \cdots N$) can be defined as those for infinitesimal rotations in the 1 - a plane. With a convenient choice of normalization, these have the non-zero elements

$$[t_a]_{1a} = -[t_a]_{a1} = i \quad (19.2.42)$$

(with no summation over a). The unbroken $O(N - 1)$ symmetry, under which the $N - 1$ Goldstone bosons transform according to the vector representation, tells us that

$$F_{ab} = \delta_{ab}F, \quad Z_{a1} = 0, \quad Z_{ab} = Z\delta_{ab}. \quad (19.2.43)$$

Then Eq. (19.2.40) requires

$$FZ = v. \quad (19.2.44)$$

It is conventional to adopt the field renormalization prescription that $Z = 1$, so $F = v$. Thus F is a measure of the strength of the symmetry breaking. As we shall now see, the parameter $1/F$ determines the strength with which the Goldstone bosons interact with each other and with other particles.

A broken symmetry tells us more about the Goldstone bosons than just that they have zero mass; it also tightly constrains their interactions at low energy. To see this in the simplest case, consider the matrix element of

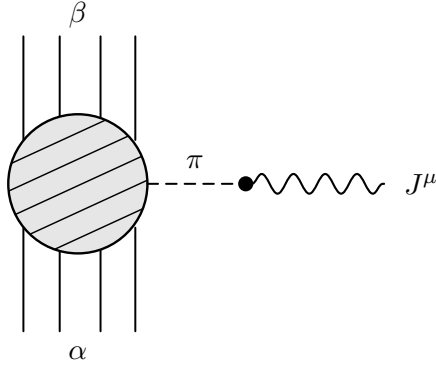


Figure 19.1. Feynman diagrams for pole terms in the matrix element of a symmetry current $J^\mu(x)$ between general states α and β , due to a Goldstone boson internal line, denoted π .

the current $J^\mu(x)$ associated with a broken symmetry. Between arbitrary states α, β :

$$\langle \beta | J^\mu(x) | \alpha \rangle = e^{iq \cdot x} \langle \beta | J^\mu(0) | \alpha \rangle, \quad (19.2.45)$$

with

$$q^\mu \equiv p_\alpha^\mu - p_\beta^\mu. \quad (19.2.46)$$

We know that $J^\mu(x)$ has a non-zero matrix element between the vacuum and a one-Goldstone-boson state $|B, \mathbf{q}\rangle$, given by Eq. (19.2.34). It follows by the usual rules of

polology (see Chapter 10) that the matrix element (19.2.45) has a pole at $q^2 \rightarrow 0$, with²

$$\langle \beta \| J^\mu(0) \| \alpha \rangle \longrightarrow \frac{iFq^\mu}{q^2} M_{\beta\alpha}, \quad (19.2.47)$$

where

$$\frac{i(2\pi)^4 \delta^4(p_\alpha - p_\beta - q) M_{\beta\alpha}}{(2\pi)^{3/2} (2p^0)^{1/2}}$$

is the S-matrix element for emitting a Goldstone boson of four-momentum q in the transition $\alpha \rightarrow \beta$. Let us therefore write

$$\langle \beta \| J^\mu(0) \| \alpha \rangle \equiv N_{\beta\alpha}^\mu + \frac{iFq^\mu}{q^2} M_{\beta\alpha}, \quad (19.2.48)$$

where $N_{\beta\alpha}^\mu$ is defined as the non-pole contributions to the matrix element of the current.

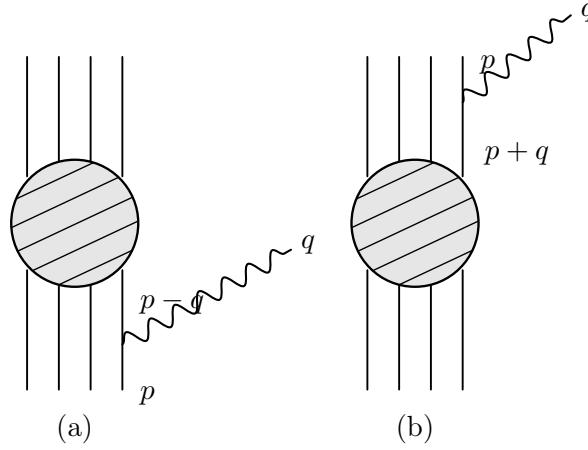


Figure 19.2. Feynman diagrams for pole terms in the matrix element of a symmetry current $J^\mu(x)$ between general states α and β , due to internal lines close to their mass shell. Solid lines indicate ‘hard’ external particles; wavy lines indicate insertions of the current $J^\mu(x)$; the cross-hatched disk represents the sum of diagrams with the indicated external lines.

From Eq. (19.2.45), we see that the conservation law $\partial_\mu J^\mu = 0$ for the current J^μ requires (19.2.48) to vanish when contracted with q_μ , and so

$$M_{\beta\alpha} = \frac{i}{F} q_\mu N_{\beta\alpha}^\mu. \quad (19.2.49)$$

One immediate consequence is that unless $N_{\beta\alpha}^\mu$ has a pole at $q \rightarrow 0$, the matrix element $M_{\beta\alpha}$ for emitting a Goldstone boson in a transition $\alpha \rightarrow \beta$ vanishes as $q \rightarrow 0$. This is called an “Adler zero” [6].

In fact, it often happens that $N_{\beta\alpha}^\mu$ *does* have a pole at $q = 0$. This is because the vertex for the current $J^\mu(x)$ might be attached to an external line of the process $\alpha \rightarrow \beta$.

²Where the Goldstone boson corresponds to an elementary field, Eq. (19.2.47) may be obtained by inspection of the class of Feynman diagrams shown in Figure 19.1. The factor $i(2\pi)^4$ in the S-matrix element for $\alpha \rightarrow \beta + B$ is cancelled by the factor $-i(2\pi)^{-4}$ associated with the B propagator. The general rules of polology tell us that the same holds even where the Goldstone boson is a composite particle.

(See Figure 19.2.) For instance, if a four-momentum q^μ is carried away by a current J^μ inserted into an outgoing or incoming particle line of four-momentum p and mass m , then the internal line that connects this vertex to the rest of the diagram will carry a four-momentum $p^\mu + q^\mu$ or $p^\mu - q^\mu$ respectively, and its propagator will therefore contribute to $N_{\beta\alpha}^\mu$ a factor

$$[(p \pm q)^2 + m^2]^{-1} = [\pm 2p \cdot q + q^2]^{-1} \longrightarrow \pm \frac{1}{2p \cdot q}. \quad (19.2.50)$$

For a fixed direction of $\mathbf{q}$, the factor $1/|\mathbf{q}|$ from Eq. (19.2.50) cancels the factor $|\mathbf{q}|$ in Eq. (19.2.49), yielding a finite (though $\mathbf{q}$ -direction-dependent) result in the limit $|\mathbf{q}| \rightarrow 0$. On the other hand, the contribution to $N_{\beta\alpha}^\mu$ of diagrams in which the current J^μ is attached to an internal line of the process $\alpha \rightarrow \beta$ has no singularity for $|\mathbf{q}| \rightarrow 0$, and therefore is cancelled by the factor $|\mathbf{q}|$ in Eq. (19.2.49). Thus Eq. (19.2.49) can be interpreted to mean that *the amplitude for emitting a very soft Goldstone boson in a process $\alpha \rightarrow \beta$ can be calculated from graphs in which the Goldstone boson is emitted only from the external lines of the process, with the vertex for emitting the Goldstone boson given by applying Eq. (19.2.49) to the transition between single-particle states.*

The effective action formalism can be used to derive interesting results for the interactions of the Goldstone bosons with each other and with other scalars. For this purpose, we note that if we define a set of renormalized Goldstone boson fields π_a for which

$$\langle B_a | \pi_b(x) | \Omega \rangle = \frac{e^{-ip_B \cdot x} \delta_{ab}}{(2\pi)^{3/2} \sqrt{2p_B^0}}, \quad (19.2.51)$$

then Eq. (19.2.39) tells us that

$$\phi_n(x) = \sum_a Z_{an} \pi_a(x) + \dots, \quad (19.2.52)$$

where “...” indicates fields that do not create Goldstone bosons. But Eq. (19.2.40) gives $Z_{an} = \sum_b F_{ab}^{-1}(it_b \bar{\phi})_n$. Thus the amplitude for any reaction among N zero-four-momentum Goldstone bosons $\pi_{a_1}, \dots, \pi_{a_N}$ is the same as would be calculated in the tree approximation from the effective interaction

$$\mathcal{H}_{\text{eff}} = \frac{1}{N!} g_{a_1 \dots a_N} \pi_{a_1} \dots \pi_{a_N}, \quad (19.2.53)$$

with

$$g_{a_1 \dots a_N} = \sum_{b_1 \dots b_N} F_{a_1 b_1}^{-1} \dots F_{a_N b_N}^{-1} (it_{b_1} \bar{\phi})_{n_1} \dots (it_{b_N} \bar{\phi})_{n_N} \frac{\partial^N V(\phi)}{\partial \phi_{n_1} \dots \partial \phi_{n_N}} \Big|_{\phi = \bar{\phi}}. \quad (19.2.54)$$

Eq. (19.2.4) at $\phi = \bar{\phi}$ then implies that the sum of all “tadpole” graphs for a single Goldstone boson line disappearing into the vacuum vanishes, and Eq. (19.2.6) tells us that the amplitudes vanish for Goldstone bosons to make transitions at zero four-momentum into

any other scalars. To go beyond these results, we can continue to differentiate with respect to the scalar fields. For instance, the derivative of Eq. (19.2.5) gives for any symmetry generator t :

$$\begin{aligned} \sum_n \frac{\partial^2 V(\phi)}{\partial \phi_n \partial \phi_m} t_{nl} + \sum_n \frac{\partial^2 V(\phi)}{\partial \phi_n \partial \phi_\ell} t_{nm} \\ + \sum_{n,k} \frac{\partial^3 V(\phi)}{\partial \phi_n \partial \phi_m \partial \phi_\ell} t_{nk} \phi_k = 0. \end{aligned} \quad (19.2.55)$$

Taking t as one of the broken symmetry generators t_a , setting $\phi = \bar{\phi}$, contracting with $(t_b \bar{\phi})_m (t_c \bar{\phi})_\ell$, and using Eq. (19.2.6) gives

$$\sum_{n,m,\ell} \frac{\partial^3 V(\phi)}{\partial \phi_n \partial \phi_m \partial \phi_\ell} \bigg|_{\phi=\bar{\phi}} (t_a \bar{\phi})_n (t_b \bar{\phi})_m (t_c \bar{\phi})_\ell = 0, \quad (19.2.56)$$

so the sum of all graphs with three external zero-four-momentum Goldstone boson lines vanishes. In particular, this means that in general processes, to leading order in small Goldstone boson energies, low energy Goldstone bosons are not emitted from external low energy Goldstone boson lines.

19.3 Spontaneously Broken Approximate Symmetries

In the previous section we dealt with exact symmetries of the action that do not leave the vacuum invariant, and are said to be spontaneously broken. We shall now consider the effect of adding small symmetry-breaking terms to the action in such a theory. Such spontaneously broken approximate symmetries are important in the theory of strong interactions, and in some areas of condensed matter physics. As we shall see, the spontaneous breakdown of an approximate symmetry does not lead to the appearance of massless Goldstone bosons, but of low-mass spinless particles, often called *pseudo-Goldstone bosons* [7].

We continue here to treat translationally invariant theories, in which the effective action is expressed as in Eq. (19.2.8) in terms of an effective potential $V(\phi)$ that depends on a set of spacetime-independent scalar field expectation values ϕ_n . For an action that obeys some set of approximate continuous symmetries with generators t_α , the effective potential may be written

$$V(\phi) = V_0(\phi) + V_1(\phi), \quad (19.3.1)$$

where $V_0(\phi)$ satisfies the invariance condition³

$$\sum_{mn} \frac{\partial V_0(\phi)}{\partial \phi_n} (t_\alpha)_{nm} \phi_m = 0 \quad (19.3.2)$$

and $V_1(\phi)$ is a small correction due to the symmetry breaking in the action. Suppose that this perturbation shifts the minimum of the potential from ϕ_0 , the minimum of $V_0(\phi)$, to

³We denote general symmetry generators as t_α , t_β , etc., in contrast with the independent broken symmetry generators, for which the subscript α takes values a , b , etc.

$\bar{\phi} = \phi_0 + \phi_1$, where ϕ_1 is small, of first order in the symmetry-breaking perturbation. The equilibrium condition for the vacuum is then

$$\left. \frac{\partial V(\phi)}{\partial \phi_n} \right|_{\phi=\phi_0+\phi_1} = 0. \quad (19.3.3)$$

The zeroth-order term on the left is just $[\partial V_0(\phi)/\partial \phi_n]_{\phi=\phi_0}$, which vanishes because ϕ_0 is defined as the minimum of $V_0(\phi)$. Thus the first-order terms must also vanish, and so:

$$\sum_m \left. \frac{\partial^2 V_0(\phi)}{\partial \phi_n \partial \phi_m} \right|_{\phi=\phi_0} \phi_{1m} + \left. \frac{\partial V_1(\phi)}{\partial \phi_n} \right|_{\phi=\phi_0} = 0. \quad (19.3.4)$$

Eq. (19.2.6) holds here with V replaced with the invariant term V_0 and $\bar{\phi}$ replaced with ϕ_0 :

$$\sum_n \left. \frac{\partial^2 V_0(\phi)}{\partial \phi_n \partial \phi_m} \right|_{\phi=\phi_0} (t_\alpha)_{n\ell} \phi_{0\ell} = 0. \quad (19.3.5)$$

Hence multiplying Eq. (19.3.4) with $(t_\alpha \phi_0)_n$ and summing over n gives

$$(t_\alpha \phi_0)_n \left. \frac{\partial V_1(\phi)}{\partial \phi_n} \right|_{\phi=\phi_0} = 0. \quad (19.3.6)$$

Recalling the interpretation of $V(\phi)$ as the generating function for one-particle-irreducible graphs, and noting that in the absence of the perturbation V_1 the Goldstone components of ϕ_n are those in the direction of $t_\alpha \phi_0$ (see Eq. (19.2.33)), the left-hand side of Eq. (19.3.6) is proportional to the sum of all ‘tadpole’ graphs, in which a pseudo-Goldstone boson disappears into the vacuum. Eq. (19.3.6) may thus be paraphrased as the condition that, to first order in V_1 , *pseudo-Goldstone bosons have no tadpoles*.

The moral of this calculation is that if we do not start with a zeroth-order vacuum expectation value that satisfies Eq. (19.3.6), then even a small perturbation will produce a large change in $\bar{\phi}$, invalidating the expansion of $\bar{\phi}$ around ϕ_0 . Fortunately, for compact Lie groups it is always possible to choose ϕ_0 to satisfy Eq. (19.3.6). To see this, note that the invariance of the potential $V_0(\phi)$ under a group of real linear transformations $\phi \rightarrow L\phi$ implies that if ϕ_* is one minimum of the potential, then so is $L\phi_*$. For continuous groups of transformations, we can always parameterize the transformations as $L(\theta)$, in such a way that

$$\frac{\partial L(\theta)}{\partial \theta^\alpha} L^{-1}(\theta) = iN_{\alpha\beta}(\theta)t_\beta, \quad (19.3.7)$$

where $N_{\alpha\beta}$ is a non-singular matrix depending on the group parameters θ_α . (Recall that in a real representation, it is it_α rather than t_α that is real.) Now consider the function $V_1(L(\theta)\phi_*)$. Where the group is compact, $L(\theta)\phi_*$ maps out a compact manifold as θ runs over the group volume, and as long as $V_1(\phi)$ is continuous, it must have a minimum on any such compact surface, say at a point $L(\theta_*)\phi_*$. The derivative of $V_1(L(\theta)\phi_*)$ with respect to θ_α is

$$\frac{\partial V_1(L(\theta)\phi_*)}{\partial \theta_\alpha} = \sum_n \left. \frac{\partial V_1(\phi)}{\partial \phi_n} \right|_{\phi=L(\theta)\phi_*} N_{\alpha\beta}(\theta) (it_\beta L(\theta)\phi_*)_n. \quad (19.3.8)$$

This must vanish at the minimum θ_* , and since $N_{\alpha\beta}$ is non-singular, this implies that

$$0 = \sum_n \left. \frac{\partial V_1(\phi)}{\partial \phi_n} \right|_{\phi=L(\theta_*)\phi_*} (t_\beta L(\theta_*)\phi_*)_n. \quad (19.3.9)$$

But then Eq. (19.3.6) is satisfied if we make the choice $\phi_0 = L(\theta_*)\phi_*$.

Eq. (19.3.6) is known as a *vacuum alignment* condition [8], because it generally has the effect of forcing the direction of the symmetry breaking by the vacuum into some sort of alignment with the symmetry-breaking terms in the Hamiltonian. For instance, consider the case of $SO(N)$ spontaneously broken to $SO(N-1)$, introduced in the previous section. In the absence of any symmetry-breaking perturbation, there is no way to tell *which* $SO(N-1)$ subgroup is left unbroken; if the dynamics of the theory leads to a ground state that is invariant under the $SO(N-1)$ subgroup of $SO(N)$ that leaves some N -vector ϕ_{0n} invariant, then by performing an $SO(N)$ rotation we can find a ground state that is invariant under the $SO(N-1)$ subgroup that leaves any other N -vector invariant. If we add a perturbation that transforms under $SO(N)$ like, say, the component $\sum_n u_n \phi_n$ of an N -vector ϕ_n (not necessarily consisting of elementary scalars), then the Hamiltonian is invariant under a specific $SO(N-1)$ subgroup of $SO(N)$, consisting of rotations that leave the vector u invariant. Without a vacuum alignment condition, we might think that the remaining exact symmetry is $SO(N-2)$, consisting of those rotations that leave invariant *both* u and the vector ϕ_0 characterizing the vacuum symmetry. But with $V_1(\phi) = \sum_n u_n \phi_n$, the condition (19.3.6) tells us that at the true vacuum, $\sum_n (t_\alpha \phi_0)_n u_n = 0$ for all $SO(N)$ generators t_α . The $SO(N)$ generators t_α span the space of all antisymmetric $N \cdot N$ matrices, so this condition requires that ϕ_0 must be in the same direction as u , and so the unbroken symmetry is $SO(N-1)$, not $SO(N-2)$.

According to the general results of Section 16.1, the mass matrix M_{ab} of the pseudo-Goldstone bosons is given to first order by

$$M_{ab}^2 = \sum_{mn} Z_{an} Z_{bm} \left. \frac{\partial^2 V(\phi)}{\partial \phi_m \partial \phi_n} \right|_{\phi=\phi_0+\phi_1}, \quad (19.3.10)$$

where Z_{an} is the field renormalization constant defined by Eq. (19.2.39). Since the mass matrix (19.3.10) vanishes in zeroth order, the first-order terms give

$$M_{ab}^2 = \sum_{mn} Z_{an} Z_{bm} \left[\left. \frac{\partial^3 V_0(\phi)}{\partial \phi_\ell \partial \phi_m \partial \phi_n} \right|_{\phi=\phi_0} \phi_{1\ell} + \left. \frac{\partial^2 V_1(\phi)}{\partial \phi_m \partial \phi_n} \right|_{\phi=\phi_0} \right], \quad (19.3.11)$$

where Z_{an} is here given by the zeroth-order approximation to (19.2.40):

$$Z_{an} = \sum_b F_{ab}^{-1} (it_b \phi_0)_n. \quad (19.3.12)$$

To calculate the mass matrix (19.3.11), we take t in Eq. (19.2.55) to be one of the broken symmetry generators t_a , set $\phi = \phi_0$, and contract with $(t_b \phi_0)_m \phi_{1\ell}$:

$$0 = \sum_{nm\ell} \left. \frac{\partial^3 V_0(\phi)}{\partial \phi_\ell \partial \phi_m \partial \phi_n} \right|_{\phi=\phi_0} \phi_{1\ell} (t_a \phi_0)_n (t_b \phi_0)_m$$

$$\begin{aligned}
& + \sum_{nm} \frac{\partial^2 V_0(\phi)}{\partial \phi_n \partial \phi_m} \Big|_{\phi=\phi_0} (t_a \phi_1)_n (t_b \phi_0)_m \\
& + \sum_{n\ell} \frac{\partial^2 V_0(\phi)}{\partial \phi_n \partial \phi_\ell} \Big|_{\phi=\phi_0} (t_a t_b \phi_0)_n \phi_{1\ell}.
\end{aligned}$$

The second term on the right-hand side vanishes according to Eq. (19.2.6), while the third may be rewritten using Eq. (19.3.4), leaving us with

$$\sum_{nm\ell} \frac{\partial^3 V_0(\phi)}{\partial \phi_\ell \partial \phi_m \partial \phi_n} \Big|_{\phi=\phi_0} \phi_{1\ell} (t_a \phi_0)_n (t_b \phi_0)_m = \frac{\partial V_1(\phi)}{\partial \phi_n} \Big|_{\phi=\phi_0} (t_a t_b \phi_0)_n. \quad (19.3.13)$$

Using this in Eq. (19.3.11) then yields a formula for the pseudo-Goldstone boson mass matrix in terms of V_1 :

$$\begin{aligned}
M_{cd}^2 = - \sum_{ab} F_{ca}^{-1} F_{db}^{-1} & \left[(t_a \phi_0)_n (t_b \phi_0)_m \frac{\partial^2 V_1(\phi)}{\partial \phi_m \partial \phi_n} \Big|_{\phi=\phi_0} \right. \\
& \left. + (t_a t_b \phi_0)_n \frac{\partial V_1(\phi)}{\partial \phi_n} \Big|_{\phi=\phi_0} \right]. \quad (19.3.14)
\end{aligned}$$

For this to be a sensible mass matrix, it had better be positive. To see that it is, it is convenient to rewrite this result in terms of derivatives with respect to the group parameters θ_α . Differentiating Eq. (19.3.8) with respect to θ_β , setting $\theta = \theta_*$, and using Eq. (19.3.9) and $\phi_0 = L(\theta_*)\phi_*$ gives

$$M_{ab}^2 = \sum_{cd\alpha\beta} N_{a\alpha}^{-1}(\theta_*) N_{b\beta}^{-1}(\theta_*) F_{ac}^{-1} F_{bd}^{-1} \frac{\partial^2 V_1(L(\theta)\phi_*)}{\partial \theta_\alpha \partial \theta_\beta} \Big|_{\theta=\theta_*}. \quad (19.3.15)$$

The matrix on the right is positive, because θ_* is the *minimum* of the function $V_1(L(\theta)\phi_*)$.

This formula has a somewhat more familiar version, in terms of the vacuum expectation value of a double commutator of symmetry generators with the symmetry-breaking perturbation. Suppose that the symmetry-breaking perturbation H_1 in the Hamiltonian is a linear combination

$$H_1 = \sum_n u_n \Phi_n \quad (19.3.16)$$

of operators Φ_n (not necessarily elementary scalar fields), which furnish a representation of the symmetry group with generators t_α , in the sense that

$$[T_\alpha, \Phi_n] = -(t_\alpha)_{nm} \Phi_m, \quad (19.3.17)$$

where T_α are the quantum mechanical generators of the symmetry group. According to the results of Section 16.3, the symmetry-breaking part of the potential is

$$V_1(\phi) = \langle H_1 \rangle_{\langle \Phi \rangle = \phi} = \sum_n u_n \phi_n, \quad (19.3.18)$$

the subscript in the middle expression indicating that the expectation value is to be taken in the state of minimum energy in which Φ_n has expectation value ϕ_n . The vacuum alignment condition (19.3.6) then reads

$$0 = \sum_n u_n (t_\alpha \phi_0)_n,$$

or using Eq. (19.3.17)

$$0 = \langle [T_\alpha, H_1] \rangle_0, \quad (19.3.19)$$

the subscript 0 now indicating that the expectation value is to be taken in the vacuum state, in which Φ_n has expectation value ϕ_{0n} . Also, Eq. (19.3.14) gives the mass matrix here as

$$M_{cd}^2 = - \sum_{ab} F_{ca}^{-1} F_{db}^{-1} \sum_n u_n (t_a t_b \phi_0)_n,$$

and, using Eq. (19.3.17), this is

$$M_{cd}^2 = - \sum_{ab} F_{ca}^{-1} F_{db}^{-1} \langle [T_a, [T_b, H_1]] \rangle_0. \quad (19.3.20)$$

This is symmetric in c and d . To see this, note that the Jacobi identity and group commutation relations may be used to write the difference of Eq. (19.3.20) and the same with c and d interchanged as a linear combination of terms $\langle [T_\alpha, H_1] \rangle_0$, which vanish according to the vacuum alignment condition (19.3.19). The mass matrix (19.3.20) is also positive, because the point $\theta = 0$ is at the *minimum* of the vacuum energy $\langle \exp(-i\theta_a T_a) H_1 \exp(i\theta_a T_a) \rangle_0$ for rotated vacuum states $\exp(i\theta_a T_a) |\Omega\rangle$.

19.4 Pions as Goldstone Bosons

The classic example of a broken symmetry in elementary particle physics is the approximate symmetry of strong interactions known as chiral $SU(2) \cdot SU(2)$. According to our present understanding, this symmetry arises because there are two quark fields, u and d , that happen to have relatively small masses. (An estimate is given in Section 19.7.) In the approximation that the u and d are massless, the Lagrangian (18.7.5) of quantum chromodynamics is

$$\mathcal{L} = -\bar{u}\gamma^\mu D_\mu u - \bar{d}\gamma^\mu D_\mu d - \dots, \quad (19.4.1)$$

where D_μ is a color-gauge-covariant derivative (see Eq. (15.1.10)) and ‘ $\dots$ ’ refers to terms involving only gluon fields and/or other quark flavors, but not u or d . This Lagrangian is invariant under the transformations

$$\begin{pmatrix} u \\ d \end{pmatrix} \longrightarrow \exp(i\vec{\theta}^V \cdot \vec{t} + i\gamma_5 \vec{\theta}^A \cdot \vec{t}) \begin{pmatrix} u \\ d \end{pmatrix}, \quad (19.4.2)$$

where $\vec{t}$ is the three-vector⁴ of isospin matrices

$$t_1 = \frac{1}{2} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad t_2 = \frac{1}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad t_3 = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

⁴We use arrows for three-vectors in isotopic spin space to distinguish them from ordinary three-vectors, which will continue to be indicated by boldface letters.

and $\vec{\theta}^V$ and $\vec{\theta}^A$ are independent real three-vectors.⁵ This Lie algebra may be written in terms of two commuting $SU(2)$ subalgebras that act respectively only on the left- and right-handed parts of the quark fields, with generators

$$\vec{t}_L = \frac{1}{2}(\mathbf{1} + \gamma_5)\vec{t}, \quad \vec{t}_R = \frac{1}{2}(\mathbf{1} - \gamma_5)\vec{t}. \quad (19.4.3)$$

satisfying the commutation relations

$$[t_{Li}, t_{Lj}] = i\epsilon_{ijk}t_{Lk}, \quad (19.4.4)$$

$$[t_{Ri}, t_{Rj}] = i\epsilon_{ijk}t_{Rk}, \quad (19.4.5)$$

$$[t_{Li}, t_{Rj}] = 0. \quad (19.4.6)$$

The underlying symmetry group is therefore identified as $SU(2) \cdot SU(2)$. It has another obvious $SU(2)$ subgroup, consisting of ordinary isospin transformations with $\vec{\theta}^A = 0$, and generators

$$\vec{t} = \vec{t}_L + \vec{t}_R. \quad (19.4.7)$$

The algebra of $SU(2) \cdot SU(2)$ may be written in terms of $\vec{t}$ and another triplet of generators:

$$\vec{x} = \vec{t}_L - \vec{t}_R = \gamma_5 \vec{t} \quad (19.4.8)$$

with commutation relations

$$[t_i, t_j] = i\epsilon_{ijk}t_k, \quad (19.4.9)$$

$$[t_i, x_j] = i\epsilon_{ijk}x_k, \quad (19.4.10)$$

$$[x_i, x_j] = i\epsilon_{ijk}t_k. \quad (19.4.11)$$

We will see that the $SU(2) \cdot SU(2)$ symmetry is spontaneously broken, while its isotopic spin subgroup generated by the $\vec{t}$ is an ordinary unbroken (though approximate) symmetry.

By Noether's method (see, e.g., Section 7.3) we may derive from the Lagrangian (19.4.1) the conserved vector and axial-vector currents

$$\vec{V}^\mu = i\bar{q}\gamma^\mu\vec{t}q, \quad \vec{A}^\mu = i\bar{q}\gamma^\mu\gamma_5\vec{t}q, \quad (19.4.12)$$

$$\partial_\mu\vec{V}^\mu = \partial_\mu\vec{A}^\mu = 0, \quad (19.4.13)$$

where q is the quark doublet,

$$q \equiv \begin{pmatrix} u \\ d \end{pmatrix}. \quad (19.4.14)$$

⁵The Lagrangian (19.4.1) has this symmetry because $\bar{\gamma}_5\bar{q}\gamma^\mu = -\bar{q}\gamma_5\gamma^\mu = +\bar{q}\gamma^\mu\gamma_5$. The Lagrangian also has two other continuous internal symmetries. One is baryon conservation, the invariance under a common phase transformation of the u and d quark fields. This is unbroken and commutes with the other symmetries, so it does not affect our discussion in this section. The other symmetry is invariance under multiplication of the quark doublet with $\exp(i\alpha\gamma_5)$. As discussed in Section 23.5, this $U(1)$ symmetry is strongly intrinsically broken by non-perturbative effects associated with instantons.

Their associated charges are the generators respectively of isospin and of the remaining symmetries

$$\vec{T} = \int d^3x \vec{V}^0, \quad (19.4.15)$$

$$\vec{X} = \int d^3x \vec{A}^0. \quad (19.4.16)$$

The currents (19.4.12) are normalized so that the quantum operators $\vec{T}$ and $\vec{X}$ satisfy the same commutation relations as the matrices $\vec{t}$ and $\vec{x}$:

$$[T_i, T_j] = i\epsilon_{ijk}T_k, \quad (19.4.17)$$

$$[T_i, X_j] = i\epsilon_{ijk}X_k, \quad (19.4.18)$$

$$[X_i, X_j] = i\epsilon_{ijk}T_k. \quad (19.4.19)$$

Acting on the quark fields, these operators induce the transformation (19.4.2), in the sense that

$$[\vec{T}, q] = -\vec{t}q, \quad (19.4.20)$$

$$[\vec{X}, q] = -\vec{x}q. \quad (19.4.21)$$

This symmetry if exact and unbroken would require any one-hadron state $||h\rangle$ to be degenerate with another state $\vec{X}||h\rangle$ of opposite parity and equal spin, baryon number, and strangeness.⁶ No such parity doubling is seen in the hadron spectrum, so we are forced to conclude that if the chiral symmetry $SU(2) \cdot SU(2)$ is a good approximation at all, then it must be spontaneously broken to its isotopic spin $SU(2)$ subgroup. In this case the operator $\vec{X}$ takes a one-hadron state $||h\rangle$ into a hadron h plus a massless pseudoscalar Goldstone boson, so there is no need for parity doubling of the hadron spectrum.

The question of whether quantum chromodynamics actually exhibits such a pattern of symmetry breaking involves all the complications of strong interaction dynamics. As we shall see in Section 19.9, there are general grounds for believing that the isospin $SU(2)$ is not spontaneously broken in quantum chromodynamics, but it is much more difficult to show that the chiral part of $SU(2) \cdot SU(2)$ is spontaneously broken. (But according to an argument given in Section 22.5, the $SU(3) \cdot SU(3)$ symmetry of quantum chromodynamics with three massless quark flavors must be spontaneously broken.) It was something of a breakthrough in the 1960s to realize that one does not have to have a detailed understanding of the mechanism of the breaking of chiral symmetry; we derive the most interesting consequences of this symmetry breaking by simply assuming that $SU(2) \cdot SU(2)$ is spontaneously broken to $SU(2)$.

The u and d quarks have small but nonzero masses, so the $SU(2) \cdot SU(2)$ symmetry is not exact. A broken *approximate* chiral symmetry entails the existence of an approximately massless Goldstone boson with the same quantum numbers as the broken symmetry

⁶One way to satisfy this condition is for the hadron to have zero mass, with two states $||\pm\rangle$ of helicity $\pm\frac{1}{2}$ and equal spin, baryon number, and strangeness; the states $||+\rangle + ||-\rangle$ and $||+\rangle - ||-\rangle$ would then have opposite parity. It is not true that an unbroken chiral symmetry necessarily implies a zero nucleon mass, unless we make further assumptions about the matrix elements of the axial-vector current. But as we shall see in Section 22.5, an exact unbroken chiral symmetry would in fact require that some baryons be massless.

generator $\vec{X}$: it must be a state of negative parity, zero spin, unit isospin, and zero baryon number and strangeness. In fact the lightest of all hadrons is the pion, which has precisely these quantum numbers, so we are led to identify the pion as the Goldstone boson associated with the spontaneous breaking of approximate chiral symmetry. As we shall see below, it is m_π^2 rather than m_π that is proportional to a linear combination of m_u and m_d , and $m_\pi^2/m_N^2 \simeq 0.022$ is very small, so the consequences we derive from spontaneously broken $SU(2) \cdot SU(2)$ should be reasonably accurate.

In exploring the consequences of chiral symmetry for pion interactions, it is very useful to note that although the chiral symmetry of the strong interactions does not depend in any way on the existence of weak interactions, the symmetry currents $\vec{V}^\mu$ and $\vec{A}^\mu$ happen to be the currents entering into strangeness-conserving semileptonic weak interactions like nuclear beta decay. As we shall see in Section 21.3, the standard model of electroweak interactions requires that the effective Lagrangian for these interactions at low energy must take the form:

$$\mathcal{L}_{\text{wk}} = -\frac{iG_{\text{wk}}}{\sqrt{2}}(V_+^\lambda + A_+^\lambda) \sum_\ell \bar{\ell} \gamma_\lambda (\mathbf{1} + \gamma_5) \nu_\ell + \text{H.c.} \quad (19.4.22)$$

where ℓ runs over the renormalized fields of the three charged leptons e , μ and τ ; ν_ℓ runs over the renormalized fields of the associated neutrinos; and $V_\pm^\lambda$ and $A_\pm^\lambda$ are the charge changing currents

$$V_\pm^\lambda = V_1^\lambda \pm iV_2^\lambda, \quad A_\pm^\lambda = A_1^\lambda \pm iA_2^\lambda. \quad (19.4.23)$$

The constant⁷ G_{wk} may be measured from the rates of beta transitions between states of zero spin within the same isotopic multiplet, such as the decays $\pi^+ \rightarrow \pi^0 + e^+ + \nu_e$ and $^{14}\text{O} \rightarrow ^{14}\text{N}^* + e^+ + \nu_e$. The momentum transfer in these transitions is very small, so parity conservation (in strong interactions) and rotational invariance tell us that only the matrix elements of $\int d^3x V_-^0 = T_1 - iT_2$ enter in the S -matrix elements for these decays. This operator has matrix elements between states in a given isospin multiplet that are just known Clebsch–Gordan coefficients, so from the rates for these ‘ $0 \rightarrow 0$ ’ processes we may calculate a value [9] for the coupling in Eq. (19.4.22): $G_{\text{wk}} \simeq 1.14959(38) \cdot 10^{-5} \text{ GeV}^{-2}$. On the other hand, in the process of pion decay, $\pi^+ \rightarrow \mu^+ + \nu_\mu$, the only current matrix element we need is the matrix element of A_-^μ between a one-pion state and the vacuum:

$$\langle \Omega | A_-^\mu(x) | \pi_j \rangle = \frac{iF_\pi \delta_{ij} p_\pi^\mu e^{ip_\pi \cdot x}}{2(2\pi)^{3/2} \sqrt{2p_\pi^0}}. \quad (19.4.24)$$

which is completely known except for the factor F_π . The rate for pion decay turns out to be

$$\Gamma(\pi \rightarrow \mu + \nu) = \frac{G_{\text{wk}}^2 F_\pi^2 m_\mu^2 (m_\pi^2 - m_\mu^2)^2}{16\pi m_\pi^3}. \quad (19.4.25)$$

⁷ G_{wk} is related to the conventional Fermi coupling constant G_F and the Cabibbo angle θ_C by $G_{\text{wk}} = G_F \cos \theta_C$; see Section 21.3.

From the known rate for $\pi^+ \rightarrow \mu^+ + \nu_\mu$, $\Gamma = (2.6033(24) \cdot 10^{-8} \text{ s})^{-1}$, and the above value of G_{wk} , one finds that⁸

$$F_\pi \simeq 184 \text{ MeV}. \quad (19.4.26)$$

Now let us consider the matrix element of $\vec{A}^\mu$ between one-nucleon states. This is of interest in its own right, and as discussed in Section 19.2, it provides information we need to calculate the emission of low energy pions in collisions of nucleons. Following the same reasoning as used for the electromagnetic current in Section 10.6, one finds that Lorentz invariance and parity conservation require this matrix element to take the form⁹

$$\begin{aligned} \langle p | A_\pm^\mu(x) | n \rangle = (2\pi)^{-3} e^{iq \cdot x} \bar{u}_p \Big[& -i\gamma^\mu \gamma_5 f(q^2) + q^\mu \gamma_5 g(q^2) \\ & + i q_\nu [\gamma^\mu, \gamma^\nu] \gamma_5 h(q^2) \Big] u_n, \end{aligned} \quad (19.4.27)$$

where $q = p_n - p_p$. In the approximation in which the $SU(2) \cdot SU(2)$ symmetry is exact, the conservation of current requires that

$$q_\mu \langle p | A_+^\mu(x) | n \rangle = 0. \quad (19.4.28)$$

Using the defining equations for the Dirac spinors u_p and u_n ,

$$\bar{u}_p(i\not{p}_p + m_N) = (i\not{p}_n + m_N)u_n = 0,$$

we see that

$$q_\mu \bar{u}_p [-i\gamma^\mu \gamma_5] u_n = -2m_N \bar{u}_p \gamma_5 u_n$$

and so Eq. (19.4.28) requires that

$$2m_N f(q^2) = q^2 g(q^2). \quad (19.4.29)$$

If $g(q^2)$ had no singularity at $q^2 = 0$, then (19.4.29) would require that either $m_N = 0$, which is certainly not true, or that $f(0) = 0$, which is not true either. In fact, the quantity $f(0)$ is measured in low energy nuclear beta decays, like neutron decay, where it is usually called g_A ; it is found to have the value

$$f(0) \equiv g_A = 1.2573(28). \quad (19.4.30)$$

The fact that neither m_N nor $f(0) = g_A$ is small requires that in the limit of exact $SU(2) \cdot SU(2)$ symmetry, $g(q^2)$ must have a pole as $q^2 \rightarrow 0$:

$$g(q^2) \longrightarrow \frac{2m_N g_A}{q^2}. \quad (19.4.31)$$

⁸It is common to encounter a pion decay constant f_π , which in terms of the F_π used here is variously defined as F_π , $F_\pi/\sqrt{2}$, or $F_\pi/2$.

⁹The currents $A_\pm^\mu$ in the standard model have charge conjugation properties that make the coefficient $h(q^2)$ vanish. It is possible that there are ‘second-class’ terms [10] in the weak currents with opposite charge conjugation properties that give a non-vanishing $h(q^2)$, but there is no evidence for such terms. As we shall see, keeping the $h(q^2)$ term here has no effect on the inferences drawn from chiral symmetry.

Such a pole is naturally provided by the massless pion that would be required by the spontaneous breakdown of an exact $SU(2) \cdot SU(2)$ symmetry. Suppose that the pion couples to the one-nucleon state as if the interaction Lagrangian were¹⁰ $-2iG_{\pi N}\vec{\pi} \cdot \bar{N}\gamma_5\vec{t}N$. Eq. (19.4.24) tells us that the matrix element of the current between a one-pion state and the vacuum is the same as if there were a term in $\vec{A}^\mu(x)$ of the form $F_\pi\partial^\mu\vec{\pi}/2$. Therefore in the limit $q^2 \rightarrow 0$ the matrix element (19.4.27) has the pole

$$\langle p||A_+^\mu(x)||n\rangle \longrightarrow \left[\frac{e^{iq\cdot x}}{(2\pi)^3}\right] [i(2\pi)^4 2G_{\pi N}\bar{u}_p(i\gamma_5)u_n] \left[\frac{-i}{(2\pi)^4 q^2}\right] [iq^\mu F_\pi/2].$$

Comparing with Eq. (19.4.27), we see that one-pion exchange gives the function $g(q^2)$ a pole

$$g(q^2) \longrightarrow \frac{G_{\pi N}F_\pi}{q^2} \quad (19.4.32)$$

for $q^2 \rightarrow 0$. Putting together Eqs. (19.4.31) and (19.4.32), we find

$$G_{\pi N} = \frac{2m_N g_A}{F_\pi}. \quad (19.4.33)$$

This is the famous Goldberger–Treiman relation [11]. It works reasonably well; taking $m_N = (m_p + m_n)/2 = 938.9$ MeV, $g_A = 1.257$, and $F_\pi = 184$ MeV gives $G_{\pi N} \simeq 12.7$, in fair agreement with the value¹¹ $G_{\pi N} = 13.5$ measured in various ways (including the effects of the one-pion pole in nucleon-nuclear scattering and the one-nucleon pole in pion-nucleon scattering.)

In the real world the pion is not massless and the $SU(2) \cdot SU(2)$ symmetry is not exact (even before being spontaneously broken.) This circumstance may be analyzed using the general formalism presented in the previous section. The Lagrangian (19.4.1) yields a symmetry-breaking term in the Hamiltonian

$$H_1 = m_u\bar{u}u + m_d\bar{d}d = (m_u + m_d)\Phi_4^+ + (m_u - m_d)\Phi_3^-, \quad (19.4.34)$$

where

$$\Phi_4^+ = \frac{1}{2}(\bar{u}u + \bar{d}d), \quad \Phi_3^- = \frac{1}{2}(\bar{u}u - \bar{d}d). \quad (19.4.35)$$

The operators Φ_4^+ and Φ_3^- are spatial scalars and, as this notation is meant to suggest, they transform under $SU(2) \cdot SU(2)$ as components of independent chiral four-vectors $\Phi_n^\pm$:

$$\vec{\Phi}^+ = i\bar{q}\gamma_5\vec{t}q, \quad \Phi_4^+ = \frac{1}{2}\bar{q}q, \quad (19.4.36)$$

$$\vec{\Phi}^- = \bar{q}\vec{t}q, \quad \Phi_4^- = -\frac{1}{2}i\bar{q}\gamma_5q. \quad (19.4.37)$$

¹⁰This is the conventional definition of the pseudoscalar pion-nucleon coupling $G_{\pi N}$. The factor 2 is introduced here to cancel the 1/2 in the isospin matrices.

¹¹The textbook [12] value is $G_{\pi N}^2/4\pi = 14.3$, or $G_{\pi N} = 13.4$. More recently, a high precision study [13] of neutron-proton charge-exchange scattering at 162 MeV has given a value $G_{\pi N}^2/4\pi = 14.6 \pm 0.3$, or $G_{\pi N} = 13.5$.

These are chiral four-vectors, in the sense that

$$[\vec{T}, \Phi_n^\pm] = - \sum_m (\vec{T})_{nm} \Phi_m^\pm, \quad (19.4.38)$$

$$[\vec{X}, \Phi_n^\pm] = - \sum_m (\vec{X})_{nm} \Phi_m^\pm, \quad (19.4.39)$$

where $\vec{T}$ and $\vec{X}$ are Hermitian $4 \cdot 4$ matrices that furnish the four-vector representation of the algebra of $SU(2) \cdot SU(2) \equiv SO(4)$:

$$(\mathcal{T}_a)_{bc} = -i\epsilon_{abc}, \quad (\mathcal{T}_a)_{b4} = (\mathcal{T}_a)_{4b} = (\mathcal{T}_a)_{44} = 0, \quad (19.4.40)$$

$$(\mathcal{X}_a)_{b4} = -(\mathcal{X}_a)_{4b} = -i\delta_{ab}, \quad (\mathcal{X}_c)_{ab} = (\mathcal{X}_c)_{44} = 0. \quad (19.4.41)$$

This notation makes it easy to see that the vacuum alignment conditions for generators T_1 , T_2 , X_1 , X_2 , and X_3 respectively take the form

$$\begin{aligned} 0 &= \langle \Phi_2^- \rangle_0 = \langle \Phi_1^- \rangle_0 = \langle \Phi_1^+ \rangle_0 = \langle \Phi_2^+ \rangle_0 \\ &= (m_u + m_d) \langle \Phi_3^+ \rangle_0 + (m_d - m_u) \langle \Phi_4^- \rangle_0. \end{aligned} \quad (19.4.42)$$

We have been assuming that in the absence of u and d quark masses, the symmetry $SU(2) \cdot SU(2)$ is spontaneously broken in such a way as to preserve unbroken the $SU(2)$ symmetry generated by $\vec{T}$ as well as parity, in which case $\langle \Phi_n^+ \rangle_0$ points in the four-direction and $\langle \Phi_n^- \rangle_0 = 0$ vanishes, so that the conditions (19.4.42) are satisfied. But with $m_u = m_d = 0$, we can find other symmetry-breaking solutions with other definitions of parity by subjecting this one to an arbitrary $SU(2) \cdot SU(2)$ transformation. Thus in the absence of u and d quark masses, there would be no way to tell in what direction $\langle \Phi_n^+ \rangle_0$ should point, or which $SU(2)$ subgroup of $SU(2) \cdot SU(2)$ is left unbroken, though in all cases $\langle \Phi_n^- \rangle_0 = 0$. The vacuum alignment condition (19.4.42) tells us that, with the symmetry broken intrinsically by the perturbation (19.4.34) and spontaneously in such a way that $\langle \Phi_n^- \rangle_0 = 0$, the vacuum must ‘line up’ in such a way that $\langle \Phi_n^+ \rangle_0 \neq 0$ points in the four-direction, so that the unbroken $SU(2)$ symmetry is ordinary isospin.

This formalism may be used to calculate the pion mass. From Eqs. (19.4.39)–(19.4.41) we find

$$[X_a, [X_b, \Phi_4^+]] = \delta_{ab} \Phi_4^+, \quad [X_a, [X_b, \Phi_3^-]] = \Phi_a^- \delta_{b3}. \quad (19.4.43)$$

Also, isospin invariance tells us that the symmetry-breaking parameter F_{ab} introduced in Section 19.2 is proportional to δ_{ab} , with a proportionality factor that according to (19.4.24) is just $F_\pi/2$, so

$$F_{ab} = \delta_{ab} F_\pi/2. \quad (19.4.44)$$

Eq. (19.3.20) thus gives the pion mass matrix as

$$M_{ab}^2 = \delta_{ab} m_\pi^2, \quad (19.4.45)$$

where

$$m_\pi^2 = 4(m_u + m_d) \langle \Phi_4^+ \rangle_0 / F_\pi^2. \quad (19.4.46)$$

It is striking that the masses of the charged and neutral pions turn out to be equal, even though we have made no assumption about the ratio of m_u and m_d . We shall see below that this ratio is not near unity; isospin is a good quantum number not because the u and d quark masses are nearly equal, but because they are *small*. Also, as promised, we see that it is the *square* of the pion mass that is proportional to the quark masses, so the quark masses should be quite small. (See Section 19.7 for an estimate.) The observed pion mass difference arises not from the u – d quark mass difference, but from electromagnetism. Indeed, the pion isospin multiplet is the only one whose mass difference has been successfully calculated on the basis of one-photon exchange alone [14].

Of course, even with quark masses taken into account, Eq. (19.4.24) can still be used to define F_π ; its divergence gives

$$\langle \Omega | \partial_\mu A_i^\mu(x) | \pi_j \rangle = \frac{F_\pi \delta_{ij} m_\pi^2 e^{ip_\pi \cdot x}}{2(2\pi)^{3/2} \sqrt{2p_\pi^0}}. \quad (19.4.47)$$

Instead of assuming that $\partial_\mu \vec{A}^\mu$ vanishes, we can now assume that it is *small*, of order m_π^2 , except where a pion pole compensates for the smallness of m_π^2 . According to Eq. (19.4.47), the behavior of matrix elements of $\partial_\mu \vec{A}^\mu$ near a one-pion pole is the same as if $\partial_\mu \vec{A}^\mu$ were $F_\pi m_\pi^2$ times a properly renormalized pion field. For instance, the one-nucleon matrix element of $\partial_\mu A_+^\mu$ should be

$$\begin{aligned} \langle p | \partial_\mu A_+^\mu(0) | n \rangle &\simeq \frac{F_\pi m_\pi^2}{2} \left[\frac{-i}{(2\pi)^4 (q^2 + m_\pi^2)} \right] \\ &\times [i(2\pi)^4 2G_{\pi N} \bar{u}_p(-i\gamma_5)u_n]. \end{aligned} \quad (19.4.48)$$

so in terms of the form-factors in (19.4.27)

$$q^2 g(q^2) - 2m_N f(q^2) \simeq -\frac{G_{\pi N} F_\pi m_\pi^2}{q^2 + m_\pi^2}. \quad (19.4.49)$$

This is expected to be valid when q^2 is of order m_π^2 , and not only in the limit $q^2 \rightarrow -m_\pi^2$, because the pion pole dominates matrix elements of $\partial_\mu A^\mu$ for all such small q^2 . Also, for such q^2 , in place of Eq. (19.4.32) we have

$$g(q^2) \simeq \frac{G_{\pi N} F_\pi}{q^2 + m_\pi^2}. \quad (19.4.50)$$

From Eqs. (19.4.49) and (19.4.50) we find that for q^2 of order m_π^2 ,

$$f(q^2) \simeq G_{\pi N} F_\pi / 2m_N. \quad (19.4.51)$$

It should be no surprise that this function is roughly constant over the range of q^2 from zero to of order m_π^2 , because it has no one-pion pole, and there is nothing else that could give it a substantial variation in such a small range of q^2 . This constant value is approximately $f(0) \equiv g_A$, and Eq. (19.4.51) thus again yields the Goldberger–Treiman relation.

We can now use the results of Section 19.2 to calculate the amplitude for emission of a single low energy pion in an arbitrary process $\alpha \rightarrow \beta$. We found that the amplitude is to

be calculated from Feynman diagrams in which the pion is emitted only from the external lines of the process, and Eq. (19.2.49) shows that these contributions are to be calculated as if the pion field interaction were $\partial_\mu \vec{\pi} \cdot \vec{A}_N^\mu / F_\pi$, in which the subscript N indicates that we are to drop the one-pion pole term in matrix elements of the axial-vector current. From Eq. (19.4.27) (and using isospin invariance) we conclude that for soft pions emitted from a nucleon line this interaction is effectively

$$-\frac{ig_A}{F_\pi} \partial_\mu \vec{\pi} \cdot \bar{N} \gamma^\mu \gamma_5 \vec{t} N.$$

Using the free-particle Dirac equation, we can see that for nucleons on the mass shell (that is, at the one-nucleon poles in Figure 19.2) this is equivalent to the pseudoscalar interaction $-2im_N g_A \vec{\pi} \cdot \bar{N} \gamma_5 \vec{t} N / F_\pi$. This provides yet another demonstration of the Goldberger–Treiman relation (19.4.33).

* * *

Our discussion here has not paralleled the historical development of these ideas. In fact, the course of historical development was chronologically almost exactly opposite to the line of argument presented here. Broken symmetries in particle physics started with the Goldberger–Treiman relation (19.4.33), which was derived [11] in 1957 on the basis of a dynamical calculation of pion decay. In order to explain the surprising success of this very approximate calculation, several theorists [15] introduced the idea of a ‘partial conservation of the axial-vector current’ (PCAC), the idea that although the axial-vector current is not conserved (as shown by the fact that pions do decay) its divergence $\partial_\mu A_\pm^\mu$ is proportional to the pion field. In itself, this assumption is meaningless—we saw in Chapter 10 that *any* field with a non-vanishing matrix element between the vacuum and one-pion states may be regarded as a pion field. Although it was not clear at the time, what was really being assumed was that the divergence of the axial-vector current is *small*, of order m_π^2 , except where a pion pole gives it a large matrix element. The problem was greatly clarified by a 1960 paper of Nambu [16], who pointed out that the axial-vector current could be regarded as exactly conserved in the limit of zero pion mass, in which case the Goldberger–Treiman relation could be derived as we have done here. In this and a subsequent paper with Jona-Lasinio [17], Nambu recognized that the appearance of this massless or nearly massless pion was a symptom of a broken exact or approximate symmetry. With other collaborators [18], Nambu also showed how to calculate the rates of emission of a single soft pion in various processes. Subsequently Goldstone [3] remarked that broken symmetries always entail massless bosons, and this was proved in 1962 by Goldstone, Salam, and myself [5], using the arguments presented here in Section 19.2.

None of this early work on pions as Goldstone bosons depended on any specific assumptions about the nature of the broken symmetry group; for instance it might have been a direct product of three commuting $U(1)$ groups, whose generators form an isotopic spin vector, or it might have been the non-compact group $SO(3, 1)$, for which a minus sign appears on the right-hand side of the commutation relation (19.4.19). The nature of the broken symmetry group became important only with the consideration of processes involving more

than one pion, starting with the Adler–Weisberger sum rule in 1965 [19], whose success showed that the broken symmetry is indeed $SU(2) \cdot SU(2)$. (Such processes are discussed in the following section.) The identification of the symmetry group $SU(2) \cdot SU(2)$ led to a shift of emphasis [20], toward the implications of this symmetry within strong interaction physics, and away from an earlier concentration [21] on the currents themselves.

All of this work was done without a specific theory of the strong interactions. One of the reasons for the rapid acceptance of quantum chromodynamics in 1973 as the correct theory of strong interactions was that it explained the $SU(2) \cdot SU(2)$ symmetry as a simple consequence of the smallness of the u and d quark masses.

19.5 Effective Field Theories: Pions and Nucleons

In Section 19.2 we learned how to calculate the amplitude for emission of a single low energy Goldstone boson B in a transition $\alpha \rightarrow \beta + B$ by applying the condition of current conservation to the matrix element of the symmetry current between the states α and β . In this calculation we never had to use any information about the details of the broken symmetry group; current conservation was all we needed. A new element enters if we wish to calculate the matrix element for the emission and/or absorption of *two* Goldstone bosons, as for instance in a Goldstone boson scattering process. Here we must apply the condition of current conservation to a matrix element like

$$\langle \beta | T \{ J_1^{\mu_1}(x_1), J_2^{\mu_2}(x_2) \} | \alpha \rangle,$$

where the states α and β contain the other particles besides the two Goldstone bosons participating in the reaction. But when we let the divergence operator $\partial/\partial x_1^{\mu_1}$ act on this matrix element we encounter a non-zero contribution from the derivative of the functions $\theta(x_1^0 - x_2^0)$ and $\theta(x_2^0 - x_1^0)$ that appear in the definition of the time-ordered product $T\{\dots\}$. This contribution is equal to the matrix element of an equal-time commutator $\delta(x_1^0 - x_2^0)[J_1^0(x_1), J_2^{\mu_2}(x_2)]$, whose value depends on the commutation relations of the group algebra. This makes such multi-Goldstone-boson processes specially interesting, because they can be used to decide experimentally on the nature of the broken symmetry group, in a way that is not possible for processes involving just a single Goldstone boson. Because of the appearance of such current commutators, this approach is known as the method of current algebra [21].

The current algebra method was used in early calculations of multi-Goldstone-boson amplitudes [22]. Unfortunately, such calculations are tedious, especially when three or more Goldstone bosons are involved, and it was also difficult to see how to deal with symmetries like the chiral symmetry of quantum chromodynamics that are not exact. For this reason a different and more physical calculational technique was introduced [23], based on the use of effective Lagrangians: We simply calculate the Goldstone boson amplitudes by the methods of perturbation theory, using some Lagrangian for the Goldstone bosons and the other particles in the states α and β that obeys the assumed broken symmetry.

Originally the justification for the effective Lagrangian procedure was based on current algebra. By using current algebra one could see that the amplitude for emission of a set of

low energy Goldstone bosons in a process $\alpha \rightarrow \beta + B_1 + B_2 + \dots$ is fixed once one knows the equal-time commutation relations of the currents associated with the broken symmetries as well as the matrix element for the process $\alpha \rightarrow \beta$ and the matrix elements of the currents between various one-particle states. We know that a Lagrangian that respects the broken symmetry will allow the construction of conserved currents with the appropriate equal-time commutators by Noether's technique, so if we simply calculate the low energy Goldstone boson amplitudes with such a Lagrangian and insert the correct values for $M_{\beta\alpha}$ and the one-particle matrix elements of the currents, we must get the same answer as provided by current algebra. In the case of interactions among Goldstone bosons alone, the states α and β may both be taken as the vacuum, and we do not need any extra information beyond the matrix element F between a Goldstone boson state and the vacuum.

In the first example of this sort [23], the starting point was the Lagrangian of the σ -model [24]. Restricting our attention for the moment to the bosonic sector of this model, its Lagrangian is the $SO(4)$ -invariant one used as an example in Section 19.2:

$$\mathcal{L} = -\frac{1}{2}\partial_\mu\phi_n\partial^\mu\phi_n - \frac{\mathcal{M}^2}{2}\phi_n\phi_n - \frac{\lambda}{4}(\phi_n\phi_n)^2, \quad (19.5.1)$$

where n is understood to be summed over the values 1, 2, 3, 4, with ϕ an isovector pseudoscalar field and ϕ_4 an isoscalar scalar field.

The immediate problem faced by any sort of effective Lagrangian is that in order to use it to calculate scattering amplitudes, we must either include all Feynman diagrams of all orders of perturbation theory, or else find some rationale for dropping higher-order diagrams. We can find no such rationale with Lagrangians like (19.5.1), in which the broken symmetry is realized through linear transformations on the various fields. Fortunately any such Lagrangian can be recast in a form that allows the use of Feynman diagrams to generate an expansion for scattering amplitudes in powers of Goldstone boson energies. To do this, we perform a symmetry transformation at every point in spacetime that eliminates the fields corresponding to the Goldstone bosons of the theory. The Goldstone boson degrees of freedom reappear in the transformed theory as the parameters of this symmetry transformation. However, since the Lagrangian is invariant under spacetime-independent symmetry transformations, it cannot have any dependence on the new Goldstone boson fields when they are constant, and so every term in the Lagrangian that involves these new Goldstone boson fields must contain at least one spacetime derivative of the field. These derivatives introduce factors of the Goldstone boson energy when we calculate S-matrix elements for Goldstone boson reactions, and as we shall see we can use the Lagrangian to construct a series for the S-matrix elements in powers of these energies.

For instance, to recast the Lagrangian (19.5.1) in a useful form, we write the four-vector ϕ_n as a chiral rotation R acting on a four-vector $\{0, 0, 0, \sigma\}$ whose first three components (the Goldstone part of ϕ_n) vanish:

$$\phi_n(x) = R_{n4}(x)\sigma(x) \quad (19.5.2)$$

with $R_{nm}(x)$ an orthogonal matrix

$$R^T(x)R(x) = \mathbf{1} \quad (19.5.3)$$

and therefore

$$\sigma(x) = \sqrt{\sum_n \phi_n(x)^2}. \quad (19.5.4)$$

The Lagrangian (19.5.1) then becomes

$$\mathcal{L} = -\frac{1}{2} \sum_{n=1}^4 (R_{n4} \partial_\mu \sigma + \sigma \partial_\mu R_{n4})^2 - \frac{1}{2} \mathcal{M}^2 \sigma^2 - \frac{\lambda}{4} \sigma^4. \quad (19.5.5)$$

Because R is an orthogonal matrix, the $\partial_\mu \sigma \partial^\mu \sigma$ term is R -independent, and the cross-term vanishes

$$\sum_n R_{n4}^2 = 1, \quad \sum_n R_{n4} \partial_\mu R_{n4} = \frac{1}{2} \partial_\mu \sum_n R_{n4}^2 = 0,$$

so $\mathcal{L}$ becomes

$$\mathcal{L} = -\frac{1}{2} \partial_\mu \sigma \partial^\mu \sigma - \frac{1}{2} \sigma^2 \sum_{n=1}^4 \partial^\mu R_{n4} \partial_\mu R_{n4} - \frac{1}{2} \mathcal{M}^2 \sigma^2 - \frac{\lambda}{4} \sigma^4. \quad (19.5.6)$$

If $\mathcal{M}^2$ is negative then σ has a non-vanishing vacuum expectation value, given in lowest order by the position of the minimum of the sum of the last two terms, at $\sigma = |\mathcal{M}|/\sqrt{\lambda}$.

In place of the field variables ϕ_n , our variables now are $\sigma = |\mathcal{M}|/\sqrt{\lambda}$ and whatever other variables are needed to parameterize the rotation R . For instance, these parameters could be simply chosen as the R_{a4} themselves (where $a, b, \dots$ from now are isovector indices running over the values 1, 2, 3), with R_{44} given by the condition that R is orthogonal. A different parameterization will give simpler final results, and was historically the first to be used for these purposes. It is to define

$$\zeta_a = \frac{\phi_a}{\phi_4 + \sigma} \quad (19.5.7)$$

and take

$$R_{a4} = \frac{2\zeta_a}{1 + \zeta^2} = -R_{4a}, \quad R_{44} = \frac{1 - \zeta^2}{1 + \zeta^2}, \quad R_{ab} = \delta_{ab} - \frac{2\zeta_a \zeta_b}{1 + \zeta^2}, \quad (19.5.8)$$

so that

$$\phi_a/\sigma = R_{a4} = \frac{2\zeta_a}{1 + \zeta^2}, \quad \phi_4/\sigma = R_{44} = \frac{1 - \zeta^2}{1 + \zeta^2}. \quad (19.5.9)$$

Then the Lagrangian (19.5.6) becomes

$$\mathcal{L} = -\frac{1}{2} \partial_\mu \sigma \partial^\mu \sigma - 2\sigma^2 \mathbf{D}_\mu \cdot \mathbf{D}^\mu - \frac{1}{2} \mathcal{M}^2 \sigma^2 - \frac{\lambda}{4} \sigma^4, \quad (19.5.10)$$

where

$$\mathbf{D}_\mu \equiv \frac{\partial_\mu \boldsymbol{\zeta}}{1 + \zeta^2}. \quad (19.5.11)$$

Whatever parameterization we use, it is clear that the fields $\boldsymbol{\zeta}$ describe particles of zero mass, whose interactions all involve field derivatives. These are (up to a normalization) our new pion fields.

Despite appearances, this Lagrangian is still invariant under $SO(4)$, but with $SO(4)$ now realized non-linearly. Under an isospin transformation with infinitesimal parameters $\boldsymbol{\theta}$, the field $\boldsymbol{\zeta}$ simply rotates like an ordinary isovector, and σ is an isoscalar:

$$\delta\boldsymbol{\zeta} = \boldsymbol{\theta} \times \boldsymbol{\zeta}, \quad \delta\sigma = 0, \quad (19.5.12)$$

so the Lagrangian (19.5.10) is manifestly isospin-invariant. On the other hand, under the broken symmetry transformations parameterized by an infinitesimal vector $\boldsymbol{\epsilon}$, the original fields transform according to

$$\delta\phi = 2\boldsymbol{\epsilon}\phi_4, \quad \delta\phi_4 = -2\boldsymbol{\epsilon} \cdot \boldsymbol{\phi}. \quad (19.5.13)$$

From (19.5.7), we then find

$$\delta\boldsymbol{\zeta} = \boldsymbol{\epsilon}(1 - \boldsymbol{\zeta}^2) + 2\boldsymbol{\zeta}(\boldsymbol{\epsilon} \cdot \boldsymbol{\zeta}), \quad \delta\sigma = 0. \quad (19.5.14)$$

The Lagrangian (19.5.10) is invariant under the broken symmetry transformation (19.5.14) because $\boldsymbol{D}_\mu$ undergoes a linear (though field-dependent) isospin rotation [25]:

$$\delta\boldsymbol{D}_\mu = 2(\boldsymbol{\zeta} \times \boldsymbol{\epsilon}) \times \boldsymbol{D}_\mu \quad (19.5.15)$$

and (19.5.10) is isospin-invariant. Because of the transformation rule (19.5.15), $\boldsymbol{D}_\mu$ is often called the *covariant derivative* of the pion field.

The transformation rules (19.5.12) and (19.5.14) specify what is called a *non-linear realization* of the group $SU(2) \cdot SU(2)$ [25]. The general theory of non-linear realizations of Lie groups is given in the next section; we shall show there that, up to field redefinitions, the transformation rules (19.5.12) and (19.5.14) provide the most general realization of $SU(2) \cdot SU(2)$ in which the isospin $SU(2)$ subgroup is realized linearly on $\boldsymbol{\zeta}$.

We see that each interaction of these new pion fields is accompanied with a spacetime derivative, so that the effective coupling is weak for low pion energies. (This remark will be made more precise below.) Therefore for sufficiently small pion energies we may use this Lagrangian in the tree approximation to reproduce the soft-pion theorems of current algebra. For this purpose, it is only necessary that the Lagrangian be $SO(4)$ -invariant. But since the σ field is an $SO(4)$ scalar, it plays no role in maintaining the $SO(4)$ invariance of the Lagrangian, and may be simply discarded¹². Of course, this procedure changes the physical content of the theory, but it does not change the amplitudes given by soft-pion theorems. The Lagrangian (19.5.10) then simplifies to

$$\mathcal{L} = -\frac{F^2}{2} \boldsymbol{D}_\mu \cdot \boldsymbol{D}^\mu = -\frac{F^2}{2} \frac{\partial_\mu \boldsymbol{\zeta} \cdot \partial^\mu \boldsymbol{\zeta}}{(1 + \boldsymbol{\zeta}^2)^2}, \quad (19.5.16)$$

where $F = 2\langle\sigma\rangle = 2|\mathcal{M}|/\sqrt{\lambda}$. (As we shall soon see, this F is the same as the constant F_π discussed in the previous section.) For many purposes it is more convenient to work with a conventionally normalized pion field

$$\boldsymbol{\pi} \equiv F\boldsymbol{\zeta} \quad (19.5.17)$$

¹²Alternatively, we can pass to the limit where $\mathcal{M}$ and λ go to infinity together, keeping the expectation value of σ constant.

for which the Lagrangian (19.5.16) reads

$$\mathcal{L} = -\frac{1}{2} \frac{\partial_\mu \pi \cdot \partial^\mu \pi}{(1 + \pi^2/F^2)^2}. \quad (19.5.18)$$

The factor $1/F$ acts as a coupling parameter that accompanies the interaction of each additional pion. Eq. (19.5.18) describes what is often called the ‘non-linear σ -model,’ for the special case of $SU(2) \cdot SU(2)$ spontaneously broken to $SU(2)$.

An important point: to derive (19.5.18) it was not really necessary to start with the ‘linear σ -model’ Lagrangian (19.5.1). Indeed, we did not need to start with *any* specific theory. Eq. (19.5.18) can be used simply because it is invariant under the $SO(4)$ transformations (19.5.12), (19.5.14), and current algebra tells us that this is all we need to get the right results for low energy pion reaction amplitudes.

Some years after the introduction of effective Lagrangians for soft pions, there emerged a different justification for the effective field theory technique [26], one that does not rely on current algebra and allows calculations that are not limited to the limit of vanishingly small Goldstone boson energies. It is based on the realization (not yet formally embodied in a theorem) that when we calculate a physical amplitude from Feynman diagrams using the most general Lagrangian that involves the relevant degrees of freedom and satisfies the assumed symmetries of the theory, we are simply constructing the most general amplitude that is consistent with general principles of relativity, quantum mechanics, and the assumed symmetries. This was the point of view underlying Volume I of this book. In the present context, the ‘relevant’ degrees of freedom are the Goldstone bosons themselves, together with the particles in the states α and β and any other particle states that can be produced from them by interactions with low energy Goldstone bosons. By invoking this justification for effective field theories, we are freed from any need to wrestle with the complications of current algebra. More importantly, the modern effective field theory approach yields results that take us beyond the extreme low energy limit and allows a systematic treatment of any intrinsic symmetry breaking.

According to this approach, in order to calculate pion interaction amplitudes to any desired order in pion energies, we must use the most general Lagrangian involving a pion field ζ that transforms according to the rules (19.5.12) and (19.5.14):

$$\mathcal{L}_{\text{eff}} = -\frac{F^2}{2} D_\mu \cdot D^\mu - \frac{c_4}{4} (D_\mu \cdot D^\mu)^2 - \frac{c'_4}{4} (D_\mu \cdot D_\nu)(D^\mu \cdot D^\nu) - \dots \quad (19.5.19)$$

The terms indicated by $\dots$ will contain higher powers of the covariant derivative D_μ , or higher covariant derivatives, whose general structure is described in the next section. The coefficients c_4 and c'_4 are dimensionless, and all higher terms have coefficients with the dimensionality of negative powers of mass.

Consider a general process involving arbitrary numbers of incoming and outgoing pions. We suppose that their energies and momenta are all at most of some order Q , which is small compared with a typical quantum chromodynamics energy scale (say, the nucleon or ρ mass). Even though Lagrangians like (19.5.19) are not renormalizable in the usual sense, we saw in Section 12.3 that such Lagrangians can yield finite results as long as they contain

all possible terms allowed by symmetries, for then there will be a counterterm available to cancel every infinity. If we define the renormalized values of the constants $F^2, c_4, c'_4, \dots$ by specifying the values of various Goldstone boson scattering amplitudes at energies of order Q , then the integrals in momentum-space Feynman diagrams will be dominated by contributions from virtual momenta which are also of order Q (because renormalization makes them finite, and there is no other possible effective cut-off in the theory.) We can then develop perturbation theory as a power series expansion in Q .

Each derivative and hence each $\mathbf{D}_\mu$ in each interaction vertex contributes one factor of Q to the order of magnitude of the diagram; each internal pion propagator contributes a factor Q^{-2} ; and each integration volume d^4q associated with the loops of the diagram contributes a factor Q^4 ; so a general connected diagram makes a contribution of order Q^ν , where

$$\nu = \sum_i V_i d_i - 2I + 4L. \quad (19.5.20)$$

Here d_i is the number of derivatives in an interaction of type i , V_i is the number of interaction vertices of type i in the diagram, I is the number of internal pion lines, and L is the number of loops. These quantities are related by the familiar topological identity (see Eq. (4.4.7)):

$$L = I - \sum_i V_i + 1, \quad (19.5.21)$$

so we can eliminate I and write

$$\nu = \sum_i V_i (d_i - 2) + 2L + 2. \quad (19.5.22)$$

The point of this is that each term here is positive; every interaction in (19.5.19) has at least two derivatives, and of course $L \geq 0$. Therefore the leading term of each process is of order Q^2 , and arises solely from *tree* graphs (that is, $L = 0$) constructed solely from the term in (19.5.19) with only two derivatives (that is, $V_i = 0$ for $d_i \neq 2$):

$$\mathcal{L}_2 = -\frac{F^2}{2} \mathbf{D}_\mu \cdot \mathbf{D}^\mu = -\frac{1}{2} \frac{\partial_\mu \boldsymbol{\pi} \cdot \partial^\mu \boldsymbol{\pi}}{(1 + \boldsymbol{\pi}^2/F^2)^2}. \quad (19.5.23)$$

For instance, the invariant amplitude M that appears in the pion-pion scattering S-matrix element

$$S = i(2\pi)^4 \delta^4(p_A + p_B - p_C - p_D) M (2\pi)^{-6} (16E_A E_B E_C E_D)^{-1/2} \quad (19.5.24)$$

is given to this order by

$$\begin{aligned} M_{abcd}^{(\nu=2)} = 4F^{-2} & \left[\delta_{ab} \delta_{cd} (-p_A \cdot p_B - p_C \cdot p_D) \right. \\ & + \delta_{ac} \delta_{bd} (p_A \cdot p_C + p_B \cdot p_D) \\ & \left. + \delta_{ad} \delta_{bc} (p_A \cdot p_D + p_B \cdot p_C) \right]. \end{aligned} \quad (19.5.25)$$

Here a, b, c, d are the isovector indices of the pions A, B, C, D , respectively. (The effect of the finite pion mass will be taken up a little later in this section.) Using (19.5.25) as the leading

term does not depend on any assumption of the smallness of the coupling constant λ in the original Lagrangian (19.5.1), or even on the validity of this formula for the Lagrangian, but only on the assumed smallness of the typical pion energy Q .

The next term in the amplitude for any Goldstone boson reaction will be of order Q^4 , and arises both from one-loop graphs involving only the Lagrangian (19.5.16), and also from tree graphs constructed solely from the interaction (19.5.16) plus a single vertex arising from the $d = 4$ terms in (19.5.19):

$$M_{abcd}^{(\nu=4)} = \frac{\delta_{ab}\delta_{cd}}{F^4} \left[-\frac{1}{2\pi^2} s^2 \ln(-s) - \frac{1}{12\pi^2} (u^2 - s^2 + 3t^2) \ln(-t) \right. \\ \left. - \frac{1}{12\pi^2} (t^2 - s^2 + 3u^2) \ln(-u) + \frac{1}{3\pi^2} (s^2 + t^2 + u^2) \ln \Lambda^2 \right. \\ \left. - \frac{1}{2} c_4 s^2 - \frac{1}{4} c'_4 (t^2 + u^2) \right] + \text{crossed terms}, \quad (19.5.26)$$

where ‘crossed terms’ denotes terms given by the interchanges $B \leftrightarrow C$ and $B \leftrightarrow D$, and s, t , and u are the Mandelstam variables

$$s = -(p_A + p_B)^2, \quad t = -(p_A - p_C)^2, \quad u = -(p_A - p_D)^2.$$

The dependence of this result on the cut-off Λ can be eliminated by a redefinition of the constants c_4 and c'_4 . The renormalized couplings are

$$c_{4R} = c_4 - \frac{2}{3\pi^2} \ln \left(\frac{\Lambda^2}{\mu^2} \right), \quad (19.5.27)$$

$$c'_{4R} = c'_4 - \frac{4}{3\pi^2} \ln \left(\frac{\Lambda^2}{\mu^2} \right), \quad (19.5.28)$$

where μ is an arbitrary renormalization scale of order Q inserted in order to make the logarithms well-defined. In terms of these renormalized couplings, the amplitude (19.5.26) takes the form

$$M_{abcd}^{(\nu=4)} = \frac{\delta_{ab}\delta_{cd}}{F^4} \left[-\frac{1}{2\pi^2} s^2 \ln \left(\frac{-s}{\mu^2} \right) \right. \\ \left. - \frac{1}{12\pi^2} (u^2 - s^2 + 3t^2) \ln \left(\frac{-t}{\mu^2} \right) \right. \\ \left. - \frac{1}{12\pi^2} (t^2 - s^2 + 3u^2) \ln \left(\frac{-u}{\mu^2} \right) - \frac{1}{2} c_{4R} s^2 - \frac{1}{4} c'_{4R} (t^2 + u^2) \right] \\ + \text{crossed terms}. \quad (19.5.29)$$

This sort of calculation can be carried to arbitrary orders in Q , always with the result that in each order we encounter a finite number of new couplings whose renormalization serves to eliminate the cut-off dependence of physical amplitudes. Notice that the ratio of the leading $\nu = 2$ terms and the $\nu = 4$ corrections is of order $Q^2/8\pi^2 F^2$, so this is likely to be

a useful expansion as long as the pion energies are all much less than an amount of order $2\pi F$.

This perturbative expansion may also be used to relate the ubiquitous parameter F to the measured pion decay amplitude F_π . Recalling the transformation rule (19.5.14) for ζ , the axial-vector current is given by

$$\epsilon \cdot A^\mu = -\frac{\partial \mathcal{L}}{\partial(\partial_\mu \zeta)} \cdot \delta \zeta$$

and so

$$A^\mu = -(1 - \zeta^2) \frac{\partial \mathcal{L}}{\partial(\partial_\mu \zeta)} - 2\zeta \zeta \cdot \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \zeta)} \right). \quad (19.5.30)$$

(This A^μ is the Noether current of the symmetry generated by 2ϵ , which on the nucleon doublet is represented by $2\gamma_5 \mathbf{t} = \gamma_5 \boldsymbol{\tau}$.) After integrating out all other fields, and using (19.5.17) to express ζ in terms of the canonically normalized pion field $\boldsymbol{\pi}$, we find

$$A^\mu = F \left[\partial^\mu \boldsymbol{\pi} \frac{1 - \boldsymbol{\pi}^2/F^2}{1 + \boldsymbol{\pi}^2/F^2} + \frac{2\boldsymbol{\pi}(\boldsymbol{\pi} \cdot \partial^\mu \boldsymbol{\pi})}{F^2(1 + \boldsymbol{\pi}^2/F^2)^2} \right] + \dots. \quad (19.5.31)$$

Using our perturbative expansion in powers of the pion energy to calculate $\langle \Omega \| A^\mu \| \boldsymbol{\pi} \rangle$, we see that in lowest order the pion decay amplitude here is

$$F_\pi = F. \quad (19.5.32)$$

Furthermore Lorentz invariance and (19.5.22) tell us that the higher-order corrections must be proportional to powers of p_π^2/F_π^2 , which vanishes for a massless pion, so (19.5.32) is actually exact in the limit $m_\pi \rightarrow 0$.

We may therefore guess that our perturbation expansion will be useful for pion energies that are less than an amount of order $2\pi F_\pi = 1200$ MeV.

In order to make contact with experiment, we must deal with the fact that the pion mass is not zero. On the mass shell it is not possible for the time component of a pion four-momentum to be less than m_π , so in counting powers of the typical pion energy and momentum Q , we must regard m_π as being of order Q . But we saw in the previous section that m_π^2 is proportional to a linear combination of quark masses, so our formula (19.5.22) for the order Q^ν of a given Feynman diagram should be rewritten to read

$$\nu = \sum_i V_i(d_i + 2m_i - 2) + 2L + 2, \quad (19.5.33)$$

where m_i is the number of factors of quark masses in the interaction of type i .

The interactions involving quark masses may be distinguished by their transformation properties under $SU(2) \cdot SU(2)$, or equivalently, under $SO(4)$. Eq. (19.4.34) shows that the terms of first order in quark masses transform as a linear combination of two scalars, the fourth component of a chiral four-vector Φ_n^+ , with coefficient $m_u + m_d$, and the third component of a different chiral four-vector Φ_n^- , with coefficient $m_u - m_d$. Thus the terms $\Delta \mathcal{L}^+$ and $\Delta \mathcal{L}^-$ in the effective Lagrangian that are of first order in $m_u + m_d$ and $m_u - m_d$ must have the chiral transformation properties of the fourth and third components of chiral

four-vectors, respectively, and of course be Lorentz scalars. One obvious candidate for a chiral four-vector whose fourth component is a scalar is the ϕ_n with which we started in this section. According to (19.5.9), its fourth component is just $\sigma(1 - \zeta^2)/(1 + \zeta^2)$. The factor σ may be dropped, as it is a chiral scalar, and therefore has no effect on the chiral transformation properties of this quantity. We can fix the normalization of this term by requiring that the coefficient of $\pi^2 = F_\pi^2 \zeta^2$ be $-m_\pi^2/2$, so that, apart from an additive constant,

$$\Delta\mathcal{L}^+ = -\frac{m_\pi^2 F_\pi^2}{2} \frac{\zeta^2}{1 + \zeta^2} = -\frac{m_\pi^2}{2} \frac{\pi^2}{1 + \pi^2/F_\pi^2}. \quad (19.5.34)$$

We shall see in the next section that this is the unique scalar function of the pion field without derivatives that transforms as the fourth component of a chiral four-vector. On the other hand, there is *no* scalar function of the pion field without derivatives that transforms as the third component of a chiral four-vector because such a function, if the third component of a chiral four-vector, would have to be odd in the pion field, and therefore pseudoscalar rather than scalar. Thus (19.5.34) is the only interaction with $d_i = 0$ and $m_i = 1$. It is striking that the isospin violating difference in the u and d quark masses has not only no effect on the pion masses, as we saw in the previous section, but also has no effect on any non-derivative multipion interaction.

We are now in a position to do a realistic calculation of the leading $\nu = 2$ terms in the pion-pion scattering amplitude. According to (19.5.33), these terms arise only from tree graphs constructed from the $d = 2$, $m = 0$ pion interaction in (19.5.23) or from the $d = 0$, $m = 1$ pion interaction in (19.5.34). To this order, the invariant amplitude M defined by (19.5.24) is now

$$M_{abcd}^{(\nu=2)} = 4F_\pi^{-2} [\delta_{ab}\delta_{cd}(s - m_\pi^2) + \delta_{ac}\delta_{bd}(t - m_\pi^2) + \delta_{ad}\delta_{bc}(u - m_\pi^2)]. \quad (19.5.35)$$

In particular, at threshold $s = 4m_\pi^2$, $t = u = 0$, so

$$\begin{aligned} M_{abcd}^{(\nu=2)}(\text{threshold}) &= 4m_\pi^2 F_\pi^{-2} [3\delta_{ab}\delta_{cd} - \delta_{ac}\delta_{bd} - \delta_{ad}\delta_{bc}] \\ &= 4m_\pi^2 F_\pi^{-2} \left[7M_{ab,cd}^{(0)} - 2M_{ab,cd}^{(2)} \right], \end{aligned} \quad (19.5.36)$$

where $M^{(0)}$ and $M^{(2)}$ are the appropriate tensors representing two-pion states with isospin $T = 0$ or $T = 2$:

$$M_{ab,cd}^{(0)} = \frac{1}{3} \delta_{ab}\delta_{cd}, \quad (19.5.37)$$

$$M_{ab,cd}^{(2)} = \frac{1}{2} \left(\delta_{ac}\delta_{bd} + \delta_{ad}\delta_{bc} - \frac{2}{3} \delta_{ab}\delta_{cd} \right), \quad (19.5.38)$$

normalized so that $\text{Tr } M^{(T)} = 2T + 1$. This result is usually expressed in terms of the scattering lengths. According to Sections 3.6 and 3.7, the scattering length a_T for two pions in a state of isospin T is given [27] by $1/32\pi m_\pi$ times the coefficient of $M^{(T)}$ in (19.5.36):

$$a_0 = \frac{7m_\pi}{8\pi F_\pi^2} = 0.16m_\pi^{-1}, \quad a_2 = -\frac{m_\pi}{4\pi F_\pi^2} = -0.046m_\pi^{-1}.$$

The pion scattering lengths are difficult to measure, but careful study of processes like $\pi + N \rightarrow \pi + \pi + N$ and $K \rightarrow \pi + \pi + e^- + \nu$ has given the results [28] $(0.26 \pm 0.05)m_\pi^{-1}$ and $(-0.028 \pm 0.012)m_\pi^{-1}$ for a_0 and a_2 , respectively, which are consistent with these theoretical values. Corrections of higher order in $m_\pi/2\pi F_\pi$ seem to improve the agreement.

This formalism can be extended to describe the interactions of pions with nucleons or other particles. The easiest approach is to suppose that we add a term involving the nucleon doublet field N to the Lagrangian with which we started:

$$\mathcal{L}_N = -\bar{N} (\not{\partial} + g[\phi_4 + 2i\mathbf{t} \cdot \boldsymbol{\phi}\gamma_5]) N, \quad (19.5.39)$$

where $\mathbf{t}$ is the isospin matrix vector for isospin 1/2 (that is, half the Pauli spinor $\boldsymbol{\tau}$.) This is invariant under chiral $SU(2) \cdot SU(2)$ transformations

$$\delta\phi = 2\epsilon\phi_4, \quad \delta\phi_4 = -2\epsilon \cdot \boldsymbol{\phi}, \quad (19.5.40)$$

$$\delta N = -2i\gamma_5\epsilon \cdot \mathbf{t} N. \quad (19.5.41)$$

Now in eliminating the non-derivative pion couplings we must express the nucleon field N as the $SO(4)$ rotation R in the representation (19.5.41) acting upon a new nucleon field $\tilde{N}$

$$N \equiv \frac{1 - 2i\gamma_5\mathbf{t} \cdot \boldsymbol{\zeta}}{\sqrt{1 + \boldsymbol{\zeta}^2}} \tilde{N} \quad (19.5.42)$$

with $\boldsymbol{\zeta}$ given again by (19.5.7). With this transformation the non-derivative term in (19.5.39) now depends only on $\tilde{N}$ and σ :

$$\bar{N}[\phi_4 + 2i\mathbf{t} \cdot \boldsymbol{\phi}\gamma_5]N = \sigma\bar{\tilde{N}}\tilde{N}.$$

On the other hand, the derivative term involves derivatives of the matrix in (19.5.41), yielding a nucleonic Lagrangian:

$$\mathcal{L}_N = -\bar{\tilde{N}} \left[\not{\partial} + g\sigma + 2i\frac{\mathbf{t} \cdot (\boldsymbol{\zeta} \times \not{\partial}\boldsymbol{\zeta})}{1 + \boldsymbol{\zeta}^2} + 2i\gamma_5\mathbf{t} \cdot \not{\boldsymbol{\pi}} \right] \tilde{N} \quad (19.5.43)$$

or in terms of the canonically normalized pion field (19.5.17):

$$\mathcal{L}_N = -\bar{\tilde{N}} \left[\not{\partial} + g\sigma + \frac{2i\mathbf{t} \cdot (\boldsymbol{\pi} \times \not{\partial}\boldsymbol{\pi})}{F_\pi^2[1 + \boldsymbol{\pi}^2/F_\pi^2]} + \frac{2i\gamma_5\mathbf{t} \cdot \not{\partial}\boldsymbol{\pi}}{F_\pi[1 + \boldsymbol{\pi}^2/F_\pi^2]} \right] \tilde{N}. \quad (19.5.44)$$

Since σ has a non-vanishing vacuum expectation value, we see that the nucleon here has a non-zero rest mass, a result that would be prohibited if the symmetry under the transformations (19.5.40), (19.5.41) were unbroken.

By its construction, the Lagrangian (19.5.44) is chiral-invariant. This can also be seen directly, using the previously obtained chiral transformation properties of σ and $\boldsymbol{\pi}$, and now also noting that under the chiral transformations (19.5.40), (19.5.41), the new nucleon field defined by (19.5.42) transforms as

$$\delta\tilde{N} = 2i\mathbf{t} \cdot [\boldsymbol{\zeta} \times \boldsymbol{\epsilon}]\tilde{N}. \quad (19.5.45)$$

That is, under a chiral transformation $\tilde{N}$ simply undergoes the same isospin rotation (19.5.15) as the quantity $\mathbf{D}_\mu$, but of course in the $T = 1/2$ representation. The isospin rotation parameter $\boldsymbol{\zeta} \times \boldsymbol{\epsilon}$ is spacetime-dependent, so derivatives of $\tilde{N}$ do not have the same chiral transformation property, but it is straightforward to check that the combination of the first and third terms in (19.5.43) behaves as a chiral-covariant derivative: that is,

$$\delta \mathcal{D}_\mu \tilde{N} = 2i \mathbf{t} \cdot [\boldsymbol{\zeta} \times \boldsymbol{\epsilon}] \mathcal{D}_\mu \tilde{N}, \quad (19.5.46)$$

where

$$\mathcal{D}_\mu \tilde{N} \equiv \left[\partial_\mu + 2i \frac{\mathbf{t} \cdot (\boldsymbol{\zeta} \times \partial_\mu \boldsymbol{\zeta})}{1 + \boldsymbol{\zeta}^2} \right] \tilde{N}. \quad (19.5.47)$$

Thus the Lagrangian (19.5.43) (and hence (19.5.44)) is obviously chiral-invariant, because it is isospin-invariant, and constructed solely from the ingredients $\tilde{N}$, $\mathcal{D}_\mu \tilde{N}$, σ , and $\mathbf{D}_\mu$, all of which transform under chiral transformations with the same isospin rotation.

Again, the specific Lagrangian with which we started here is not important. As in the case of the pure pion theory considered earlier, the important thing is the chiral invariance of the Lagrangian. For chiral invariance, the Lagrangian must conserve isospin, and be constructed only from the ingredients $\tilde{N}$, $\mathcal{D}_\mu \tilde{N}$, and $\mathbf{D}_\mu$ (together with higher covariant derivatives of these objects). The most general chiral-invariant Lagrangian that is bilinear in the new nucleon field and involves no more than one derivative therefore takes the form

$$\mathcal{L}_{N,0} = -\tilde{N} [\not{\mathcal{D}} + m_N + 2ig_A \gamma_5 \mathbf{t} \cdot \not{\mathbf{D}}] \tilde{N} \quad (19.5.48)$$

or in terms of the pion field (19.5.17):

$$\mathcal{L}_{N,0} = -\tilde{N} \left[\not{\mathcal{D}} + m_N + \frac{2i \mathbf{t} \cdot (\boldsymbol{\pi} \times \not{\boldsymbol{\pi}})}{F_\pi^2 [1 + \boldsymbol{\pi}^2 / F_\pi^2]} + \frac{2ig_A \gamma_5 \mathbf{t} \cdot \not{\boldsymbol{\pi}}}{F_\pi [1 + \boldsymbol{\pi}^2 / F_\pi^2]} \right] \tilde{N}. \quad (19.5.49)$$

Note that we have inserted an arbitrary constant g_A in the last term in (19.5.48) because this term is chiral symmetric by itself, and so chiral symmetry cannot dictate its coefficient. (This is in contrast with the third term in (19.5.43), whose coefficient is fixed by the condition that it and the first term together give a chiral invariant.) We can check that the constant g_A inserted here is indeed the axial-vector coupling of beta decay by constructing the extra term in the axial-vector current that arises from the nucleonic Lagrangian (19.5.48). Alternatively, integrating by parts and using the Dirac equation, we find that the pion-nucleon interaction $-2ig_A \tilde{N} \gamma_5 \mathbf{t} \cdot (\not{\boldsymbol{\pi}}) \tilde{N} / F_\pi$ is equivalent on the nucleon mass shell to an interaction $-4im_N g_A \tilde{N} \gamma_5 \mathbf{t} \cdot \boldsymbol{\pi} \tilde{N} / F_\pi$, corresponding to a pion-nucleon coupling constant

$$G_{\pi N} = 2m_N g_A / F_\pi,$$

which is just the Goldberger–Treiman relation (19.4.33).

Incidentally, if we had used the Lagrangian (19.5.44) in this calculation, we would have obtained the Goldberger–Treiman relation with $g_A = 1$. However, this result is in fact an artifact of the particular form (19.5.39) of the interaction with which we started. We might

have included a non-renormalizable derivative coupling term¹³

$$\mathcal{L}'_N = ig'\bar{N} [(\mathbf{t} \cdot \boldsymbol{\phi} \not{\partial} \phi_4 - \phi_4 \mathbf{t} \cdot \not{\partial} \boldsymbol{\phi}) \gamma_5 + \mathbf{t} \cdot (\boldsymbol{\phi} \times \not{\partial} \boldsymbol{\phi})] N. \quad (19.5.50)$$

It is straightforward to show that in terms of the transformed fields σ , $\tilde{N}$, and $\boldsymbol{\zeta}$, this takes the form

$$\mathcal{L}'_N = -8i\sigma^2 g' \tilde{N} \mathbf{t} \cdot \not{\boldsymbol{\zeta}} \gamma_5 \tilde{N}.$$

Since σ has a non-vanishing vacuum expectation value, this makes a contribution to the pion-nucleon coupling constant and to g_A proportional to g' . Hence the value $g_A = 1$ is *not* dictated by the broken $SU(2) \cdot SU(2)$ symmetry alone; by including the interaction (19.5.50) and adjusting g' , we can make g_A anything we like.

Now let us consider how to use this Lagrangian to calculate amplitudes for reactions involving both pions and nucleons. We will have to give special attention to the nucleon propagators, since a nucleon can never be a ‘soft’ particle like a pion. A nucleon line that enters a diagram with an on-shell four-momentum p of order m_N , and then from interactions with soft pions absorbs a net four-momentum q with components much less than m_N , will have a propagator

$$\frac{-i(\not{p} + \not{q}) + m_N}{(p + q)^2 + m_N^2} \xrightarrow{q \rightarrow 0} \frac{-i\not{p} + m_N}{2p \cdot q}. \quad (19.5.51)$$

(The neglected terms may be taken into account by including higher derivative terms in the nucleon Lagrangian.) Suppose again that all external pion four-momenta have components at most of order Q and define all renormalized couplings at renormalization points of order Q , so that integrals converge in such a way that internal pion lines also have four-momenta Q . Then (19.5.51) shows that internal nucleon lines make a contribution of order $1/Q$. A general Feynman diagram for such a process will make a contribution to the invariant amplitude of order Q^ν , where now

$$\nu = \sum_i V_i (d_i + 2m_i) - 2I_\pi - I_N + 4L. \quad (19.5.52)$$

Here V_i is the number of vertices associated with interactions of type i , d_i is the number of derivatives in each such interaction, m_i is the number of quark mass factors in each interaction, I_π and I_N are the numbers of internal pion and nucleon lines, respectively, and L is the number of loops in the graph. We use the familiar topological relations for connected graphs:

$$L = I_\pi + I_N - \sum_i V_i + 1 \quad (19.5.53)$$

and

$$2I_N + E_N = \sum_i V_i n_i, \quad (19.5.54)$$

¹³The left- and right-handed parts of the nucleon doublet transform according to the representations $(\frac{1}{2}, 0)$ and $(0, \frac{1}{2})$ of $SU(2) \cdot SU(2)$, respectively. The bilinear $\bar{N} \gamma_\mu \gamma_5 N$ is thus a sum of terms quadratic in $(\frac{1}{2}, 0)$ or $(0, \frac{1}{2})$ terms, so it transforms like a direct sum of the $(1, 0)$, $(0, 0)$, and $(0, 1)$ representations. In Eq. (19.5.50) we have coupled the $(1, 0) + (0, 1)$ terms in $\bar{N} \gamma_\mu \gamma_5 N$ to the antisymmetric tensor formed from the $SU(2) \cdot SU(2)$ four-vectors ϕ_n and $\partial_\mu \phi_n$. Its invariance under the transformations (19.5.40), (19.5.41) can of course be checked directly.

where n_i is the number of nucleon fields in interactions of type i , and E_N is the number of external nucleon lines. Eliminating the quantities I_N and I_π , this gives

$$\nu = \sum_i V_i \left(d_i + 2m_i + \frac{n_i}{2} - 2 \right) + 2L - E_N + 2. \quad (19.5.55)$$

The important point here is that the coefficient $d_i + 2m_i + \frac{1}{2}n_i - 2$ in the first term is always positive or zero. We have already seen that $d_i + 2m_i \geq 2$ for the purely pionic interactions with $n_i = 0$, and inspection of (19.5.49) shows that $d_i \geq 1$ for the pion-nucleon interactions with $n_i = 2$ and $m_i = 0$. Any interaction with $n_i = 2$ and $m_i \geq 1$ or $n_i \geq 4$ clearly also has $d_i \geq 2$. Hence the leading terms for $Q \ll 2\pi F_\pi$ are tree graphs (that is, $L = 0$) for which all interactions have

$$d_i + 2m_i + \frac{n_i}{2} - 2 = 0.$$

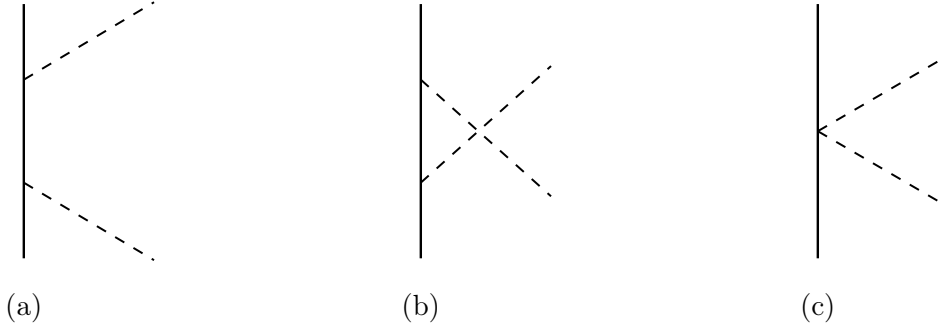


Figure 19.3. Feynman diagrams used with an effective chiral Lagrangian to calculate the scattering of a soft pion from a nucleon. Dashed lines are pions; solid lines are nucleons.

The interactions that satisfy this condition are just those shown explicitly in Eqs. (19.5.23), (19.5.34), and (19.5.49), plus possible interactions with $d_i = 0$ and $n_i = 4$:

$$\left(\bar{N} \Gamma_\alpha \tilde{N} \right) \left(\bar{N} \Gamma^\alpha \tilde{N} \right), \quad (19.5.56)$$

where Γ_α and Γ^α are any matrices in spin and isospin space that yield Lorentz, space inversion, and isospin-invariant four-fermion interactions. These last interactions are important for multinucleon processes [29], which will not be considered here.

Let us now apply this method to pion-nucleon scattering. For pion energies roughly of order m_π , the pion-nucleon scattering amplitude is given by the Feynman graphs of Figure 19.3, all of which make contributions of order m_π to the invariant amplitude. However, at threshold the leading contribution is from Figure 19.3(c), the others being suppressed by an extra factor of m_π/m_N . This is because at threshold in the rest frame the incoming and outgoing pion four-momenta q, q' and nucleon four-momenta p, p' are given by

$$q = q' = (m_\pi, \mathbf{0}) = \left(\frac{m_\pi}{m_N} \right) p = \left(\frac{m_\pi}{m_N} \right) p'. \quad (19.5.57)$$

Thus the invariant amplitude at threshold from either Figure 19.3(a) or 19.3(b) is proportional to

$$\bar{u} \gamma_5 \not{q} \frac{-i(\not{p} \pm \not{q}) + m_N}{(p \pm q)^2 + m_N^2} \gamma_5 \not{q} u$$

$$\begin{aligned}
&= \frac{m_\pi^2 \bar{u} \gamma_5 \{-i \not{p}(1 \pm m_\pi/m_N) + m_N\} \gamma_5 u}{\mp 2m_\pi m_N - m_\pi^2} \\
&= \frac{\mp m_\pi}{2m_N \pm m_\pi} \bar{u} \{i \not{p}(1 \pm m_\pi/m_N) + m_N\} u = \frac{m_\pi^2}{2m_N \pm m_\pi},
\end{aligned}$$

where u is a Dirac spinor with $\bar{u}u = 1$. In contrast, the diagram of Figure 19.3(c) gives a contribution to M of order m_π . We may write this amplitude as a $2 \cdot 2$ matrix in the nucleon isospin indices:

$$M_{ba} = -\frac{2i}{F_\pi^2} t_c \epsilon_{abc} \bar{u}(-i \not{q} - i \not{q}') u,$$

where q' and q are the final and initial pion four-momenta and b and a are the corresponding isovector indices. Using (19.5.57) and the momentum-space Dirac equation $(i \not{p} + m_N)u = 0$, this becomes

$$M_{ba} = -\frac{4im_\pi}{F_\pi^2} t_c \epsilon_{abc} = -\frac{4m_\pi}{F_\pi^2} \mathbf{t} \cdot [\mathbf{t}^{(\pi)}]_{ba}, \quad (19.5.58)$$

where $[t_c^{(\pi)}]_{ba} \equiv -i\epsilon_{bac}$ is the pion isovector matrix. The matrix $\mathbf{t} \cdot \mathbf{t}^{(\pi)}$ has eigenvalues in states of total isospin T equal to $\frac{1}{2}[T(T+1) - 2 - \frac{3}{4}]$, so in the two isospin states, $T = 1/2$ and $T = 3/2$, the invariant amplitudes are [30]

$$M_{1/2} = 4m_\pi/F_\pi^2, \quad M_{3/2} = -2m_\pi/F_\pi^2.$$

These results are usually expressed in terms of the scattering lengths, which are defined as the invariant amplitudes divided by $4\pi(1 + m_\pi/m_N)$:

$$a_{1/2} = \frac{m_\pi}{\pi F_\pi^2(1 + m_\pi/m_N)} = 0.15m_\pi^{-1}, \quad (19.5.59)$$

$$a_{3/2} = -\frac{m_\pi}{2\pi F_\pi^2(1 + m_\pi/m_N)} = -0.075m_\pi^{-1}. \quad (19.5.60)$$

These are in reasonable agreement with the experimental values [12] $a_{1/2} = (0.173 \pm 0.003)m_\pi^{-1}$ and $a_{3/2} = (-0.101 \pm 0.004)m_\pi^{-1}$. (The results (19.5.59) and (19.5.60) are only supposed to be valid to lowest order in m_π/m_N , but we retain the factor $1 + m_\pi/m_N$, because it arises just from the definition of the scattering lengths.)

The corrections to the scattering lengths of next order in m_π arise from several sources. There are the Born diagram graphs of Figures 19.3(a) and 19.3(b), which are nominally of leading order but as we have seen at threshold are suppressed by an extra factor of m_π/m_N . There are additional tree graphs containing a vertex with two derivatives. Of special interest are tree graphs containing a vertex with no derivatives that arises from a symmetry-breaking interaction proportional to m_π^2 . These interactions must have the chiral and spacetime transformation properties of the operators in (19.4.34): the fourth and third components of two different chiral four-vectors Φ_n^+ and Φ_n^- . There are two obvious candidates for such operators that are bilinear in the nucleon fields: a Φ_4^+ term,

$$\bar{N}N = \left(\frac{1 - \zeta^2}{1 + \zeta^2}\right) \bar{\tilde{N}}\tilde{N} - 4i \left(\frac{\zeta}{1 + \zeta^2}\right) \cdot \bar{\tilde{N}}\gamma_5 \mathbf{t} \tilde{N},$$

and a Φ_3^- term,

$$\overline{N}t_3N = \overline{\tilde{N}}t_3\tilde{N} - 2\left(\frac{\zeta_3}{1+\zeta^2}\right)\overline{\tilde{N}}\mathbf{t} \cdot \boldsymbol{\zeta}\tilde{N} - i\left(\frac{\zeta_3}{1+\zeta^2}\right)\overline{\tilde{N}}\gamma_5\tilde{N}.$$

Chiral symmetries act on $\tilde{N}$ like ordinary isospin rotations, so they cannot mix up scalar and pseudoscalar nucleon bilinears. Thus there are really two independent Φ_4^+ operators:

$$\left(\frac{1-\zeta^2}{1+\zeta^2}\right)\overline{\tilde{N}}\tilde{N} \quad (19.5.61)$$

and

$$i\left(\frac{\zeta}{1+\zeta^2}\right) \cdot \overline{\tilde{N}}\gamma_5\mathbf{t}\tilde{N}, \quad (19.5.62)$$

and two independent Φ_3^- operators:

$$\overline{\tilde{N}}t_3\tilde{N} - 2\left(\frac{\zeta_3}{1+\zeta^2}\right)\overline{\tilde{N}}\mathbf{t} \cdot \boldsymbol{\zeta}\tilde{N} \quad (19.5.63)$$

and

$$i\left(\frac{\zeta_3}{1+\zeta^2}\right)\overline{\tilde{N}}\gamma_5\tilde{N}. \quad (19.5.64)$$

We shall show in the next section that these are the only operators with the transformation properties of the terms in (19.4.34) that are bilinear in the nucleon fields. The operators (19.5.62) and (19.5.64) evidently provide isospin-conserving and isospin-violating corrections to the Goldberger–Treiman formula for the pion-nucleon couplings. The other two operators, (19.5.61) and (19.5.63), contribute directly to both the nucleon mass and to low energy pion-nucleon scattering. From their contribution to the nucleon mass, we see that these latter terms enter into the effective Lagrangian in the form (now replacing $\boldsymbol{\zeta}$ with the conventionally normalized pion field):

$$\begin{aligned} \delta\mathcal{L}_{\text{eff}} = & -\frac{\delta m_p + \delta m_n}{2}\left(\frac{1-\boldsymbol{\pi}^2/F_\pi^2}{1+\boldsymbol{\pi}^2/F_\pi^2}\right)\overline{\tilde{N}}\tilde{N} \\ & -(\delta m_p - \delta m_n)\left[\overline{\tilde{N}}t_3\tilde{N} - \frac{2}{F_\pi^2}\left(\frac{\pi_3}{1+\boldsymbol{\pi}^2/F_\pi^2}\right)\overline{\tilde{N}}\mathbf{t} \cdot \boldsymbol{\pi}\tilde{N}\right], \end{aligned} \quad (19.5.65)$$

where δm_p and δm_n are the contributions of the quark mass terms (19.4.34) to the proton and neutron masses. This makes a contribution to the pion-nucleon scattering amplitude (again written as a matrix in the isospin space of the nucleon):

$$\delta M_{ba} = \frac{2[\delta m_p + \delta m_n]}{F_\pi^2}\delta_{ab} + \frac{2[\delta m_p - \delta m_n]}{F_\pi^2}(t_a\delta_{3b} + t_b\delta_{3a}). \quad (19.5.66)$$

The first term in (19.5.65) is often known as the ‘ σ -term’. The second term is an isospin violating correction to the σ -term, which shows up only in processes involving neutral pions, such as the charge-exchange processes $\pi^+ + n \rightarrow \pi^0 + p$ and $\pi^- + p \rightarrow \pi^0 + n$. Even though it is not possible to measure the nucleon mass shifts $\delta m_{p,n}$ directly, we shall see in Section 19.7 that an $SU(3)$ symmetry allows their difference to be calculated from hyperon mass

differences. This yields the result $\delta m_p - \delta m_n = -2.5$ MeV. Unfortunately, we still do not have a firm theoretical estimate of the coefficient $\delta m_p + \delta m_n$ of the first term in (19.5.65).

Eq. (19.4.34) shows that in quantum chromodynamics the coefficients $\delta m_p + \delta m_n$ and $\delta m_p - \delta m_n$ in (19.5.66) are respectively proportional to $m_u + m_d$ and $m_u - m_d$. We shall see in Section 19.7 that m_u and m_d are not at all degenerate, so these coefficients are roughly of the same order of magnitude; the isospin-violating corrections to the σ -term are not much smaller than the σ -term itself. We see again that the reason that isospin conservation is such a good approximation in hadronic physics is not that the u and d quark masses are nearly equal, but just that they are small. The use here of Lagrangians like (19.5.19) or (19.5.49) is one example of the method of *effective field theories*, already introduced in Section 12.3. Similar techniques have been employed to deal with meson and baryon interactions including strange particles (see Section 19.7), with quark and lepton interactions at energies below the scale of electroweak symmetry breaking (see Section 21.3), and even with superconductivity (see Section 21.6). In all these cases effective field theories provide the most convenient method for working out the consequences of symmetries and the general principles underlying quantum field theory.

* * *

The low energy theorems provided by broken symmetries when combined with dispersion relations yield useful sum rules. Let's see how this works for chiral symmetry, ignoring the small u and d quark masses and the pion mass. Consider the forward scattering of a massless pion with isovector index a and four-momentum q on a nucleon of four-momentum p , yielding a pion with isovector index b . The scattering amplitude M , defined by $S_{fi} = -2\pi i \delta^4(P_f - P_i) M_{fi}$, is given to first order in q by the Feynman diagrams of Figure 19.3 as

$$\begin{aligned}
 -2\pi i M = & \frac{1}{(2\pi)^6 (2q^0)} \bar{u} \left[\right. \\
 & \left(-i \frac{(2\pi)^4 2g_A \gamma_5 t_b \not{q}}{F_\pi} \right) \left(\frac{-i}{(2\pi)^4} \frac{-i(\not{p} + \not{q}) + m_N}{(p+q)^2 + M^2} \right) \left(i \frac{(2\pi)^4 2g_A \gamma_5 t_a \not{q}}{F_\pi} \right) \\
 & + \left(i \frac{(2\pi)^4 2g_A \gamma_5 t_a \not{q}}{F_\pi} \right) \left(\frac{-i}{(2\pi)^4} \frac{-i(\not{p} - \not{q}) + m_N}{(p-q)^2 + M^2} \right) \left(-i \frac{(2\pi)^4 2g_A \gamma_5 t_b \not{q}}{F_\pi} \right) \\
 & \left. - \frac{4i t_c \epsilon_{abc} \not{q}}{F_\pi^2} \right] u. \tag{19.5.67}
 \end{aligned}$$

To this order in q , the nucleon propagator in the first two terms inside the square brackets may be approximated by

$$\frac{-i(\not{p} \pm \not{q}) + m_N}{(p \pm q)^2 + M^2} \approx \frac{-i\not{p} + m_N}{\pm 2p \cdot q}.$$

Eq. (19.5.67) can then be further simplified, using the relations $\not{q}\not{q} = q^2 = 0$, $\bar{u}\gamma^\mu u = -ip^\mu/m_N$, and $[t_a, t_b] = i\epsilon_{abc}t_c$. Also, in the laboratory frame (which for small q is the

same as the center-of-mass frame) we have $p \cdot q = -m_N \omega$, where $\omega \equiv q^0$ is the pion energy in the nucleon rest frame. Finally, the conventional forward scattering amplitude (whose absolute square is the differential cross section for forward scattering) is given by Eq. (3.6.9) as $f = -4\pi^2 \omega M$. Putting this all together, we find the forward scattering amplitude at pion energy ω is given for $\omega \ll m_N$ by

$$f_{ba}(\omega) \longrightarrow -i \frac{\omega}{\pi F_\pi^2} (1 - g_A^2) \epsilon_{abc} t_c. \quad (19.5.68)$$

In particular, for π^+ -proton scattering, we must contract (19.5.68) with $v_b^* v_a$, where $\mathbf{v}$ is the normalized isovector $(1, i, 0)/\sqrt{2}$, and set $t_3 = +\frac{1}{2}$. The forward low energy π^+ -proton scattering amplitude is then

$$f_{\pi^+p}(\omega) \longrightarrow -\frac{\omega}{2\pi F_\pi^2} (1 - g_A^2). \quad (19.5.69)$$

Now, the dispersion relation for the forward scattering of a massless π^+ on a proton is given by Eq. (10.8.24) (with subtraction polynomial $P(E) \propto E$) as

$$\begin{aligned} f_{\pi^+p}(\omega) = R + \frac{i\omega}{4\pi} \sigma_{\pi^+p}(\omega) \\ + \frac{\omega}{4\pi^2} \int_0^\infty \left[\frac{\sigma_{\pi^+p}(E)}{E - \omega} - \frac{\sigma_{\pi^-p}(E)}{E + \omega} \right] dE, \end{aligned} \quad (19.5.70)$$

where R is a possible subtraction constant, and ω and E are pion energies in the proton rest frame. Comparison of this dispersion relation with the low energy limit (19.5.69) shows that $R = 0$, and that

$$g_A^2 = 1 + \frac{F_\pi^2}{2\pi} \int_0^\infty [\sigma_{\pi^+p}(E) - \sigma_{\pi^-p}(E)] dE/E. \quad (19.5.71)$$

This is the celebrated Adler–Weisberger sum rule [19], for the case of exact chiral symmetry, with $m_\pi = 0$. A similar sum rule may be derived for the scattering of pions on any baryon or meson, including the pion itself [30a].

19.6 Effective Field Theories: General Broken Symmetries

The techniques described in the previous section for constructing effective Lagrangians for the case of $SU(2) \cdot SU(2)$ broken to $SU(2)$ were soon generalized [31] to the case of a general group G broken to an arbitrary subgroup H . Consider a quantum field theory whose Lagrangian is invariant under an arbitrary compact Lie group G of ordinary linear spacetime-independent transformations g of the fields $\psi_n(x)$:

$$\psi_n(x) \longrightarrow \sum_m g_{nm} \psi_m(x). \quad (19.6.1)$$

We suppose that this symmetry group is spontaneously broken to some subgroup $H \subset G$ of symmetry transformations that leave all vacuum expectation values invariant: for $h \in H$,

$$\sum_m h_{nm} \langle \Omega | \psi_m(x) | \Omega \rangle = \langle \Omega | \psi_n(x) | \Omega \rangle. \quad (19.6.2)$$

(Not all of the ψ s need be scalars, but of course only the scalars will have non-vanishing vacuum expectation values.) In the example at the beginning of the previous section, G was the group $SO(4)$, the fields ψ_n furnished a four-vector representation of this group, and their vacuum expectation values broke $SO(4)$ down to the subgroup $H = SO(3)$ of rotations that leave the four-axis invariant. In the general case the fields ψ_n may furnish a reducible rather than an irreducible representation of G , as for instance became the case when we introduced the nucleon fields in the previous section.

We next express a general ‘point’ ψ_n in field space as a G transformation acting on fields $\tilde{\psi}_n$ from which the massless Goldstone mode has been eliminated:

$$\psi_n(x) = \sum_m \gamma_{nm}(x) \tilde{\psi}_m(x). \quad (19.6.3)$$

For instance, in the previous section γ was the $SO(4)$ rotation we called R , and the condition satisfied by $\tilde{\psi}_n$ was that the first three components of the four-vector should vanish.

In order to formulate this condition more generally, let us suppose that we are working with a real representation of G . (There is no loss of generality here, because we are not insisting on an irreducible representation, so if we start with a set of fields forming a complex representation of G we can always take the ψ_n to be the real and imaginary parts of these complex fields.) As shown in Section 19.2, the massless eigenvectors of the mass matrix are just the independent linear combinations of the vectors $\sum_m [t_\alpha]_{nm} \langle \Omega | \psi_m(0) | \Omega \rangle$, where t_α are the generators of G . (The reality of the representation means that it_α is real, and the fact that G is compact means that we can choose the representation to be unitary and hence orthogonal, so that the t_α are imaginary and antisymmetric.) The condition that $\tilde{\psi}_n$ does not contain Goldstone modes may thus be formulated as

$$\sum_{nm} \tilde{\psi}_n(x) [t^\alpha]_{nm} \langle \Omega | \psi_m(0) | \Omega \rangle = 0. \quad (19.6.4)$$

The number of independent conditions here is the dimensionality of the group G minus the dimensionality of the subgroup H whose generators annihilate $\langle \Omega | \psi_m(0) | \Omega \rangle$. As in the previous section, the Goldstone bosons whose fields are eliminated in this way will reappear in the $\dim(G) - \dim(H)$ fields needed to parameterize the transformation $\gamma_{nm}(x)$. In general the $\tilde{\psi}_n$ include all the heavy fields of the theory, including those (like nucleons in the previous section) that have different spacetime symmetry properties from the Goldstone bosons.

It is necessary to show that we may always choose the transformation $\gamma_{nm}(x)$ so as to satisfy Eq. (19.6.4). For this purpose, consider the quantity

$$V_\psi(g) \equiv \sum_{nm} \psi_n g_{nm} \langle \Omega | \psi_m(0) | \Omega \rangle, \quad (19.6.5)$$

where g runs over the whole group G in the real orthogonal representation furnished by the ψ_n . This is obviously a continuous real function of g , and since the group is compact $V_\psi(g)$ is also a bounded function of g . At each spacetime point x , $V_{\psi(x)}(g)$ therefore reaches a maximum value for some group element, which we shall call $\gamma(x)$. At $g = \gamma(x)$, $V_{\psi(x)}(g)$

must be stationary with respect to arbitrary variations in g . But any infinitesimal shift in a group element g may always be written as a linear combination

$$\delta g = i \sum_{\alpha} \epsilon_{\alpha} g t^{\alpha},$$

with real infinitesimal coefficients ϵ_{α} that may depend on g . Hence the condition that $V_{\psi(x)}(g)$ be stationary with respect to the variation δg at $g = \gamma(x)$ reads

$$\begin{aligned} 0 &= \delta V_{\psi(x)}(\gamma(x)) = i \sum_{\alpha} \epsilon_{\alpha} \sum_{nm\ell} \psi_n(x) \gamma_{nm}(x) t_{m\ell}^{\alpha} \langle \Omega | \psi_{\ell}(0) | \Omega \rangle, \\ &= i \sum_{\alpha} \epsilon_{\alpha} \sum_{nm\ell} [\gamma^{-1}(x)]_{\ell n} \psi_n(x) t_{\ell m}^{\alpha} \langle \Omega | \psi_m(0) | \Omega \rangle. \end{aligned}$$

This must be satisfied for all variations, and thus for all ϵ_{α} , so we see that Eq. (19.6.4) is satisfied for $\tilde{\psi}(x) = \gamma^{-1}(x) \psi(x)$, as was to be shown. In the purely bosonic theory with Lagrangian (19.4.1), $\tilde{\psi}_n$ was the four-vector field $(0, 0, 0, \sigma)$. This points in the direction of the vacuum expectation value $\langle \Omega | \psi_n | \Omega \rangle$, but this is not always the case.

Incidentally, it is not necessary in dealing with complex fields to formulate the condition on $\tilde{\psi}(x)$ explicitly in terms of its real and imaginary parts. If a set of fields $\chi(x)$ transform according to some complex representation of G with Hermitian generators T^{α} , then in the real representation

$$\psi(x) = \begin{pmatrix} \text{Re } \chi(x) \\ \text{Im } \chi(x) \end{pmatrix}$$

the generators are

$$it^{\alpha} = \begin{pmatrix} -\text{Im } T^{\alpha} & -\text{Re } T^{\alpha} \\ \text{Re } T^{\alpha} & -\text{Im } T^{\alpha} \end{pmatrix}.$$

The condition (19.6.4) may then be expressed directly in terms of T^{α} and $\chi(x)$ as

$$\text{Im} \left\{ \tilde{\chi}(x)^{\dagger} T^{\alpha} \langle \Omega | \chi(0) | \Omega \rangle \right\} = 0.$$

Because the Lagrangian is assumed to be only invariant under spacetime-independent G transformations, it will be found after the transformation (19.6.3) to depend on $\gamma(x)$ as well as on $\tilde{\psi}(x)$, though always with at least one derivative in each γ -dependent term. As already remarked, the spacetime-dependent parameters needed to specify $\gamma(x)$ will play the role of Goldstone boson fields. We now have to consider how to parameterize $\gamma(x)$.

It must be recognized from the outset that the choice of γ in Eq. (19.6.3) is generally not unique. Because $\langle \Omega | \psi_m(0) | \Omega \rangle$ is H -invariant in the sense of Eq. (19.6.2), the quantity $V_{\psi}(g)$ defined by Eq. (19.6.5) is invariant under right multiplication of g by any element h of the unbroken subgroup H :

$$V_{\psi}(g) = V_{\psi}(gh) \quad \text{for} \quad h \in H. \quad (19.6.6)$$

It follows that if γ maximizes $V_{\psi}(g)$, then so does γh , so that condition (19.6.4) is satisfied for $\tilde{\psi} = h^{-1} \gamma^{-1} \psi$ as well as $\tilde{\psi} = \gamma^{-1} \psi$. Hence γ is only defined so far up to right

multiplication by an element of H . (For instance in the example of the previous section, the four-dimensional rotation R could have been multiplied on the right with any rotation acting only on the first three components of four-vectors.) We may think of two group elements γ_1 and γ_2 as equivalent if $\gamma_1 = \gamma_2 h$ with $h \in H$. This is an equivalence relation because it is reflexive (if γ_1 is equivalent to γ_2 then γ_2 is equivalent to γ_1), symmetric (γ is equivalent to itself), and transitive (if γ_1 is equivalent to γ_2 and γ_2 is equivalent to γ_3 then γ_1 is equivalent to γ_3). The elements of the group G can therefore be sorted into disjoint ‘equivalence classes’, each consisting of elements γ that differ only by right multiplication by an element of H . These are known as the *right cosets* of G with respect to H . What we need is a parameterization of the space (known as G/H) of right cosets.

For this purpose we need only choose one representative group element from each right coset. For the $SO(4)$ symmetry broken to $SO(3)$ of the previous section, it was convenient to choose these representative elements as the rotations R parameterized by the three-vector $\boldsymbol{\zeta}$. There is another choice that is available for any compact group G broken to any subgroup H . Let us first adapt our notation for the generators of G to the pattern of symmetry breaking. The complete set of independent generators of H will be called t_i . According to Eq. (19.6.2) these satisfy

$$\sum_m (t_i)_{nm} \langle \Omega | \psi_m | \Omega \rangle = 0. \quad (19.6.7)$$

Since H is a subgroup, the t_i form a subalgebra

$$[t_i, t_j] = i \sum_k C_{ijk} t_k. \quad (19.6.8)$$

We will take the x_a to be the *other* independent generators of G , in any basis with totally antisymmetric structure constants. (Such a basis always exists for compact groups; see Section 15.2.) Since x_a does not appear on the right-hand side of Eq. (19.6.8), the structure constants C_{ija} all vanish, and since they are totally antisymmetric, it follows that $C_{iaj} = 0$, so

$$[t_i, x_a] = i \sum_b C_{iab} x_b. \quad (19.6.9)$$

However, it is *not* necessarily the case that commutators of x s with each other are linear combinations of t s; this depends on the nature of G and H , and also on how H is embedded in G . (Where C_{abc} vanishes, the coset space G/H is known as a *symmetric space*.) In general we may write

$$[x_a, x_b] = i \sum_i C_{abi} t_i + i \sum_c C_{abc} x_c. \quad (19.6.10)$$

Any set of generators with commutation relations of the form (19.6.8)–(19.6.10) is called a *Cartan decomposition* of the Lie algebra. An example is provided by the chiral $SU(2) \cdot SU(2)$ Lie algebra (19.4.9)–(19.4.11), for which the C_{abc} did happen to vanish.

Because t_i and x_a span the Lie algebra of G , any finite element of G may be expressed in the form

$$g = \exp \left[i \sum_a \xi_a x_a \right] \exp \left[i \sum_i \theta_i t_i \right], \quad (19.6.11)$$

where ξ_a and θ_i are a set of real parameters. But the transformation $\gamma(x)$ in Eq. (19.6.3) is only defined so far up to right-multiplication with an element of H , so we may standardize our definition of γ by taking it in the form

$$\gamma(x) = \exp \left[i \sum_a \xi_a(x) x_a \right]. \quad (19.6.12)$$

The $\xi_a(x)$ may be identified (apart from normalization) with the Goldstone boson fields.

In what follows we will simply assume that representative elements have been chosen from each right coset, and expressed as continuous functions $\gamma(\xi)$ of some parameters ξ_a . Eq. (19.6.12) in general provides one explicit example, but we will not limit ourselves to this parameterization.

Now, suppose we use Eq. (19.6.3) to replace all fields $\psi(x)$ in the Lagrangian with $\gamma(x)\tilde{\psi}(x)$. The derivatives of the fields are given by

$$\partial_\mu \psi(x) = \gamma(x) \left[\partial_\mu \tilde{\psi}(x) + (\gamma^{-1}(x) \partial_\mu \gamma(x)) \tilde{\psi}(x) \right]. \quad (19.6.13)$$

Therefore when we express the Lagrangian in terms of $\tilde{\psi}$ rather than ψ , the Goldstone boson fields appear through the dependence of $\gamma^{-1}(x) \partial_\mu \gamma(x)$ on $\xi_a(x)$ and its derivatives.

Any variation of a group element like $\gamma(x)$ may be written as the group element times a linear combination of the generators of the group. In our case, we may write this as

$$\gamma^{-1}(x) \partial_\mu \gamma(x) = i \sum_a x_a D_{a\mu}(x) + i \sum_i t_i E_{i\mu}(x), \quad (19.6.14)$$

where $D_{a\mu}$ and $E_{i\mu}$ take the forms

$$D_{a\mu}(x) = \sum_b D_{ab}(\xi(x)) \partial_\mu \xi_b(x), \quad (19.6.15)$$

$$E_{i\mu}(x) = \sum_b E_{ib}(\xi(x)) \partial_\mu \xi_b(x). \quad (19.6.16)$$

The Goldstone boson fields will thus enter the Lagrangian through the appearance of the quantities $D_{a\mu}(x)$ and $E_{i\mu}(x)$ (and their derivatives). Note that for exact broken symmetries every interaction of the ξ s must involve at least one derivative, so mass terms $m_{ab}^2 \xi_a \xi_b$ cannot appear in the Lagrangian, and all interactions vanish when any Goldstone boson four-momentum vanishes.

Even where we do not know the details of the underlying Lagrangian, we can learn a great deal about the way that the quantities $D_{a\mu}$ and $E_{i\mu}$ appear in the transformed Lagrangian from their transformation properties. Under an arbitrary element g of the group G , the original field ψ transforms according to Eq. (19.6.1):

$$\psi(x) \longrightarrow \psi'(x) = g\psi(x) = g\gamma(\xi(x))\tilde{\psi}(x). \quad (19.6.17)$$

Now, $g\gamma(\xi)$ is an element of G , so it must be in the same right coset as some $\gamma(\xi')$, and may therefore be written in the form

$$g\gamma(\xi(x)) = \gamma(\xi'(x))h(\xi(x), g), \quad (19.6.18)$$

where h is some element of the unbroken subgroup H . Using this in Eq. (19.6.17), we find that $\psi'(x)$ is of the form (19.6.3)

$$\psi'(x) = \gamma(\xi'(x))\tilde{\psi}'(x), \quad (19.6.19)$$

with $\xi'(x)$ defined by Eq. (19.6.18), and

$$\tilde{\psi}'(x) = h(\xi(x), g)\tilde{\psi}(x). \quad (19.6.20)$$

In the example at the beginning of the previous section, $\tilde{\psi}$ consisted of a single isoscalar σ , and since this was invariant under the unbroken isospin subgroup, it was also chiral invariant. More generally, we find here that $\tilde{\psi}(x)$ is not necessarily invariant under general G transformations, but that its G transformation depends only on its transformation under the unbroken subgroup H . We saw this in the previous section when we introduced the nucleon fields; under a general infinitesimal broken chiral symmetry, the fields $\tilde{N}$ transformed according to the field-dependent isospin rotation (19.5.45).

The transformation rules $\xi \rightarrow \xi'$ and $\tilde{\psi} \rightarrow \tilde{\psi}'$ specified by Eqs. (19.6.18) and (19.6.20) make no reference to the particular linear G transformation properties of the original fields ψ_n . Indeed, we did not need to start with a Lagrangian that was invariant under linear G transformations in order to deduce these transformation rules. Given any set of fields on which a group G acts linearly or non-linearly, with a subgroup H that leaves one special set of field values (their vacuum expectation values) invariant, we can always express these fields (in at least a finite neighborhood of the special field values) in terms of a set ξ_a and $\tilde{\psi}_n$, which transform under G according to the standard realization $\xi_a \rightarrow \xi'_a$, $\tilde{\psi} \rightarrow \tilde{\psi}'$ defined by Eqs. (19.6.18) and (19.6.20). The essential uniqueness of this realization was first proved [25] for $SU(2) \cdot SU(2)$ broken to $SU(2)$, and then for general Lie groups broken to any subgroup [31]. For an easier argument, we may note that for any set of fields that transforms according to a non-linear realization of an arbitrary compact Lie group G , it is always possible to define functions of these fields (and perhaps additional fields¹⁴) that transform linearly under G [32]. Starting from this linear realization, the arguments of this section allow us to construct fields ξ and $\tilde{\psi}$ with the transformation rules (19.6.18) and (19.6.20).

Although the transformation of $\xi(x)$ and $\tilde{\psi}(x)$ is complicated for general $g \in G$, it is much simpler when g is itself a member h of the unbroken subgroup H . It is usual and always possible to choose the Goldstone boson fields $\xi_a(x)$ to transform according to some *linear* representation $\mathcal{D}_{ab}(h)$ of the unbroken subgroup H :

$$h\gamma(\xi)h^{-1} = \gamma(\mathcal{D}(h)\xi). \quad (19.6.21)$$

For instance, in the parameterization of the coset space $SO(4)/SO(3)$ used in the previous section, the Goldstone boson fields ξ_a formed an isospin three-vector. For the more generally

¹⁴For instance, the polar and azimuthal angles θ and φ furnish a non-linear realization of $SO(3)$; by adding an additional variable r we can construct quantities x_1 , x_2 , and x_3 that transform linearly under $SO(3)$.

applicable parameterization based on Eq. (19.6.12), the commutation relations (19.6.9) show that the x_a transform linearly under H :

$$hx_b h^{-1} = \sum_a \mathcal{D}_{ab}(h) x_a, \quad (19.6.22)$$

from which Eq. (19.6.21) follows immediately. Comparing Eq. (19.6.21) with Eq. (19.6.18), we see that for $g = h \in H$,

$$\xi'_a(x) = \sum_b \mathcal{D}_{ab}(h) \xi_b(x), \quad (19.6.23)$$

$$\tilde{\psi}'_n(x) = \sum_m h_{nm} \tilde{\psi}_m(x). \quad (19.6.24)$$

In other words, ξ_a and $\tilde{\psi}_n$ transform under H according to the representations $\mathcal{D}_{ab}(h)$ and h_{nm} itself, respectively.

We must now consider how to construct the most general Lagrangian that is invariant under the transformations (19.6.18) and (19.6.20). The transformation $\xi \rightarrow \xi'$ for general $g \in G$ is non-linear and rather complicated, and rules out the appearance of $\xi_a(x)$ in the Lagrangian except in quantities like $D_{a\mu}(x)$ and $E_{i\mu}(x)$. Fortunately, these quantities obey very simple transformation rules. Differentiate Eq. (19.6.18) with respect to x^μ and multiply on the left with its inverse. This gives

$$\begin{aligned} \gamma^{-1}(\xi(x)) \partial_\mu \gamma(\xi(x)) &= [g\gamma(\xi(x))]^{-1} \partial_\mu [g\gamma(\xi(x))] \\ &= h^{-1}(\xi(x), g) \gamma^{-1}(\xi'(x)) \partial_\mu [\gamma(\xi'(x)) h(\xi(x), g)] \\ &= h^{-1}(\xi(x), g) [\gamma^{-1}(\xi'(x)) \partial_\mu \gamma(\xi'(x))] h(\xi(x), g) \\ &\quad + h^{-1}(\xi(x), g) \partial_\mu h(\xi(x), g), \end{aligned}$$

and so

$$\begin{aligned} \gamma^{-1}(\xi'(x)) \partial_\mu \gamma(\xi'(x)) &= h(\xi(x), g) [\gamma^{-1}(\xi(x)) \partial_\mu \gamma(\xi(x))] h^{-1}(\xi(x), g) \\ &\quad - [\partial_\mu h(\xi(x), g)] h^{-1}(\xi(x), g). \end{aligned} \quad (19.6.25)$$

Eq. (19.6.12) gives the quantity $\gamma^{-1} \partial_\mu \gamma$ as a linear combination of xs and ts . Also, the second term on the right-hand side of Eq. (19.6.25) is a linear combination of ts alone. Since the x_a and t_i are all independent, the coefficients of each generator on both sides of Eq. (19.6.25) must all be equal:

$$\sum_a x_a D'_{a\mu} = h(\xi, g) \left(\sum_a x_a D_{a\mu} \right) h^{-1}(\xi, g), \quad (19.6.26)$$

$$\begin{aligned} \sum_i t_i E'_{i\mu} &= h(\xi, g) \left(\sum_i t_i E_{i\mu} \right) h^{-1}(\xi, g) \\ &\quad + i [\partial_\mu h(\xi, g)] h^{-1}(\xi, g), \end{aligned} \quad (19.6.27)$$

or in more detail

$$\begin{aligned}
D'_{a\mu}(x) &= \sum_b \mathcal{D}_{ab}(h(\xi(x), g)) D_{b\mu}(x), \\
E'_{i\mu}(x) &= \sum_j \mathcal{E}_{ij}(h(\xi(x), g)) E_{j\mu}(x) \\
&\quad - \sum_b \mathcal{H}_{ib}(h(\xi(x), g)) \partial_\mu \xi_b(x),
\end{aligned} \tag{19.6.28}$$

where $D_{a\mu}$ and $E_{i\mu}$ are defined by Eqs. (19.6.15) and (19.6.16), and

$$[\partial_\mu h(\xi(x), g)] h^{-1}(\xi(x), g) = i \sum_{ib} \mathcal{H}_{ib}(\xi(x)) t_i \partial_\mu \xi_b(x),$$

$$ht_j h^{-1} = \sum_i \mathcal{E}_{ij}(h) t_i,$$

while $\mathcal{D}_{ab}$ is defined by Eq. (19.6.22). We see that the quantities $D_{a\mu}(x)$ are ‘covariant derivatives’ of the Goldstone boson fields, transforming under a general element $g \in G$ much like the fields $\tilde{\psi}$: both are subjected to the H transformation $h(\xi(x), g)$, though in different representations. For instance, in the $SO(4)$ theory of the previous section, the covariant derivative of the pion field ζ was the quantity $\partial_\mu \zeta / (1 + \zeta^2)$, which is transformed under an infinitesimal chiral transformation by an isospin rotation (19.5.15).

On the other hand, $E_{i\mu}(x)$ transforms inhomogeneously, much like a gauge field. The extra term in Eq. (19.6.28) allows a cancellation of the inhomogeneous term in derivatives of $\tilde{\psi}$. Differentiating Eq. (19.6.20), we have

$$\partial_\mu \tilde{\psi}'(x) = h(\xi(x), g) \left[\partial_\mu \tilde{\psi}(x) + h^{-1}(\xi(x), g) \partial_\mu h(\xi(x), g) \tilde{\psi}(x) \right].$$

Combining this with Eq. (19.6.27) gives

$$\left(\mathcal{D}_\mu \tilde{\psi}(x) \right)' = h(\xi(x), g) \mathcal{D}_\mu \tilde{\psi}(x), \tag{19.6.29}$$

where $\mathcal{D}_\mu \tilde{\psi}$ is a covariant derivative of the heavy particle fields:

$$\mathcal{D}_\mu \tilde{\psi}(x) \equiv \partial_\mu \tilde{\psi}(x) + i \sum_i t_i E_{i\mu}(x) \tilde{\psi}(x). \tag{19.6.30}$$

(Eq. (19.5.47) provides an example of such a covariant derivative in the case of $SO(4)$ broken to $SO(3)$.) *Any Lagrangian that is invariant under the unbroken subgroup H and is constructed from $\tilde{\psi}$, $\mathcal{D}_\mu \tilde{\psi}$, and $D_{a\mu}$ will thus also be invariant under the full group G .*

Renormalizability is not an issue here, so we can also consider quantities involving more than one spacetime derivative. One such is $\mathcal{D}_\nu \mathcal{D}_\mu \tilde{\psi}$, obtained by just repeating the operation (19.6.30):

$$\mathcal{D}_\nu \mathcal{D}_\mu \tilde{\psi} = \left[\partial_\nu + i \sum_j t_j E_{j\nu} \right] \left[\partial_\mu + i \sum_i t_i E_{i\mu} \right] \tilde{\psi}. \tag{19.6.31}$$

This transforms just like $\mathcal{D}_\mu \tilde{\psi}$

$$\left(\mathcal{D}_\nu \mathcal{D}_\mu \tilde{\psi}\right)' = h(\xi, g) \mathcal{D}_\nu \mathcal{D}_\mu \tilde{\psi}. \quad (19.6.32)$$

Another covariant quantity can be constructed from the covariant derivative of the covariant Goldstone boson derivative (19.6.15), which is defined just like (19.6.30), but with t_i replaced with the corresponding matrix in the representation of H furnished by the x_a :

$$\mathcal{D}_\nu D_{a\mu} \equiv \partial_\nu D_{a\mu} + \sum_b C_{iab} E_{i\nu} D_{b\mu}, \quad (19.6.33)$$

which transforms just like $D_{a\mu}$:

$$(\mathcal{D}_\nu D_{a\mu})' = \sum_b \mathcal{D}_{ab}(h, g) \mathcal{D}_\nu D_{b\mu}. \quad (19.6.34)$$

Also, by antisymmetrizing the derivative of $E_{i\mu}$ we can eliminate the inhomogeneous term in (19.6.27). This yields a ‘curvature’

$$R_{i,\mu\nu} \equiv \partial_\nu E_{i\mu} - \partial_\mu E_{i\nu} - i C_{ijk} E_{j\mu} E_{k\nu}. \quad (19.6.35)$$

This is not really new; it is easy to see that the antisymmetric covariant derivative of the ‘matter’ field $\tilde{\psi}$ is proportional to this curvature

$$[\mathcal{D}_\nu, \mathcal{D}_\mu] \tilde{\psi} = \sum_i t_i R_{i,\mu\nu} \tilde{\psi}. \quad (19.6.36)$$

Of course, we can continue this process and construct yet more covariants by taking higher and higher covariant derivatives of $\mathcal{D}_\mu \tilde{\psi}$ and $D_{a\mu}$.

It is useful to note that the quantities $D_{a\mu}$ and $E_{i\mu}$ may be calculated once and for all for given groups G and H and a given parameterization of the coset space G/H , and do not depend on the assumed transformation properties of the fields ψ_n or the specific matrices x_a, t_i used to represent the Lie algebra of the broken symmetry group. For instance, in the exponential parameterization (19.6.12), the first few terms in the power series for these covariants are easily calculated from Eqs. (19.6.12) and (19.6.14) as

$$\begin{aligned} D_{a\mu} &= \partial_\mu \xi_a + \frac{1}{2} \sum_{bc} C_{abc} \xi_b \partial_\mu \xi_c \\ &+ \frac{1}{6} \sum_{bcd} \left[\sum_e C_{cde} C_{bea} + \sum_i C_{cdi} C_{bia} \right] \xi_b \xi_c \partial_\mu \xi_d \\ &+ O(\xi^3 \partial_\mu \xi), \end{aligned} \quad (19.6.37)$$

$$\begin{aligned} E_{i\mu} &= \frac{1}{2} \sum_{ab} C_{abi} \xi_a \partial_\mu \xi_b + \frac{1}{6} \sum_{abcd} C_{acd} C_{bdi} \xi_a \xi_b \partial_\mu \xi_c \\ &+ O(\xi^3 \partial_\mu \xi). \end{aligned} \quad (19.6.38)$$

All we need to know in order to construct the most general Lagrangian involving the Goldstone boson fields ξ_a and a set of heavy particle fields is the pattern of spontaneous symmetry breaking, of G to H , and the transformation of the heavy particle fields under H , not G : the Lagrangian is taken to be the most general H -invariant function of $\tilde{\psi}$, $\mathcal{D}_\mu \tilde{\psi}$, $D_{a\mu}$, and higher covariant derivatives.

Now consider the case where the symmetry under G is not only spontaneously broken, but not even exact to begin with. Suppose that there is a term $\Delta\mathcal{L}$ in the initial Lagrangian that is *not* invariant under the group G of linear transformations $\psi \rightarrow g\psi$, but transforms as a linear combination of the components of some representation (reducible or irreducible) of G . That is, we take

$$\Delta\mathcal{L} = \sum_A c_A \mathcal{O}_A, \quad (19.6.39)$$

where $\mathcal{O}_A$ transforms under G according to some representation $D[g]_{AB}$:

$$\mathcal{O}_A \longrightarrow \sum_B D[g]_{AB} \mathcal{O}_B. \quad (19.6.40)$$

If we now replace the fields ψ with the set $\xi_a, \tilde{\psi}$, this term in the Lagrangian will still take the form (19.6.39), but Eq. (19.6.40) now reads

$$\mathcal{O}_A \left[f_a(\xi, g), D[h(\xi, g)]\tilde{\psi} \right] = \sum_B D[g]_{AB} \mathcal{O}_B[\xi, \tilde{\psi}], \quad (19.6.41)$$

where $f_a(\xi, g)$ is the result of applying the transformation g to ξ_a , defined by Eq. (19.6.18):

$$g\gamma(\xi) = \gamma(f(\xi, g))h(\xi, g). \quad (19.6.42)$$

As we shall see, it is easy to find sets of operators satisfying (19.6.41), and the solution is unique up to a specification of certain H -invariant functions of the $\tilde{\psi}$.

First, consider the case $\xi_a = 0$, with $g = \gamma(\xi')$. In this case $g\gamma(\xi) = \gamma(\xi')$, so

$$f_a(0, \gamma(\xi')) = \xi'_a, \quad h(0, \gamma(\xi')) = 1.$$

Applying this in Eq. (19.6.41), we find (now dropping the prime):

$$\mathcal{O}_A[\xi_a, \tilde{\psi}] = \sum_B D[\gamma(\xi)]_{AB} \mathcal{O}_B[0, \tilde{\psi}]. \quad (19.6.43)$$

This gives the operators $\mathcal{O}_A[\xi_a, \tilde{\psi}]$ for all ξ_a if we know them for $\xi_a = 0$.

Now take $\xi_a = 0$, with g an element h of the unbroken subgroup H . Here we have $g\gamma(\xi) = h$, so

$$f_a(0, h) = 0, \quad h(0, h) = h.$$

Applying this in Eq. (19.6.41), we find here

$$\mathcal{O}_A[0, h\tilde{\psi}] = \sum_B D[h]_{AB} \mathcal{O}_B[0, \tilde{\psi}]. \quad (19.6.44)$$

In other words, the $\mathcal{O}_A$ for $\xi = 0$ must have the same transformation under linear H transformations as are found in the representation (19.6.40) of G . (However, there is nothing in this that fixes the normalization of the different irreducible H representations found in the G representation (19.6.40); some of them may even be absent altogether.)

Finally, we note that any set of $\mathcal{O}$ operators satisfying (19.6.44) allows the construction using Eq. (19.6.43) of operators satisfying the general G -transformation rules (19.6.40). Using Eqs. (19.6.43) and (19.6.44), the left-hand side of Eq. (19.6.41) becomes

$$\begin{aligned}\mathcal{O}_A [f_a(\xi, g), h(\xi, g)\tilde{\psi}] &= \sum_B D [\gamma(f(\xi, g))]_{AB} \mathcal{O}_B[0, h(\xi, g)\tilde{\psi}] \\ &= \sum_{BC} D [\gamma(f(\xi, g))]_{AB} D[h(\xi, g)]_{BC} \mathcal{O}_C[0, \tilde{\psi}] \\ &= \sum_B D [\gamma(f(\xi, g))h(\xi, g)]_{AB} \mathcal{O}_B[0, \tilde{\psi}] \\ &= \sum_B D[g\gamma(\xi)]_{AB} \mathcal{O}_B[0, \tilde{\psi}] \\ &= \sum_B D[g]_{AB} \mathcal{O}_B[\xi, \tilde{\psi}],\end{aligned}$$

as was to be shown.

As an example, consider the group $SO(4)$ spontaneously broken to $SO(3)$ as in the previous section, and suppose we want to construct operators that transform according to the four-vector representation of $SO(4)$ out of the Goldstone boson field $\xi(x)$ and the other fields $\tilde{\psi}(x)$. According to (19.6.43), these must take the form

$$\mathcal{O}_n[\xi, \tilde{\psi}] = R_{nm}(\xi) \mathcal{O}_m[0, \tilde{\psi}],$$

where R is the $SO(4)$ rotation defined by Eqs. (19.5.8) and (19.5.9):

$$R_{a4} = \frac{2\xi_a}{1 + \xi^2}, \quad R_{44} = \frac{1 - \xi^2}{1 + \xi^2}.$$

The condition that R is an orthogonal matrix is satisfied if we choose its other components as

$$R_{ab} = \delta_{ab} - \frac{2\xi_a\xi_b}{1 + \xi^2}, \quad R_{4a} = -\frac{2\xi_a}{1 + \xi^2}.$$

Thus any scalar (as opposed to pseudoscalar) operators that transform as the fourth and third components of chiral four-vectors must appear in the effective Lagrangian in the forms

$$\Phi_4^+[\xi, \tilde{\psi}] = -\left(\frac{2\xi}{1 + \xi^2}\right) \cdot \Phi^+[0, \tilde{\psi}] + \left(\frac{1 - \xi^2}{1 + \xi^2}\right) \Phi_4^+[0, \tilde{\psi}]$$

and

$$\begin{aligned}\Phi_3^-[\xi, \tilde{\psi}] &= \left[\Phi_3^-[0, \tilde{\psi}] - \left(\frac{2\xi_3}{1 + \xi^2}\right) \xi \cdot \Phi^-[0, \tilde{\psi}] \right] \\ &\quad + \left(\frac{2\xi_3}{1 + \xi^2}\right) \Phi_4^-[0, \tilde{\psi}].\end{aligned}$$

The only condition on the operators $\Phi^+[0, \tilde{\psi}]$, $\Phi_4^+[0, \tilde{\psi}]$, $\Phi^-[0, \tilde{\psi}]$, and $\Phi_4^-[0, \tilde{\psi}]$ is that they should transform under Lorentz and isospin transformations respectively as a pseudoscalar isovector, a scalar isoscalar, a scalar isovector, and a pseudoscalar isoscalar, but they do not have to be related in any way. For processes involving only pions we use an effective Lagrangian that involves no fields except the pion field ξ , so the operators $\Phi^+[0, \tilde{\psi}]$, $\Phi^-[0, \tilde{\psi}]$, and $\Phi_4^-[0, \tilde{\psi}]$ must all vanish, while $\Phi_4^+[0, \tilde{\psi}]$ is just a numerical constant. We see as promised in the previous section that the only non-derivative chiral symmetry breaking operator in the effective Lagrangian is proportional to $(1 - \xi^2)/(1 + \xi^2)$, and hence to the operator (19.5.34). In order to include symmetry-breaking operators bilinear in the nucleon field, parity and isospin conservation require that we take

$$\begin{aligned}\Phi^+[0, \tilde{\psi}] &\propto \bar{N} \gamma_5 \mathbf{t} \tilde{N}, & \Phi_4^+[0, \tilde{\psi}] &\propto \bar{N} \tilde{N}, \\ \Phi^-[0, \tilde{\psi}] &\propto \bar{N} \mathbf{t} \tilde{N}, & \Phi_4^-[0, \tilde{\psi}] &\propto \bar{N} \gamma_5 \tilde{N}.\end{aligned}$$

Thus as claimed in the previous section the only non-derivative symmetry-breaking terms involving nucleon bilinears are of the form (19.5.61)–(19.5.64).

The phenomenological Lagrangian may be used to calculate the amplitudes for processes involving particles of small three-momentum in much the same way as was done for the chiral theory of pions and nucleons in the previous section. In counting powers of a characteristic small momentum Q , Goldstone boson internal lines always contribute factors of order Q^{-2} . Also, an internal line for one of the heavy particles (of any spin) described by the fields $\tilde{\psi}$ that has absorbed a net four-momentum q from the Goldstone boson field will have a propagator

$$\frac{N(p+q)}{(p+q)^2 + M^2} \longrightarrow \frac{N(p)}{2p \cdot q}$$

(where N is some polynomial, depending on the particle spin and field type, and M is the particle mass) so it will contribute a factor of order Q^{-1} . The same argument that led to (19.5.55) applies here, and gives the number of powers of Q in a given Feynman diagram as

$$\nu = \sum_i V_i (d_i + h_i/2 - 2) + 2L - E_h + 2, \quad (19.6.45)$$

where V_i is the number of vertices of type i , d_i is the number of derivatives (or for approximate broken symmetries, derivatives or Goldstone boson masses) at a vertex of type i , h_i is the number of heavy particle lines at a vertex of type i , L is the number of loops, and E_h is the number of external lines of heavy particles. In general the coefficient $d_i + h_i/2 - 2$ is non-negative for all interactions allowed by chiral symmetries. The interactions among Goldstone bosons alone always have $d_i \geq 2$, because they must either be at least bilinear in the covariant derivative (19.6.15), or proportional to symmetry-breaking parameters (like quark masses in the previous section) which are of order of the squares of the Goldstone boson masses. Interactions between Goldstone bosons and heavy particles must involve the covariant derivatives (19.6.15) or (19.6.30), and so must have $d_i \geq 1$ as well as $h_i \geq 2$. The only interactions for which $d_i + h_i/2 < 2$ would be trilinear interactions among the heavy particles, which we exclude because we are only considering graphs in which all heavy

hadrons remain non-relativistic. With $d_i + h_i/2 \geq 2$ for all interactions, the leading graphs are those constructed solely from interactions with $d_i + h_i/2 = 2$ and no loops. Corrections are again provided by interactions with more derivatives and/or more heavy-particle fields, and/or one or more loops.

In the absence of intrinsic symmetry breaking, the part of the Lagrangian that involves only the Goldstone boson fields has a unique $d_i = 2$ term

$$\mathcal{L}_{\text{GB}} = -\frac{1}{2} \sum_{ab} F_{ab}^2 D_{a\mu} D_b^\mu, \quad (19.6.46)$$

with F_{ab}^2 some real positive-definite matrix. For (19.6.12) and a large class of other parameterizations of the cosets, $\gamma(\xi)$ approaches $1 + i\xi_a x_a$ for $\xi_a \rightarrow 0$, so Eq. (19.6.14) shows that the linear term in $D_{a\mu}$ is just $\partial_\mu \xi_a$. The canonically normalized Goldstone boson fields π_a with a kinematic Lagrangian $-\frac{1}{2} \sum_a \partial_\mu \pi_a \partial^\mu \pi_a$ are then

$$\pi_a = \sum_b F_{ab} \xi_b. \quad (19.6.47)$$

In the $SO(4)$ theory of the previous section, the generators x_a and the fields ξ_a transform according to an irreducible representation of the unbroken subgroup $H = SO(3)$, so in this case $F_{ab}^2 = F_\pi^2 \delta_{ab}$, with F_π a single constant that characterizes the energy scale of the spontaneous symmetry breakdown. In the general case we can choose the generators (without changing the structure constants) so that F_{ab}^2 is diagonal, with diagonal elements all equal within each irreducible representation of H , but otherwise independent.

* * *

We have seen how to construct theories of Goldstone boson fields by starting with fields belonging to linear representations of the broken symmetry group, such as the $SU(2) \cdot SU(2)$ four-vector $\phi_n(x)$ with which we started in Section 19.5. At very low energy the only significant degrees of freedom are the Goldstone bosons, so we generally throw away the non-Goldstone parts of the field, such as the field $\sigma(x)$ given by Eq. (19.5.4). But there are circumstances in which it is necessary to return to full linear representations of the broken symmetry group, in which the Goldstone boson fields are just one part.

This happens when, by varying the temperature or turning on an external field of some sort, a system is brought close to a second-order transition, in which it smoothly goes from broken to unbroken symmetry. On one side of this transition the symmetry is broken, so we have massless Goldstone bosons, and various other massive excitations, not generally forming complete multiplets that would furnish linear representations of the broken symmetry group. On the other side of the transition the symmetry is unbroken, so here we have complete linear multiplets, not generally massless. If this transition is continuous, then very near the phase transition the Goldstone boson must form part of a complete linear multiplet of nearly massless excitations. Barring accidents or other symmetries, this multiplet will form an *irreducible* representation of the broken symmetry group. This irreducible multiplet of fields, which become massless only just at the phase transition, is known as the *order parameter*.

This is a more precise definition of order parameters than usual. Often one speaks of an order parameter as any set of fields whose expectation values break the symmetry, but this is too vague—there is no one set of fields whose expectation values can be blamed for a broken symmetry. For instance, the $SU(2) \cdot SU(2)$ symmetry of quantum chromodynamics with massless u and d quarks is broken by the expectation value of $\bar{u}u + \bar{d}d$, which is the fourth component of a chiral four-vector, but quartic and higher powers of quark fields also have non-vanishing vacuum expectation values, and these belong to other representations of $SU(2) \cdot SU(2)$. In contrast, at a smooth phase transition at which $SU(2) \cdot SU(2)$ becomes unbroken the Goldstone boson becomes a member of just one massless representation of $SU(2) \cdot SU(2)$, and this provides an unambiguous definition of the order parameter.

It is important to identify the correct order parameters, both in order to calculate the critical exponents discussed in Section 18.5, and to deal with configurations like vortex lines or magnetic monopoles, where, as we shall see in Sections 21.6 and 23.3, there is a singularity at which the broken symmetry becomes unbroken. It is often assumed that the order parameter for the chiral $SU(2) \cdot SU(2)$ symmetry of quantum chromodynamics is a chiral four-vector, but this is not known to be the case [32a].

19.7 Effective Field Theories: $SU(3) \cdot SU(3)$

The approximate $SU(2)$ isospin symmetry of nuclear physics was extended in the early 1960s by Gell-Mann [33] and Ne’eman [34] to an even less exact $SU(3)$ symmetry, which grouped the known baryons and mesons in various irreducible representations: an octet of $\frac{1}{2}^+$ baryons $p, n, \Lambda^0, \Sigma^{\pm,0}, \Xi^{\pm,0}$, an octet of 0^- mesons $K^{\pm,0}, \pi^{\pm,0}, \eta^0, \bar{K}^{\mp,0}$, an octet of 1^- mesons $K^{*\pm,0}, \rho^{\pm,0}, \omega, \bar{K}^{*-0,0}$, and a decuplet of $\frac{3}{2}^+$ baryons $\Delta^{++,+0,-}, \Sigma^{*+0,-}, \Xi^{*0,-}, \Omega^-$. (The η and Ω^- were not discovered until later.) After the successes of the chiral $SU(2) \cdot SU(2)$ symmetry in the mid-1960s, it became natural to suppose that the strong interactions also respect an approximate $SU(3) \cdot SU(3)$ symmetry, which like $SU(2) \cdot SU(2)$ is spontaneously broken to its diagonal subgroup, the $SU(3)$ of Gell-Mann and Ne’eman. Then after the advent of quantum chromodynamics it became clear that this symmetry arises because there are not two but three fairly light quarks: the u and d , and a third quark s which like d has charge $-1/3$. In this case the $SU(3) \cdot SU(3)$ symmetry consists of independent $SU(3)$ transformations (analogous to Eq. (19.4.2)) on the left- and right-handed parts of the u, d, s quark fields:

$$\begin{pmatrix} u \\ d \\ s \end{pmatrix} \longrightarrow \exp \left[i \sum_a (\theta_a^V \lambda_a + \theta_a^A \lambda_a \gamma_5) \right] \begin{pmatrix} u \\ d \\ s \end{pmatrix}. \quad (19.7.1)$$

Here the λ_a are a complete set of traceless Hermitian matrices:

$$\begin{aligned}
\lambda_1 &= \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, & \lambda_2 &= \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, & \lambda_3 &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \\
\lambda_4 &= \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, & \lambda_5 &= \begin{pmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{pmatrix}, & \lambda_6 &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \\
\lambda_7 &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}, & \lambda_8 &= \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix}.
\end{aligned} \tag{19.7.2}$$

They are normalized so that $\text{Tr}(\lambda_a \lambda_b) = 2\delta_{ab}$. The generators of $SU(3) \cdot SU(3)$ are thus represented on the quark fields by the generators $t_a = \lambda_a$ of the unbroken $SU(3)$ symmetry and the broken symmetry generators $x_a = \lambda_a \gamma_5$.

To define the Goldstone boson fields and work out their transformation properties, we note that in the representation furnished by the quark fields, any $SU(3) \cdot SU(3)$ transformation may be written as $\exp(-i\gamma_5 \sum_a \xi_a \lambda_a)$ times a transformation $\exp(i \sum_a \theta_a \lambda_a)$ belonging to the unbroken $SU(3)$ subgroup of $SU(3) \cdot SU(3)$. (The minus sign accompanying ξ is inserted for convenience in later comparing components of this field with the pion fields introduced in Section 19.5.) Hence in this representation, each right coset of $SU(3) \cdot SU(3)/SU(3)$ is represented by the matrix of the form $\gamma(\xi) = \exp(-i\gamma_5 \sum_a \xi_a \lambda_a)$ that it contains, and aside from normalization, the parameters ξ_a of these right cosets may be taken as our Goldstone boson fields. According to Eq. (19.6.18), the transformation rule of these fields is dictated by

$$\begin{aligned}
&\exp \left[i \sum_a (\theta_a^V \lambda_a + \theta_a^A \lambda_a \gamma_5) \right] \exp \left[-i\gamma_5 \sum_a \xi_a(x) \lambda_a \right] \\
&= \exp \left[-i\gamma_5 \sum_a \xi'_a(x) \lambda_a \right] \exp \left[i \sum_a \theta_a(x) \lambda_a \right],
\end{aligned} \tag{19.7.3}$$

with $\theta_a(x)$ some function of θ^V , θ^A , and $\xi(x)$. Also, according to Eq. (19.6.3), the Goldstone-free quark fields $\tilde{q}(x)$ are here defined by

$$q(x) \equiv \begin{pmatrix} u(x) \\ d(x) \\ s(x) \end{pmatrix} = \exp \left[-i\gamma_5 \sum_a \xi_a(x) \lambda_a \right] \tilde{q}(x), \tag{19.7.4}$$

and have the transformation rule given by Eq. (19.6.24):

$$\tilde{q}'(x) = \exp \left[i \sum_a \theta_a(x) \lambda_a \right] \tilde{q}(x). \tag{19.7.5}$$

We could proceed to introduce covariant derivatives (19.6.15) and (19.6.30), and use them to construct chiral-invariant Lagrangian densities, but for chiral symmetries there is a simpler approach available.

Note that the parts of Eq. (19.7.3) that are proportional to $(1 + \gamma_5)$ and $(1 - \gamma_5)$ read

$$\begin{aligned} & \exp \left(i \sum_a \theta_a^L \lambda_a \right) \exp \left(-i \sum_a \xi_a(x) \lambda_a \right) \\ &= \exp \left(-i \sum_a \xi'_a(x) \lambda_a \right) \exp \left(i \sum_a \theta_a(x) \lambda_a \right) \end{aligned} \quad (19.7.6)$$

and

$$\begin{aligned} & \exp \left(i \sum_a \theta_a^R \lambda_a \right) \exp \left(i \sum_a \xi_a(x) \lambda_a \right) \\ &= \exp \left(i \sum_a \xi'_a(x) \lambda_a \right) \exp \left(i \sum_a \theta_a(x) \lambda_a \right), \end{aligned} \quad (19.7.7)$$

where $\theta^L = \theta^V + \theta^A$ and $\theta^R = \theta^V - \theta^A$. By multiplying Eq. (19.7.7) on the right by the inverse of Eq. (19.7.6), we find the simple transformation rule

$$U'(x) = \exp \left(i \sum_a \lambda_a \theta_a^R \right) U(x) \exp \left(-i \sum_a \lambda_a \theta_a^L \right), \quad (19.7.8)$$

where $U(x)$ is the unitary unimodular matrix

$$U(x) = \exp \left(2i \sum_a \xi_a(x) \lambda_a \right). \quad (19.7.9)$$

In other words, $U(x)$ transforms according to the $(\bar{\mathbf{3}}, \mathbf{3})$ representation of $SU(3) \cdot SU(3)$. This is still a non-linear realization of $SU(3) \cdot SU(3)$ because the components of $U(x)$ are not independent; they are subject to the nonlinear constraints $U^\dagger U = \mathbf{1}$ and $\det U = 1$.

The unique $(SU(3) \cdot SU(3))$ -invariant term in the Goldstone boson Lagrangian that is of second order in spacetime derivatives is

$$\mathcal{L}_{2 \text{ deriv}} = -\frac{1}{16} F^2 \text{Tr} \left\{ \partial_\mu U \partial^\mu U^\dagger \right\}, \quad (19.7.10)$$

with F^2 a positive constant to be determined. We can express the ξ_a in terms of conventionally normalized pseudoscalar meson fields by writing

$$\sum_a \lambda_a \xi_a = \frac{\sqrt{2}}{F} \begin{pmatrix} \frac{\pi^0}{\sqrt{2}} + \frac{\eta^0}{\sqrt{6}} & \pi^+ & K^+ \\ \pi^- & -\frac{\pi^0}{\sqrt{2}} + \frac{\eta^0}{\sqrt{6}} & K^0 \\ \bar{K}^- & \bar{K}^0 & -\sqrt{\frac{2}{3}} \eta^0 \end{pmatrix} \equiv \frac{\sqrt{2}}{F} B, \quad (19.7.11)$$

so that the kinetic term in Eq. (19.7.10) is of the usual form

$$\mathcal{L}_{\text{kin}} = -\frac{1}{2}\partial_\mu\pi^0\partial^\mu\pi^0 - \partial_\mu\pi^+\partial^\mu\pi^- - \partial_\mu K^+\partial^\mu\bar{K}^- - \partial_\mu K^0\partial^\mu\bar{K}^0 - \frac{1}{2}\partial_\mu\eta^0\partial^\mu\eta^0.$$

To determine the constant F , we note by comparing Eq. (19.7.4) with Eq. (19.5.42), that for ξ infinitesimal, the components ξ_1 , ξ_2 , and ξ_3 are the same as the Goldstone boson fields ζ_1 , ζ_2 , and ζ_3 introduced in Section 19.5. Then, comparing (19.7.11) with (19.5.17) and (19.5.32), we see that the constant F defined by Eq. (19.7.10) is the same as the $F_\pi = 184$ MeV introduced in Section 19.4.

The $SU(3) \cdot SU(3)$ symmetry of quantum chromodynamics is broken by the quark mass term, which in terms of the quark field $\tilde{q}(x)$ defined by Eq. (19.7.4) is

$$\mathcal{L}_{\text{mass}} = -\bar{q}M_qq = -\tilde{\bar{q}} \exp\left(-i\sqrt{2}\gamma_5 B/F_\pi\right) M_q \exp\left(-i\sqrt{2}\gamma_5 B/F_\pi\right) \tilde{q}, \quad (19.7.12)$$

where

$$M_q = \begin{pmatrix} m_u & 0 & 0 \\ 0 & m_d & 0 \\ 0 & 0 & m_s \end{pmatrix}. \quad (19.7.13)$$

Eq. (19.7.12) contains a purely bosonic part, obtained by replacing the quark bilinear with its vacuum expectation value, given by the unbroken $SU(3)$ and parity symmetries as¹⁵

$$\langle \tilde{\bar{q}}_n \gamma_5 \tilde{q}_m \rangle_0 = 0, \quad \langle \tilde{\bar{q}}_n \tilde{q}_m \rangle_0 = -v\delta_{nm}. \quad (19.7.14)$$

The Goldstone boson mass term in the Lagrangian is then

$$\begin{aligned} -\frac{v}{F_\pi^2} \text{Tr}\{B, \{B, M_q\}\} = & -\frac{v}{F_\pi^2} \left[4m_u \left(\frac{\pi^0}{\sqrt{2}} + \frac{\eta^0}{\sqrt{6}} \right)^2 + 4(m_u + m_d)\pi^+\pi^- \right. \\ & + 4(m_u + m_s)K^+\bar{K}^- + 4m_d \left(-\frac{\pi^0}{\sqrt{2}} + \frac{\eta^0}{\sqrt{6}} \right)^2 \\ & \left. + 4(m_d + m_s)K^0\bar{K}^0 + \frac{8}{3}m_s(\eta^0)^2 \right]. \end{aligned} \quad (19.7.15)$$

From this, we can read off that [35]

$$\begin{aligned} m_{\pi^+}^2 = m_{\pi^0}^2 &= \frac{4v}{F_\pi^2}(m_u + m_d), \\ m_{K^+}^2 &= \frac{4v}{F_\pi^2}(m_u + m_s), \\ m_{K^0}^2 &= \frac{4v}{F_\pi^2}(m_d + m_s), \\ m_{\eta^0}^2 &= \frac{4v}{F_\pi^2} \left(\frac{4m_s + m_d + m_u}{3} \right), \end{aligned} \quad (19.7.16)$$

¹⁵The fact that the conservation laws of parity, charge, and strangeness respected by the vacuum are the same as those respected by the quark mass matrix is a result of the vacuum alignment condition discussed in Section 19.3. Likewise, if the quark fields are defined so that the quark masses are all positive, then with $v > 0$ the minus sign in Eq. (19.7.3) is required by the condition that the true vacuum be at a minimum rather than a maximum of the vacuum energy for vacuum states rotated by $SU(3) \cdot SU(3)$ transformations, for as we shall see, it is this sign that will yield positive masses for the pseudo-Goldstone boson octet.

and there is also a term that mixes the π^0 and η^0 :

$$m_{\pi\eta}^2 = \frac{4v}{\sqrt{3}F_\pi^2}(m_u - m_d). \quad (19.7.17)$$

The same results can also be obtained by operator methods. Noether's theorem allows us to construct generators T_a and X_a , for which

$$[T_a, q] = -\lambda_a q, \quad [X_a, q] = -\gamma_5 \lambda_a q. \quad (19.7.18)$$

We can write the Goldstone boson fields in a real basis as π_a , with $\pi^\pm = (\pi_1 \pm i\pi_2)/\sqrt{2}$, $\pi^0 = \pi_3$, $K^\pm = (\pi_4 \pm i\pi_5)/\sqrt{2}$, $K^0 = (\pi_6 + i\pi_7)/\sqrt{2}$, $\bar{K}^0 = (\pi_6 - i\pi_7)/\sqrt{2}$, and $\eta^0 = \pi_8$. The $SU(3)$ symmetry that survives the spontaneous symmetry breaking tells us that the F -matrix for these bosons takes the form $F_{ab} = F_\pi \delta_{ab}$. The $SU(3) \cdot SU(3)$ symmetry is also intrinsically broken by the mass term in the Hamiltonian $H_1 = m_u \bar{u}u + m_d \bar{d}d + m_s \bar{s}s = \bar{q} M_q q$, where M_q is the quark mass matrix (19.7.13). The mass matrix of the pseudo-Goldstone bosons is given here by (19.3.20) as

$$M_{ab}^2 = -F_\pi^{-2} \langle \bar{q} \{ \lambda_a, \{ \lambda_b, M_q \} \} q \rangle_0. \quad (19.7.19)$$

Since this is already of first order in quark masses, to this order we may use the unbroken $SU(3)$ relations $\langle \bar{u}u \rangle_0 = \langle \bar{d}d \rangle_0 = \langle \bar{s}s \rangle_0 \equiv -v$, and find the results (19.7.16) and (19.7.17), as before.

The masses $m_{K^+} = 493.65$ MeV and $m_{K^0} = 497.7$ MeV of the kaon doublet are quite close, and considerably larger than the pion mass, so we can see from (19.7.16) that the m_u and m_d must be considerably smaller than m_s . In calculating quantities that are sensitive to m_u and m_d , like the kaon mass difference or the pion masses, we should also take into account another small correction, the effect of electromagnetism. The electromagnetic current is $J^\mu = ie \bar{q} \gamma^\mu Q q$, where Q is a diagonal matrix with elements $2/3$, $-1/3$, and $-1/3$. Its commutators with the $SU(3) \cdot SU(3)$ generators are

$$[T_a, J^\mu] = -\frac{ie}{2} \bar{q} \gamma^\mu [Q, \lambda_a] q, \quad [X_a, J^\mu] = -\frac{ie}{2} \bar{q} \gamma^\mu \gamma_5 [Q, \lambda_a] q.$$

We see that J^μ commutes with X_3, X_6, X_7 , and X_8 as well as T_3, T_6, T_7 , and T_8 , so the electromagnetic part of the Hamiltonian is invariant under the $SU(2) \cdot SU(2) \cdot U(1) \cdot U(1)$ subgroup with these generators. Therefore in the limit of zero quark masses, *electromagnetic effects give no mass to the neutral pseudo-Goldstone bosons* [36] $\pi^0, K^0, \bar{K}^0$, and η^0 , the Goldstone bosons associated with the spontaneous breakdown of the symmetries with generators X_3, X_6, X_7 , and X_8 . Also, in the zero-quark-mass limit there is an unbroken $SU(2)$ symmetry generated by T_6, T_7 , and $\sqrt{3}T_8 - T_3$, under which the K^+ and π^+ transform as a doublet, so that for zero quark masses *the electromagnetic corrections to the K^+ and π^+ masses are equal* [36]. Since the quark mass terms and electromagnetic corrections are all small, it is reasonable to treat these effects as additive corrections to the effective quark Hamiltonian. We see then that with electromagnetism taken into account, the mass

formulas (19.7.16) should be corrected to read [37]

$$\begin{aligned}
m_{K^\pm}^2 &= \frac{4v(m_u + m_s)}{F_\pi^2} + \Delta, \\
m_{K^0}^2 &= m_{\bar{K}^0}^2 = \frac{4v(m_d + m_s)}{F_\pi^2}, \\
m_{\pi^\pm}^2 &= \frac{4v(m_d + m_u)}{F_\pi^2} + \Delta, \\
m_{\pi^0}^2 &= \frac{4v(m_d + m_u)}{F_\pi^2}, \\
m_\eta^2 &= \frac{4v(m_u + m_d + 4m_s)}{3F_\pi^2},
\end{aligned} \tag{19.7.20}$$

where Δ is the common electromagnetic correction to the K^+ and π^+ squared masses.

These formulas impose one linear relation on the five pseudo-Goldstone masses, a version of the Gell-Mann–Okubo relation:¹⁶

$$3m_\eta^2 + 2m_{\pi^+}^2 - m_{\pi^0}^2 = 2m_{K^+}^2 + 2m_{K^0}^2. \tag{19.7.21}$$

Taking the kaon and pion masses from experiment, this relation yields an η mass of 566 MeV, in comparison with the experimental value of 547 MeV. The small discrepancy is usually attributed to the mixing of the η state with a heavier pseudoscalar particle, the η' at 958 MeV.

From the mass formulas (19.7.20) we may derive formulas for the quark mass ratios in terms of the pion and kaon masses [37]:

$$\frac{m_d}{m_s} = \frac{m_{K^0}^2 + m_{\pi^+}^2 - m_{K^+}^2}{m_{K^0}^2 + m_{K^+}^2 - m_{\pi^+}^2}, \quad \frac{m_u}{m_s} = \frac{2m_{\pi^0}^2 - m_{K^0}^2 - m_{\pi^+}^2 + m_{K^+}^2}{m_{K^0}^2 + m_{K^+}^2 - m_{\pi^+}^2}. \tag{19.7.22}$$

Using the mass values $m_{\pi^+} = 139.57$ MeV, $m_{\pi^0} = 134.974$ MeV, $m_{K^+} = 493.65$ MeV, and $m_{K^0} = 497.7$ MeV gives the ratios $m_d/m_s = 0.050$ and $m_u/m_s = 0.027$. Thus the ratio of the d and u quark masses is closer to 2 than 1. (A 1996 calculation [38a] using various other pieces of information as well as pseudoscalar meson masses has given values $m_d/m_s = 0.053 \pm 0.002$ and $m_u/m_s = 0.029 \pm 0.003$.)

None of this gives definite values for the individual quark masses. Indeed, these masses are not well defined until we define a renormalization prescription for the quark bilinears.

¹⁶This relation was originally derived [38] on the basis of the approximate $SU(3)$ symmetry of Gell-Mann and Ne'eman generated by the T_a , ignoring mass differences within isotopic multiplets. From this derivation, one cannot tell whether this relation should be applied to the pseudoscalar meson masses themselves, or as here to their squares. Applying this relation to the pseudoscalar masses themselves would not in fact work very well; it gives an η mass of 613 MeV. The fact that the Gell-Mann–Okubo relation works well for the squares of the meson masses, but not for their first powers, supports the interpretation of these particles as Goldstone bosons of a spontaneously broken approximate $SU(3) \cdot SU(3)$ symmetry. Gell-Mann and Okubo also used the approximate $SU(3)$ symmetry generated by the T_a to derive relations among the masses of other particles, such as (ignoring isospin violating effects) the relation $2m_N + 2m_\Xi = 3m_\Lambda + m_\Sigma$ among the masses of the lightest baryon octet. For such multiplets, the average mass is so much larger than the mass differences within the multiplet that it makes little difference whether one applies the relation to the masses or their squares.

Often this prescription is fashioned so that m_s equals the mass difference between isomultiplets differing by one unit of strangeness in some $SU(3)$ multiplet. For instance, the lightest vector meson octet consists of an isotopic doublet K^* at 892 MeV, interpreted as a bound state of an s antiquark and a u or d ; its antidoublet, consisting of an s and a $\bar{u}$ or $\bar{d}$; and a $T = 1$ ρ at 770 MeV and a $T = 0$ ω at 783 MeV, both interpreted as bound states of a u or d and a $\bar{u}$ or $\bar{d}$. If we wish to attribute the difference between the K^* mass and the average of the ρ and ω masses to the relatively large s quark mass, then we must renormalize the quark bilinear so that

$$m_s - \frac{1}{2}(m_u + m_d) = m_{K^*} - \frac{1}{2}(m_\rho + m_\omega) = 120 \text{ MeV},$$

yielding $m_s = 125$ MeV. With the above quark mass ratios, this gives $m_d = 6.0$ MeV and $m_u = 3.3$ MeV. However, these estimates of mass values are much less reliable than the values of the mass ratios given in Eq. (19.7.22). Often m_s is estimated [38b] as 180 MeV rather than 125 MeV.

The mass term (19.7.12) includes meson-meson interactions. Using Eqs. (19.7.14), (19.7.11), and (19.7.9), we may write the purely bosonic part of this term in the Lagrangian as

$$\mathcal{L}_{\text{mass,bosonic}} = \frac{1}{2}v \text{Tr}\{M_q(U^\dagger + U)\}. \quad (19.7.23)$$

The form of this term could have been deduced from general symmetry considerations, which also allow us to find the allowed terms of higher order in M_q . Suppose we invent an external $3 \cdot 3$ field χ , and replace the mass term (19.7.12) in the underlying quantum chromodynamics Lagrangian with the χ -quark coupling term

$$\mathcal{L}_\chi = -\bar{q} \left[\frac{1}{2}(1 + \gamma_5)\chi + \frac{1}{2}(1 - \gamma_5)\chi^\dagger \right] q, \quad (19.7.24)$$

which becomes the same as Eq. (19.7.12) when we make the replacements $\chi = \chi^\dagger = M_q$. The point of this procedure is that the Lagrangian (19.7.24) becomes formally invariant under $SU(3) \cdot SU(3)$ if we give χ the formal transformation rule

$$\chi \longrightarrow \exp \left(i \sum_a \lambda_a \theta_a^R \right) \chi \exp \left(-i \sum_a \lambda_a \theta_a^L \right). \quad (19.7.25)$$

Thus we can work out the allowed bosonic terms involving quark masses by writing the most general $SU(3) \cdot SU(3)$ Lagrangian (up to some given order in derivatives and M_q) involving U and χ , requiring also invariance under the parity transformation

$$U(\mathbf{x}, t) \longleftrightarrow U^\dagger(-\mathbf{x}, t), \quad \chi \longleftrightarrow \chi^\dagger, \quad (19.7.26)$$

and then making the replacements

$$\chi = \chi^\dagger = M_q. \quad (19.7.27)$$

For instance, the interaction $\text{Tr}(U^\dagger \chi + U \chi^\dagger)$ is invariant under $SU(3) \cdot SU(3)$ and parity, and becomes of the same form as Eq. (19.7.23) when we give χ the values (19.7.27).

Using this technique, Gasser and Leutwyler [39] have given the complete effective Lagrangian for the pseudoscalar octet of fourth order in momenta or meson masses (with quark masses counted as being of second order in meson masses) as

$$\begin{aligned}
\mathcal{L}_4 = & L_1 \left[\text{Tr}\{\partial_\mu U^\dagger \partial^\mu U\} \right]^2 \\
& + L_2 \text{Tr}\{\partial_\mu U \partial_\nu U^\dagger\} \text{Tr}\{\partial^\mu U \partial^\nu U^\dagger\} \\
& + L_3 \text{Tr}\{\partial_\mu U \partial^\mu U^\dagger \partial_\nu U \partial^\nu U^\dagger\} \\
& + L_4 \text{Tr}\{\partial_\mu U \partial^\mu U^\dagger\} \text{Tr}\{M_q(U + U^\dagger)\} \\
& + L_5 \text{Tr}\{\partial_\mu U \partial^\mu U^\dagger (M_q U + U^\dagger M_q)\} \\
& + L_6 \left[\text{Tr}\{M_q(U + U^\dagger)\} \right]^2 \\
& + L_7 \left[\text{Tr}\{(U^\dagger - U)M_q\} \right]^2 \\
& + L_8 \text{Tr}\{(UM_q)^2 + (U^\dagger M_q)^2\},
\end{aligned} \tag{19.7.28}$$

where $L_1, \dots, L_8$ are constants to be determined by comparison with experiment. The complete effective Lagrangian up to fourth order in meson masses and momenta is

$$\mathcal{L}_{\text{eff}} = \mathcal{L}_2 + \mathcal{L}_4, \tag{19.7.29}$$

where $\mathcal{L}_2$ is the sum of the terms (19.7.10) and (19.7.23):

$$\mathcal{L}_2 = -\frac{1}{16}F^2 \text{Tr}\{\partial_\mu U \partial^\mu U^\dagger\} + \frac{1}{2}v \text{Tr}\{M_q(U^\dagger + U)\}. \tag{19.7.30}$$

Electroweak interactions may be included by replacing the derivatives ∂_μ with suitable gauge-covariant derivatives D_μ , and adding a few additional terms to $\mathcal{L}_{\text{eff}}$.

Following the same power-counting arguments as in Sections 19.5 and 19.6, to calculate S -matrix elements to fourth order in meson masses and momenta we must include both tree graphs to all orders in $\mathcal{L}_2$ and up to first order in $\mathcal{L}_4$, and also one-loop graphs constructed from $\mathcal{L}_2$ alone. Gasser and Leutwyler and others have carried out a comprehensive study of meson dynamics (and associated electroweak interactions) using this effective Lagrangian [40].

* * *

The quark mass term (19.7.12) naturally affects other $SU(3)$ multiplets. For any multiplet other than the pseudoscalar octet this can be treated as a first-order perturbation, and therefore gives a shift in the mass matrix of a generic multiplet $\|i\rangle$ equal to

$$\delta m_{ij} = \langle i | \bar{q} M_q q | j \rangle, \tag{19.7.31}$$

and $SU(3)$ may be used [41] to relate the various matrix elements $\langle i | \bar{q}_r q_s | j \rangle$. In this way, it is straightforward to show that

$$\frac{\delta m_p - \delta m_n}{m_u - m_d} = \frac{m_\Xi - m_\Sigma}{m_s}. \tag{19.7.32}$$

Using this together with Eq. (19.7.22) gives the quark mass contribution to the nucleon mass splitting $\delta m_p - \delta m_n \simeq -2.5$ MeV. This result may be used in Eqs. (19.5.65) and (19.5.66) to calculate the leading violations of isospin symmetry in low energy pion-nucleon interactions. Of course, as defined here $\delta m_p - \delta m_n$ is not the full proton-neutron mass difference, which receives an important contribution also from photon emission and absorption. Because the neutron is electrically neutral, this electromagnetic term is almost certainly positive, in agreement with the fact that the observed proton-neutron mass difference is -1.3 MeV, leaving $+1.2$ MeV to be accounted for by electromagnetism. Unfortunately, accurate calculation of this electromagnetic mass difference has proved difficult.

19.8 Anomalous Terms in Effective Field Theories¹⁷

In implementing the effective field theory program described in the previous three sections, it is necessary that the action should contain all possible terms allowed by the assumed symmetries of the theory. The methods described in these sections allow us to identify all *manifestly* invariant terms, either by using the general formalism of covariant derivatives described in Section 19.6, or for chiral symmetries by using a linearly transforming field subject to non-linear constraints, like the $U(x)$ of Section 19.7. But it is possible that there may be other terms in the action of the effective field theory that are anomalous in the sense of being given by integrals of four-dimensional Lagrangian densities that are *not* invariant, but whose variation under the broken symmetry is a spacetime derivative, preserving the invariance of the corresponding term in the action.

As we shall see in Section 22.7, such a term was discovered for $SU(3) \cdot SU(3)$ by Wess and Zumino [42], in a study of ‘anomalies’ due to quark loop graphs. However, the structure of this term can be understood without knowing anything about the underlying theory of quarks and gluons.

The easiest way to describe the Wess–Zumino term is by an extension of spacetime to five dimensions, introduced for this purpose by Witten [43]. As long as we require the fields in an effective field theory to approach a common limit as $x^\mu \rightarrow \infty$ in any direction, we can think of spacetime as having the topology of a sphere S_4 , with the point at infinity included as an ordinary point. As remarked in Section 19.6, when a group G is broken to a subgroup H , the possible values of the Goldstone boson fields ξ_a at any spacetime point may be regarded as defining a point in the coset space G/H (the space of elements of G , with two elements identified if they differ by multiplication on the right by elements of H), so a set of functions $\xi_a(x)$ represents a mapping of the spacetime S_4 sphere into G/H . Depending on the topology of G/H , it may be possible smoothly to deform any four-sphere in G/H to a point; that is, it may be possible to extend any $\xi_a(x)$ to a continuous function $\xi_a(x; s)$ defined for $0 \leq s \leq 1$, for which $\xi_a(x; 0) = \xi_a(x)$ and $\xi_a(x; 1)$ is any fixed point (say, $\xi_a = 0$) on the original sphere. Where this is true, it is expressed mathematically as the statement that the homotopy group $\pi_4(G/H)$ is trivial. (For a discussion of homotopy groups, see Section 23.2.) It is known that this is the case for $SU(N) \cdot SU(N)$ spontaneously broken to

¹⁷This section lies somewhat out of the book’s main line of development, and may be omitted in a first reading.

$SU(N)$, where the coset space $SU(N) \cdot SU(N)/SU(N)$ has the topology of $SU(N)$ itself.¹⁸ Hence in the case of physical interest, where $G = SU(3) \cdot SU(3)$ and $H = SU(3)$, we may extend the Goldstone boson fields, or equivalently $U(x)$, (see Eq. (19.7.9)) to a unitary unimodular matrix $U(y)$ defined in a five-dimensional ball B_5 with coordinates x^μ and s , whose surface is the four-dimensional sphere of spacetime.

Now consider the following function formed from $U(y)$:

$$\omega(y) \equiv -\frac{i}{240\pi^2} \epsilon^{ijklm} \text{Tr} \left\{ U^{-1} \frac{\partial U}{\partial y^i} U^{-1} \frac{\partial U}{\partial y^j} U^{-1} \frac{\partial U}{\partial y^k} U^{-1} \frac{\partial U}{\partial y^l} U^{-1} \frac{\partial U}{\partial y^m} \right\}, \quad (19.8.1)$$

where indices i, j , etc. run over the five coordinate directions for the coordinates x^μ and s . (The phase and numerical coefficient are chosen for later convenience.) This is manifestly invariant under the chiral transformations (19.7.8). Also, because ϵ^{ijklm} is a tensor density, the integral of $\omega(y)$ over the five-ball is manifestly independent (up to a sign) of the choice of five-dimensional coordinates y^i . Furthermore, this integral depends only on the values taken by $U(y)$ on the ball's surface; that is, in spacetime. To check this last point, note that when we make an infinitesimal variation $\delta U(y)$ in $U(y)$ in the interior of the ball, $\omega(y)$ changes by a derivative:

$$\delta\omega(y) = -\frac{i}{48\pi^2} \epsilon^{ijklm} \frac{\partial}{\partial y^m} \text{Tr} \left\{ U^{-1} \frac{\partial U}{\partial y^i} U^{-1} \frac{\partial U}{\partial y^j} U^{-1} \frac{\partial U}{\partial y^k} U^{-1} \frac{\partial U}{\partial y^l} U^{-1} \delta U \right\}, \quad (19.8.2)$$

(for this calculation, see Section 23.4), so a change in $U(y)$ that does not affect its value in spacetime also does not affect the integral of (19.8.1) over the five-ball B_5 whose surface is spacetime. We can therefore include this integral as a term in the action:

$$I_{\text{WZW}}[U] = n \int_{B_5} d^5 y \omega(y) \quad (19.8.3)$$

with n a coefficient that so far is arbitrary.

This term may be written as the four-dimensional integral of a Lagrangian density, but not of an $(SU(3) \cdot SU(3))$ -invariant Lagrangian density. Using Eq. (19.7.9), the leading term in $\omega(x)$ in the limit of small meson fields is

$$\omega(x) \longrightarrow \frac{8\sqrt{2}}{15\pi^2 F_\pi^5} \epsilon^{ijklm} \text{Tr} \left\{ \frac{\partial B}{\partial y^i} \frac{\partial B}{\partial y^j} \frac{\partial B}{\partial y^k} \frac{\partial B}{\partial y^l} \frac{\partial B}{\partial y^m} \right\}, \quad (19.8.4)$$

where B is the matrix (19.7.11) of Goldstone boson fields. It follows then from Gauss's theorem that

$$I_{\text{WZW}}[U] = \frac{8\sqrt{2}n}{15\pi^2 F_\pi^5} \epsilon^{\mu\nu\rho\sigma} \int_{S_4} d^4 x \text{Tr} \left\{ B \frac{\partial B}{\partial x^\mu} \frac{\partial B}{\partial x^\nu} \frac{\partial B}{\partial x^\rho} \frac{\partial B}{\partial x^\sigma} \right\} + O\left(\frac{B^6}{F_\pi^6}\right). \quad (19.8.5)$$

Although chiral invariant, this cannot be written as the integral of a chiral-invariant density over spacetime, because any chiral invariant density would have to be constructed out of the first and higher covariant derivatives of the Goldstone boson fields, and so, when

¹⁸This is shown by the fact that we can use $SU(N)$ elements $U(x)$ to represent the right cosets of $SU(N)$ in $SU(N) \cdot SU(N)$.

expanded in powers of the Goldstone boson fields, such an invariant density would begin with a term involving only derivatives of B , not B itself.

As Witten noted, the inclusion of this term in the effective action resolved what otherwise would have been a conflict between the $SU(3) \cdot SU(3)$ effective field theory and experiment. Because no $\epsilon^{\mu\nu\rho\sigma}$ terms appear in the effective Lagrangians (19.7.28) and (19.7.30) (or in higher-order terms of this sort) parity conservation imposes the requirement that these terms are *even* in the Goldstone boson fields, ruling out processes like $K + \bar{K} \rightarrow 3\pi$. Not only is there no symmetry in the underlying theory of quantum chromodynamics that would account for such a selection rule—there is even experimental evidence against it, for as Witten pointed out, the ϕ meson is observed to decay both into $K + \bar{K}$ and 3π final states. Eq. (19.8.5) shows that this unwanted selection rule is removed by the Wess–Zumino–Witten term in the action.

Remarkably, the coefficient of the Wess–Zumino–Witten term is not a freely adjustable parameter. As Witten showed [43], this is because, although this term is not changed by smooth deformations of the function $U(y)$ in B_5 that do not affect $U(x)$ on the spacetime boundary S_4 , the Wess–Zumino–Witten term *can* be changed by a discontinuous change in $U(y)$ that leaves $U(x)$ unchanged on the boundary. We may think of the five-ball B_5 as half of a five-sphere S_5 , with the spacetime S_4 as the border between B_5 and the other half B'_5 . (Think of S_5 as analogous to the surface of the Earth, with the spacetime S_4 as the equator and B_5 and B'_5 as the northern and southern hemispheres.) Because S_4 is also the boundary of the other half of S_5 , we could just as well have written a Wess–Zumino–Witten term as

$$I'_{\text{WZW}}[U] = -n \int_{B'_5} d^5 y \omega(y) \quad (19.8.6)$$

with a minus sign inserted because the boundary of B'_5 is the four-sphere with opposite orientation. It is not possible to require that the terms (19.8.3) and (19.8.6) should be equal for arbitrary Goldstone boson fields without setting $n = 0$, but in order that the weighting factor $\exp(iI)$ in path integrals should be unaffected by the difference between the terms (19.8.3) and (19.8.6), it is only necessary to require that this difference should be 2π times an integer. That is,

$$I_{\text{WZW}}[U] - I'_{\text{WZW}}[U] = n \int_{S_5} d^5 y \omega(y) = 2\pi \cdot \text{integer}. \quad (19.8.7)$$

With the normalization factors we have inserted in the definition (19.8.1) of $\omega(y)$, the integral of $\omega(y)$ over any five-sphere turns out to have the value [44] 2π . It follows that *the coefficient n must be an integer*.

The example of the Wess–Zumino–Witten term raises the question whether there may be other anomalous terms in the action, not necessarily related to quark loops, that also are invariant under $SU(3) \cdot SU(3)$, despite not being four-dimensional integrals of $(SU(3) \cdot SU(3))$ -invariant Lagrangian densities. Fortunately, the answer is no. It has been shown [45] that for a general group G broken to an arbitrary subgroup H (with $\pi_4(G/H) = 0$), any term $F[\xi]$ in the action of the Goldstone boson fields $\xi_a(x)$ may always be written as the integral of a G -invariant five-form Ω over a five-ball B_5 whose boundary is the spacetime

four-sphere S_4 :

$$F[\xi] = \int_{B_5} d^5 y \epsilon^{ijklm} \frac{\partial \xi_a}{\partial y^i} \frac{\partial \xi_b}{\partial y^j} \frac{\partial \xi_c}{\partial y^k} \frac{\partial \xi_d}{\partial y^l} \frac{\partial \xi_e}{\partial y^m} \Omega_{abcde}(\xi(y)). \quad (19.8.8)$$

In order that this should be independent of the particular way that $\xi_a(x)$ is extended into the interior of the five-ball, it is necessary that Ω should be exact, in the sense that there it is the external derivative of a four-form

$$\Omega_{abcde}(\xi) = (\partial/\partial \xi_{[a}) \mathcal{L}_{bcde]}(\xi) \quad (19.8.9)$$

(with square brackets as usual indicating antisymmetrization with respect to the enclosed indices), so that

$$F[\xi] = \int_{S_4} d^4 x \epsilon^{\mu\nu\rho\sigma} \frac{\partial \xi^a}{\partial x^\mu} \frac{\partial \xi^b}{\partial x^\nu} \frac{\partial \xi^c}{\partial x^\rho} \frac{\partial \xi^d}{\partial x^\sigma} \mathcal{L}_{abcd}(\xi). \quad (19.8.10)$$

It follows that Ω is also closed; that is, it has a vanishing exterior derivative

$$(\partial/\partial \xi_{[f}) \Omega_{abcde]}(\xi) = 0. \quad (19.8.11)$$

Where the four-form $\mathcal{L}_{abcd}(\xi)$ is also G -invariant, the functional (19.8.10) is just one of the ordinary manifestly G -invariant terms in the action, discussed in the previous three sections. The anomalous terms in the action arise from the possibility that, although every term in $\Omega(y)$ is G -invariant and the exterior derivative of a four-form, some terms may not be the exterior derivatives of G -invariant four-forms. Thus the new terms in the action may be identified with the closed G -invariant five-forms that are independent, in the sense that no real linear combination of them is the exterior derivative of a G -invariant four-form. These are known in mathematics as the generators of the *de Rham cohomology group* $H^5(G/H; \mathbb{R})$. (The group multiplication rule here is just simple addition.) The de Rham cohomology groups have been calculated for manifolds of various topologies [46]. In particular, $H^5(SU(N) \cdot SU(N)/SU(N); \mathbb{R})$ has a single generator, given by Eq. (19.8.1). Thus without knowing anything about the underlying theory of quarks and gluons, we can learn everything about the anomalous terms in the Goldstone boson action, with the single exception of the value of the integer n . We will see in Section 22.7 that in $SU(N_c)$ gauge theories this integer equals the number N_c of colors, which in quantum chromodynamics is $n = 3$.

19.9 Unbroken Symmetries

We have seen how the properties of Goldstone bosons and their low energy interactions may be deduced from an assumption that the theory is invariant (or approximately invariant) under a group G spontaneously broken to a subgroup H . But in applying these methods to the cases where G is $SU(2) \cdot SU(2)$ or $SU(3) \cdot SU(3)$, we had to take the pattern of symmetry breaking, of $SU(2) \cdot SU(2)$ or $SU(3) \cdot SU(3)$ to their non-chiral $SU(2)$ or $SU(3)$ subgroups, from experiment. In Section 22.5 we shall show that the $SU(3) \cdot SU(3)$ symmetry for massless u , d , and s quarks must in fact be spontaneously broken in quantum chromodynamics, but it is more difficult to show on the basis of quantum chromodynamics

that the $SU(2) \cdot SU(2)$ symmetry with only u and d massless is also spontaneously broken.¹⁹ On the other hand, there is an intuitive argument that their non-chiral $SU(2)$ or $SU(3)$ subgroups are *not* broken, based on a conjecture known as the *persistent mass condition* [48], which states that composite particles will not be massless if the particles of which they are composed are massive. Non-chiral symmetries like isospin conservation are not violated if we give the quarks equal masses, and then if they were spontaneously broken we would have the massless Goldstone bosons formed as composites of massive quarks, in contradiction with the persistent mass condition. In what follows we shall present a proof by Vafa and Witten [49] that in gauge theories like quantum chromodynamics those non-chiral symmetries that are not violated by quark masses can not be spontaneously broken. This result is of more than academic interest; as we shall see in Section 21.4, it is possible that the spontaneous breakdown of electroweak gauge symmetries is described by a ‘technicolor’ theory similar to quantum chromodynamics, and in testing this idea it is important to know what symmetries are left unbroken in this theory.

Consider a gauge theory like quantum chromodynamics, with a number of fermion ‘flavors’ in identical representations of the gauge group. If all fermions have masses, then the theory will be invariant under all unitary global non-chiral transformations on the fermion flavors that commute with the fermion mass matrix. For instance, if n_1 fermions are degenerate with a common mass m_1 , n_2 fermions are degenerate with some other common mass m_2 , and so on, then this global symmetry group is $U(n_1) \cdot U(n_2) \cdots$. (As a special case, if there are no degeneracies we have a global symmetry under $U(1) \cdot U(1) \cdots$; an example is the conservation of baryon number, strangeness, etc. in quantum chromodynamics.) *These symmetries cannot be spontaneously broken.*

To prove this, let us consider a general Greens function for r fermion and antifermion fields.²⁰ In the path-integral formalism this is²¹

$$\begin{aligned} & \langle \Omega \| T \left\{ \Psi_{u_1 k_1}(x_1) \cdots \Psi_{u_r k_r}(x_r) \Psi_{v_1 l_1}^\dagger(y_1) \cdots \Psi_{v_r l_r}^\dagger(y_r) \right\} \| \Omega \rangle \\ &= \frac{1}{Z} \int [dA][d\psi][d\psi^\dagger] \psi_{u_1 k_1}(x_1) \cdots \psi_{u_r k_r}(x_r) \psi_{v_1 l_1}^\dagger(y_1) \cdots \psi_{v_r l_r}^\dagger(y_r) \\ & \quad \cdot \exp \left(iI_{\text{gauge}}[A] + iI_{\text{Dirac}}[\psi, \psi^\dagger; A] \right), \end{aligned} \tag{19.9.1}$$

¹⁹Weingarten [47] has used lattice methods to show that whether or not chiral symmetry is broken, the lightest particle in quantum chromodynamics with massless u and d quarks must have the quantum numbers of the pion. As we will see in Section 22.5, the existence of anomalies due to fermion loops in quantum chromodynamics together with the assumption of quark trapping requires that some hadron be massless, so it follows that the pion is massless, which strongly suggests that chiral symmetry is spontaneously broken.

²⁰In the original work of Vafa and Witten [49], they first proved the absence of symmetry breaking in the case $r = 1$ with $x = y$, then observed that the absence of symmetry breaking in this vacuum expectation value did not rule out the possibility of a spontaneous symmetry breakdown occurring in other Greens functions, and so went on to different methods of proof.

²¹As we saw in Section 15.5, both numerator and denominator are proportional to the infinite volume of the gauge group, which cancels in the ratio (19.9.1). The presence of this infinite factor detracts from the rigor of the following arguments, but if we were to remove it by introducing ghosts then some of the steps below that depend on the positivity of the action would raise difficulties. One way to deal with this problem is to replace the spacetime continuum with a finite lattice of points, in which case the gauge group has a finite volume, and no gauge fixing or ghosts are needed.

where $k_1 \dots k_r$ and $l_1 \dots l_r$ are Dirac spin indices, $u_1 \dots u_r$ and $v_1 \dots v_r$ are flavor indices, $I_{\text{gauge}}[A]$ is the action for a pure gauge theory, $I_{\text{Dirac}}[\psi, \psi^\dagger; A]$ is the action for the Dirac fields in the presence of a gauge field $A_\alpha^\mu(x)$, and Z is the vacuum–vacuum amplitude

$$Z \equiv \int [dA][d\psi][d\psi^\dagger] \exp \left(iI_{\text{gauge}}[A] + iI_{\text{Dirac}}[\psi, \psi^\dagger; A] \right). \quad (19.9.2)$$

It will be necessary here to work in a Euclidean spacetime, with $x^4 = x_4 = ix^0$, $y^4 = y_4 = iy^0$, and $A_\alpha^4 = A_{4\alpha} = iA_{0\alpha}$, all real. (See Appendix A of Chapter 23.) In this case, the Dirac action is

$$I_{\text{Dirac}}[\psi, \psi^\dagger; A] = i \int d^3x \int dx^4 \psi^\dagger [\not{D} + M] \psi, \quad (19.9.3)$$

where M is the fermion mass matrix, and $\not{D}$ is the Euclidean gauge-covariant derivative contracted with the Euclidean Dirac matrices

$$\not{D} = \sum_{i=1}^4 (\partial_i - it^\alpha A_{i\alpha}) \gamma_i, \quad (19.9.4)$$

where as usual $\gamma_4 = i\gamma^0$. Because the action is quadratic in fermion fields, we can explicitly perform the integral over these fields

$$\begin{aligned} & \langle \Omega | T \left\{ \Psi_{u_1 k_1}(x_1) \dots \Psi_{u_r k_r}(x_r) \Psi_{v_1 l_1}^\dagger(y_1) \dots \Psi_{v_r l_r}^\dagger(y_r) \right\} | \Omega \rangle \\ &= \frac{1}{Z} \int [dA] \text{Det}(\not{D} + M) \exp(iI_{\text{gauge}}[A]) \\ & \quad \cdot \left[(\not{D} + M)_{x_1 u_1 k_1, y_1 v_1 l_1}^{-1} \dots (\not{D} + M)_{x_r u_r k_r, y_r v_r l_r}^{-1} \pm \text{permutations} \right], \end{aligned} \quad (19.9.5)$$

where ‘ $\pm$ permutations’ indicates that we must sum over all $r!$ permutations of the ψ fields, with a minus sign for odd permutations, and

$$Z = \int [dA] \text{Det}(\not{D} + M) \exp(iI_{\text{gauge}}[A]). \quad (19.9.6)$$

The expression (19.9.5) is manifestly invariant under any unitary transformation on flavor indices that commutes with the mass matrix M . It does not matter what non-perturbative effects are produced by the functional integral over gauge fields and ghosts; these fields are inert under the symmetries in question here, so that the remaining functional integral in Eq. (19.9.5) cannot break these symmetries.²² But for this argument to be convincing, we must show that the expression (19.9.5) is well defined.

This is not merely an academic question of mathematical rigor. As we saw in Section 19.1, the sign that a symmetry is not spontaneously broken is not just that the ground state is invariant under the symmetry transformations, but also that the symmetry of the ground

²²It is more difficult to show that P , C , and T are not spontaneously broken in quantum chromodynamics with massive quarks, because these symmetries act non-trivially on the gauge fields. Vafa and Witten [50] have applied the methods of Ref. [49] to show that P is not spontaneously broken in quantum chromodynamics.

state is stable under small perturbations. If we break the symmetries of M by adding a small perturbation δM , and if the expression (19.9.5) becomes singular as $\delta M \rightarrow 0$, then factors of δM in symmetry-breaking matrix elements may be cancelled by the singularities in the matrix elements for $\delta M = 0$. Precisely this happens for chiral symmetries, which arise in a symmetry limit where some of the eigenvalues of M vanish. In this case, as we approach the symmetry limit of zero mass, the factors of mass in the numerators of symmetry-breaking expectation values are cancelled by factors of mass in the denominators of the propagators $(\not{D} + M)^{-1}$ wherever $\not{D}$ has zero eigenvalues. For this reason, the arguments here will apply only to the non-chiral symmetries of theories with mass matrices M whose eigenvalues are all non-zero.

To set a bound on the matrix element (19.9.5) in this case, note first that the differential operator (19.9.4) is here antihermitian, so as long as the hermitian matrix M has no null eigenvalues, $\not{D} + M$ has a well-defined inverse. It is still necessary to show that the remaining integration over the gauge fields in Eq. (19.9.5), including that in Z , does not make this expression singular when M satisfies the conditions for a symmetry, for instance that n of its eigenvalues are equal for a $U(n)$ symmetry. As we shall see, this will insure that when a fermion mass matrix M that is invariant under some global symmetry transformation is perturbed by adding a small term δM that breaks this symmetry, the change in the expectation value (19.9.4) under this symmetry transformation vanishes in the limit $\delta M \rightarrow 0$.

It is difficult to show that the coordinate-space Greens functions Eq. (19.9.5) are non-singular, so let us consider instead the matrix element for smeared fields

$$\Psi_u[f] \equiv \int d^4x f^{k\dagger}(x) \Psi_{uk}(x), \quad (19.9.7)$$

where $f^k(x)$ are arbitrary smooth square-integrable functions. In these terms, Eq. (19.9.5) reads

$$\begin{aligned} & \langle \Omega | T \left\{ \Psi_{u_1}[f_1] \cdots \Psi_{u_r}[f_r] \Psi_{v_1}^\dagger[g_1] \cdots \Psi_{v_r}^\dagger[g_r] \right\} | \Omega \rangle \\ &= \frac{1}{Z} \int [dA] \text{Det}(\not{D} + M) \exp(iI_{\text{gauge}}[A]) \\ & \quad \cdot \left[(\not{D} + M)_{f_1 u_1, g_1 v_1}^{-1} \cdots (\not{D} + M)_{f_r u_r, g_r v_r}^{-1} \pm \text{permutations} \right], \end{aligned} \quad (19.9.8)$$

where

$$(\not{D} + M)_{f u, g v}^{-1} \equiv \int d^4x \int d^4y f^{k\dagger}(x) (\not{D} + M)_{x u k, y v l}^{-1} g^l(y). \quad (19.9.9)$$

In this basis, the fermion propagators for a given gauge field are not only well defined, but bounded uniformly in $A_\alpha^\mu(x)$.²³

$$\left| (\not{D} + M)_{f u, g v}^{-1} \right| \leq \sum_a \frac{1}{|m_a|}, \quad (19.9.10)$$

²³To see this, we may expand in the eigenvectors of M

$$M_{uv} c_v^a = m_a c_u^a, \quad c_u^{a*} c_u^b = \delta_{ab},$$

where m_a are the eigenvalues of M , and for convenience we are now normalizing the functions $f_i(x)$ and $g_j(y)$ so that

$$\int f_i^\dagger(x) f_i(x) d^4x = \int g_j^\dagger(y) g_j(y) d^4y = 1. \quad (19.9.11)$$

(The summation convention is suspended here.) Furthermore, the weight function for the average over gauge fields is positive:²⁴

$$\begin{aligned} & \text{Det}(\not{D} + M) \exp(iI_{\text{gauge}}[A]) \\ &= \exp\left(-\frac{1}{4} \int d^3x \int d^4x \sum_{i=1}^4 \sum_{j=1}^4 F_{ij}^2\right) \cdot \sqrt{\text{Det}[(i\not{D})^2 + M^2]}. \end{aligned} \quad (19.9.12)$$

With a positive weight function, the average of any function is bounded by the bound of that function. Using the bound (19.9.10) in Eq. (19.9.8), the matrix element is bounded by

$$\left| \langle \Omega \| T \left\{ \Psi_{u_1}[f_1] \cdots \Psi_{u_r}[f_r] \Psi_{v_1}^\dagger[g_1] \cdots \Psi_{v_r}^\dagger[g_r] \right\} \| \Omega \rangle \right| \leq r! \left[\sum_a \frac{1}{|m_a|} \right]^r. \quad (19.9.13)$$

Thus there are no singularities as the symmetry-breaking terms δM in M go to zero that could cancel factors of δM . To see this more explicitly, we can use the same methods to show that if we perturb M by a small symmetry-breaking term δM , then the term in Eq. (19.9.8) of first order in δM is bounded by

$$\begin{aligned} & \left| \delta \langle \Omega \| T \left\{ \Psi_{u_1}[f_1] \cdots \Psi_{u_r}[f_r] \Psi_{v_1}^\dagger[g_1] \cdots \Psi_{v_r}^\dagger[g_r] \right\} \| \Omega \rangle \right| \\ & \leq r r! \sum_{a,b} \frac{|(\delta M)_{ab}|}{|m_a m_b|} \left[\sum_a \frac{1}{|m_a|} \right]^{r-1}, \end{aligned} \quad (19.9.14)$$

so this vanishes when $\delta M \rightarrow 0$.

In the real world none of the quark masses are zero or degenerate, so it follows immediately from the above arguments that the $U(1)$ symmetries like conservation of baryon number, strangeness, etc., are neither spontaneously nor intrinsically broken. The other strong-interaction symmetries like isospin or $SU(3)$ are more problematic. These symmetries would be completely unbroken if the u and d quarks, or u , d , and s quarks, had equal

and, adapting a trick of Vafa and Witten [49], write Eq. (19.9.9) as

$$\begin{aligned} (\not{D} + M)_{fu,gv}^{-1} &= \sum_a c_u^{a*} c_v^a \int d^4x \int d^4y f^{k\dagger}(x) (\not{D} + m_a)_{xk,yl}^{-1} g^l(y) \\ &= \sum_a \pm c_u^{a*} c_v^a \int_0^\infty \exp(-|m_a|\tau) \int d^4x \int d^4y f^{k\dagger}(x) (\exp(\mp \tau \not{D}))_{xk,yl} g^l(y), \end{aligned}$$

where $\pm$ is the sign of m_a . The matrix $c_u^{a*} c_v^a$ and the operator $(\exp(\mp \tau \not{D}))_{xk,yl}$ are both unitary, so

$$\left| c_u^{a*} c_v^a \int d^4x \int d^4y f^{k\dagger}(x) (\exp(\mp \tau \not{D}))_{xk,yl} g^l(y) \right|^2 \leq \int d^4x |f(x)|^2 \int d^4y |g(y)|^2 = 1.$$

Using this inside the integral yields Eq. (19.9.10).

²⁴In the fermion determinant, we use the fact that $\text{Det}(\not{D} + M) = \text{Det} \gamma_5 (\not{D} + M) \gamma_5 = \text{Det}(-\not{D} + M)$.

non-zero masses, but as we saw in Section 19.7, these masses are not at all degenerate; the isospin and $SU(3)$ symmetries arise because the quark masses are small, not equal. These symmetries do remain unbroken if we give two or three quarks equal masses and let the masses become arbitrarily small, so isospin or $SU(3)$ will be good symmetries in any process that is not sensitive to the small quark masses. But not all processes are insensitive to these masses, because for two or three massless quarks, the pion or the pseudoscalar meson octet to which it belongs becomes massless. This is not a problem for isospin, because as we saw in Section 19.5, the pion triplet remains degenerate to first order in quark masses even though $m_u \neq m_d$. For $SU(3)$ the quark mass differences produce first-order mass differences among the pion, kaon, and eta, so that processes dominated by one-meson poles *can* show large departures from $SU(3)$.

19.10 The $U(1)$ Problem

The success of quantum chromodynamics in explaining the pattern of strong-interaction symmetries seemed at first to be marred by one failure. As we saw in Section 19.5, the broken $SU(2) \cdot SU(2)$ symmetry of the strong interactions is a natural consequence of the smallness of the u and d quark masses. But the Lagrangian of quantum chromodynamics with small u and d quark masses also has another chiral symmetry [51], the $U(1)_A$ symmetry under transformations

$$u \longrightarrow \exp(i\gamma_5\theta)u, \quad d \longrightarrow \exp(i\gamma_5\theta)d. \quad (19.10.1)$$

Such a symmetry if unbroken would, like $SU(2) \cdot SU(2)$, impose a parity doubling on the hadron spectrum, but no such parity doubling is observed. On the other hand, a broken $U(1)_A$ symmetry would imply the existence of an isoscalar 0^- Goldstone boson with a mass comparable to that of the pion, and this Goldstone boson is also not observed. It is true that the η meson is an isoscalar 0^- boson, but it is considerably heavier than the pion, and as we saw in Section 19.7, it is well understood as one of the Goldstone bosons of $SU(3) \cdot SU(3)$. With the s as well as the u and d quarks regarded as relatively light, the Lagrangian of quantum chromodynamics would have a $U(1)$ chiral symmetry in addition to $SU(3) \cdot SU(3)$, under the transformations

$$u \longrightarrow \exp(i\gamma_5\theta)u, \quad d \longrightarrow \exp(i\gamma_5\theta)d, \quad s \longrightarrow \exp(i\gamma_5\theta)s. \quad (19.10.2)$$

The spontaneous breakdown of such a symmetry would require the existence of *two* isoscalar 0^- mesons: one the η , and the other with a mass comparable to that of the pion.

This prediction can be made more explicit [52]. If we include the transformation (19.10.2) among the spontaneously broken symmetries of massless quantum chromodynamics, then in the presence of quark masses we encounter a term (19.7.12):

$$\mathcal{L}_{\text{mass}} = -\bar{q}M_q q = -\bar{\tilde{q}} e^{-i\sqrt{2}\gamma_5 B/F_\pi} M_q e^{-i\sqrt{2}\gamma_5 B/F_\pi} \tilde{q}, \quad (19.10.3)$$

where again

$$M_q = \begin{bmatrix} m_u & 0 & 0 \\ 0 & m_d & 0 \\ 0 & 0 & m_s \end{bmatrix}, \quad (19.10.4)$$

but now B includes a Goldstone boson field ζ for the broken symmetry (19.10.2):

$$B = \begin{bmatrix} \frac{1}{\sqrt{2}}\pi^0 + \frac{1}{\sqrt{6}}\eta^0 & \pi^+ & K^+ \\ \pi^- & -\frac{1}{\sqrt{2}}\pi^0 + \frac{1}{\sqrt{6}}\eta^0 & K^0 \\ \bar{K}^- & \bar{K}^0 & -\sqrt{\frac{2}{3}}\eta^0 \end{bmatrix} + \frac{F_\pi}{\sqrt{3}F_\zeta} \begin{bmatrix} \zeta & 0 & 0 \\ 0 & \zeta & 0 \\ 0 & 0 & \zeta \end{bmatrix}, \quad (19.10.5)$$

with F_ζ the unknown coupling of the $U(1)_A$ Goldstone boson to the corresponding current (and the factor $\sqrt{3}$ inserted for future convenience.) We can again use the unbroken $SU(3)$ symmetry to write the vacuum expectation value of the quark bilinear in the form (19.7.14), and expanding in powers of the boson fields, we find a Goldstone boson mass term in the Lagrangian of the form

$$\begin{aligned} -\frac{v}{F_\pi^2} \text{Tr}\{B, \{B, M_q\}\} = & -\frac{v}{F_\pi^2} \left[4m_u \left(\frac{1}{\sqrt{2}}\pi^0 + \frac{1}{\sqrt{6}}\eta^0 + \frac{F_\pi}{\sqrt{3}F_\zeta}\zeta \right)^2 \right. \\ & + 4(m_u + m_d)\pi^+\pi^- + 4(m_u + m_s)K^+\bar{K}^- \\ & + 4m_d \left(-\frac{1}{\sqrt{2}}\pi^0 + \frac{1}{\sqrt{6}}\eta^0 + \frac{F_\pi}{\sqrt{3}F_\zeta}\zeta \right)^2 \\ & + 4(m_d + m_s)K^0\bar{K}^0 \\ & \left. + 4m_s \left(-\sqrt{\frac{2}{3}}\eta^0 + \frac{F_\pi}{\sqrt{3}F_\zeta}\zeta \right)^2 \right]. \end{aligned} \quad (19.10.6)$$

The charged and strange meson masses are the same as before, but now the neutral non-strange mesons have a mass matrix

$$M_0^2 = 8v \begin{bmatrix} \frac{m_u + m_d}{2F_\pi^2} & \frac{m_u - m_d}{2\sqrt{3}F_\pi^2} & \frac{m_u - m_d}{\sqrt{6}F_\pi F_\zeta} \\ \frac{m_u - m_d}{2\sqrt{3}F_\pi^2} & \frac{m_u + m_d + 4m_s}{6F_\pi^2} & \frac{m_u + m_d - 2m_s}{3\sqrt{2}F_\pi F_\zeta} \\ \frac{m_u - m_d}{\sqrt{6}F_\pi F_\zeta} & \frac{m_u + m_d - 2m_s}{3\sqrt{2}F_\pi F_\zeta} & \frac{m_u + m_d + m_s}{3F_\zeta^2} \end{bmatrix} \quad (19.10.7)$$

with rows and columns listed in the order π^0 , η^0 , and ζ .

In the limit where m_u and m_d vanish, this has two eigenvectors with eigenvalue zero:

$$u_a = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad u_b = \frac{1}{\sqrt{F_\pi^2 + 2F_\zeta^2}} \begin{pmatrix} 0 \\ F_\pi \\ \sqrt{2}F_\zeta \end{pmatrix}. \quad (19.10.8)$$

In this orthonormal basis, the mass-squared matrix to first order in m_u and m_d is

$$m_{aa}^2 \simeq u_a^\top M_0^2 u_a = \frac{4v(m_u + m_d)}{F_\pi^2}, \quad (19.10.9)$$

$$m_{bb}^2 \simeq u_b^\top M_0^2 u_b = \frac{12v(m_u + m_d)}{F_\pi^2 + 2F_\zeta^2}, \quad (19.10.10)$$

$$m_{ab}^2 = m_{ba}^2 \simeq \frac{\sqrt{3}v(m_u - m_d)}{2F_\pi \sqrt{F_\pi^2 + 2F_\zeta^2}}. \quad (19.10.11)$$

The effect of the off-diagonal term (19.10.11) is to decrease the product of eigenvalues by a negligible fractional amount $(m_u - m_d)^2/64(m_u + m_d)^2$, while of course leaving the sum of the eigenvalues unchanged, so the eigenvalues are given to a good approximation by the diagonal elements (19.10.9) and (19.10.10). Comparing Eq. (19.10.9) with Eq. (19.7.16), we see that the particle corresponding to the eigenvector u_a is the π^0 . The other particle, corresponding to u_b , has a mass

$$m_b \simeq \sqrt{m_{bb}^2} \simeq \frac{\sqrt{3}m_\pi F_\pi}{\sqrt{F_\pi^2 + 2F_\zeta^2}} \leq \sqrt{3}m_\pi. \quad (19.10.12)$$

Thus a broken $U(1)_A$ symmetry would require a neutral pseudoscalar Goldstone boson with mass less than $\sqrt{3}m_\pi$, in addition to the pion itself. It hardly needs to be said that no such strongly-interacting particle exists.²⁵ We shall see in Section 23.5 that this problem was eventually solved by the discovery of non-perturbative effects that violate the extra $U(1)_A$ symmetry.

Problems

1. Apply the general theory of broken global symmetries to the case where an $SO(3)$ symmetry group (with generators t_1, t_2, t_3) is spontaneously broken to its $SO(2)$ subgroup (with generator t_3). How do the Goldstone boson fields transform under infinitesimal $SO(3)$ transformations? (Use the exponential parameterization of the coset space $SO(3)/SO(2)$.) Evaluate the covariant derivative $D_{a\mu}$ of the Goldstone boson field, and of a general field with a non-vanishing value q for the unbroken symmetry generator t_3 . What is the most general $SO(3)$ -invariant Lagrangian involving Goldstone bosons alone with no more than two derivatives? Use this Lagrangian to calculate the terms in the invariant amplitude for elastic scattering of the Goldstone bosons to lowest order in their energy. What is the most general $SO(3)$ -invariant Lagrangian with two non-Goldstone field factors and at most one derivative? What is the most general function of the Goldstone boson field alone (with no derivatives) that transforms like the three-component of an $SO(3)$ three-vector. Add this term to the Lagrangian, with a coefficient chosen to give the Goldstone boson a mass m , and recalculate the lowest order amplitude for Goldstone boson scattering.

²⁵Note that taking $F_\zeta \gg F_\pi$ would give the extra neutral scalar a very light mass, but its interactions would be so weak that it might have escaped detection. It seems unlikely that a very large ratio of F_π to F_ζ could arise in quantum chromodynamics, but something like this happens in theories with extra field variables that have been proposed to avoid parity violation by instantons; see Section 23.6.

2. Consider a theory with an $SU(N)$ global symmetry, spontaneously broken to $SU(N-1)$. Suppose we add a small symmetry-breaking perturbation, belonging to the defining representation $\mathbf{N}$ of $SU(N)$. Taking vacuum alignment into account, what symmetry group is completely unbroken? What about the case where the symmetry-breaking perturbation belongs to the adjoint representation of $SU(N)$?
3. Calculate the pion-pion scattering amplitude to one-loop order, taking account of a finite pion mass.
4. Derive the ‘Adler sum rule,’ the analog for the case of pion-pion scattering of the Adler–Weisberger sum rule.
5. Calculate the $SU(2) \cdot SU(2)$ transformation properties of the pion field ξ_a in the case where the cosets of $SU(2) \cdot SU(2)/SU(2)$ are parameterized as $\exp(i\xi_a x_a)$.
6. Use Eq. (19.7.31) and $SU(3)$ symmetry to derive the relation (19.7.32).

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